The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 November 2018.

INCH-POUND

MIL-PRF-19500P
w/AMENDMENT 4
18 May 2018
SUPERSEDING
MIL-PRF-19500P
w/AMENDMENT 3
6 April 2015

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICES, GENERAL SPECIFICATION FOR

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general performance requirements for semiconductor devices. Product assurance is provided by effective screening, conformance inspection, and process controls to mitigate risk. Mission assurance and standardization of parts are the highest priorities. This specification establishes a heritage program of semiconductor devices the military and space community can rely on to be dependable and available. Detail requirements and characteristics are specified in the specification sheets. Revisions to this specification and specification sheets are structured to assure the interchangeability of devices of the same part type regardless of manufacturing date code or conformance inspection (CI) completion date. Four quality levels for encapsulated devices are provided for in this specification, differentiated by the prefixes JAN, JANTX, JANTXV, and JANS. Eight radiation hardness assurance (RHA) levels are provided for the JANTXV and JANS quality levels. These are designated by the letters M, D, P, L, R, F, G, and H following the quality level portion of the prefix. Two quality levels for unencapsulated devices are provided for in this specification, differentiated by the prefixes JANHC and JANKC.

1.2 Description. This specification contains the performance requirement and verification methods for semiconductor devices. The main body specifies the performance requirements and requires the manufacturer to verify that their devices are capable of meeting those performance requirements. Appendix A contains definitions of terms used throughout the specification. Appendix B contains abbreviations and symbols. Appendix C contains the Quality Management (QM) Program. Appendix D contains the quality system. Appendix E contains the standard verification system for qualified products. Appendix F has been cancelled. Appendix G contains discrete semiconductor die/ship lot acceptance. Appendix H contains critical interface and materials for semiconductor devices.

1.3 Identification. The part or identifying number (PIN) schemes are as follows:

a. The PIN for encapsulated semiconductor devices furnished under this specification is formulated as follows:

JANQQQ	A	XN	YYYY	ZZZ
JAN brand and quality level (see 1.3.1)	RHA designator (see 1.3.4)	Component designation (see 1.3.5)	Identification number (see 1.3.6)	Suffix letters (see 1.3.7)

b. The PIN for unencapsulated semiconductor devices furnished under this specification is formulated as follows:

JANQCW	A	XN	YYYY	ZZ
JAN brand quality level and identifiers (see 1.3.2 and 1.3.3)	RHA designator (see 1.3.4)	Component designation (see 1.3.5)	Identification number (see 1.3.6)	Suffix letters (see 1.3.7)

1.3.1 Quality level for encapsulated devices. The quality levels for encapsulated devices includes the JAN brand and associated modifiers as applicable (denoted by "QQQ" in 1.3.a). These quality levels from the lowest level to the highest level are JAN, JANTX, JANTXV, and JANS in accordance with appendix E. JANS is intended for space applications. In specification sheets where the JAN level has been removed or omitted, it is acceptable via this document to manufacture and qualify the JAN assurance level once the qualifying activity has been notified and qualification has been extended to the JAN level. In these cases, the manufacturer will also notify the preparing activity to include the JAN level in the next revision of the applicable specification sheet.

1.3.2 Quality level for unencapsulated devices. The quality levels for unencapsulated devices includes the JAN brand and associated modifiers as applicable (denoted by "QC" in 1.3.b). JANKC is intended for space applications and JANHC is intended for standard military applications.

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1.3.3 Manufacturers and critical interface identifiers. "W" is the place holder for the applicable letter which identifies the manufacturer and the critical interface of a semiconductor die (see 1.3.b).

1.3.4 RHA designator. The RHA designator is a letter which identifies the applicable RHA level (denoted by "A" in 1.3). The RHA levels from lowest to highest are M, D, P, L, R, F, G, and H. (See appendix E, table E-II.)

1.3.5 Component designation. Semiconductor devices are identified by the prefix "XN". The "X" will usually be a number that is one less than the number of active element terminations (see 1.3).

1.3.6 Identification number. It is recommended that each type of semiconductor device intended for standardization be assigned an identification, serially by the JEDEC Solid State Technology Association, JEDEC, 3103 North 10th Street, Suite 240-S, Arlington, VA 22201-2107, www.jedec.org. The assignment will provide the component designation and the identification number.

1.3.7 Suffix letters. The following suffix letters may be incorporated in the military part number as applicable.

A, B, C, etc. (except L, M, R, S, U, P)	Indicates a modified version which is substitutable for the basic numbered (non-suffix) device.
M	Indicates matching of specified parameters of separate devices.
R	Indicates reverse polarity packaging of the basic numbered device.
L or S	Indicates that the terminal leads are longer or shorter, respectively, than those of the basic numbered device.
P	Indicates particle impact noise detection (PIND) screened devices (JANTX, and JANTXV only).
U	Indicates unleaded or surface mounted devices (different package configurations may also include a suffix letter).
UR	Indicates unleaded or surface mounted (round end-cap diodes).
US	Indicates unleaded or surface mounted (square end-cap diodes).
T1, T2, T3, U1, U2, UB, etc.	Indicates package identifier (see H.5)
-1	Indicates metallurgical bond.

Suffix letter(s), except for P, must be used and marked on the device only when specific device types are covered by the applicable specification sheet requiring the suffix letters (see 3.10.6). Suffix letters must be used in the order of appearance in the table. Tolerance must be listed before any case outline designator. A reverse designation must be listed after any part tolerance listing.

1.3.8 Device substitutions. A device of a higher product assurance level may be substituted for the same basic PIN device of a lower product assurance level. Product assurance levels, in descending order of assurance are: JANS, JANTXV, JANTX, and JAN (for chips, JANKC may replace JANHC). RHA devices tested to a higher total dose requirement may be substituted for the same basic PIN device with a lower total dose requirement (see appendix E, table E-II). For axial leaded diodes where the same PIN, both with and without a dash one (-1) suffix exists, the dash one device is considered to be a higher assurance level unless otherwise specified in the specification sheet and may be substituted for the non-dash one part. Non-dash one devices are inactive for new design (whenever dash-one devices exist). For those devices selected to voltage tolerance (e.g., Zener diodes, transient voltage suppressors) the tighter tolerance device may be substituted for one of the more relaxed tolerance devices. For those devices selected to temperature coefficient (e.g., voltage reference diodes) the tighter tolerance device may be substituted for one of the more relaxed tolerance devices. Those devices (including unencapsulated devices) having higher voltage ratings may be substituted for lower voltage ratings, provided all other parameters are equal (e.g., 600 PIV for 100 PIV diode). Devices having suffix letters L or S or no lead length designator, may be substituted for each other in applications where the alternate lead length will fit. Devices having suffix letters U, US, or URS, may be substituted for each other in applications where the alternate end cap will fit. JANTX 'P' devices are substitutable for JANTX devices and JANTXV 'P' devices are substitutable for JANTXV devices. The part being used for substitution may retain its original marking, or the PIN may be remarked. Lot records must maintain traceability of any remarking.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

(See Assist database for list of specification sheets at http://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

JEDEC - SOLID STATE TECHNOLOGY ASSOCIATION (JEDEC)

JESD-31 - General Requirement for Distributors of Commercial and Military Semiconductor Devices.
JESD-625 - Requirements for Handling Electrostatic-Discharge-Sensitive (ESDS) Devices.

(Application for copies should be addressed to JEDEC, 3103 North 10th Street, Suite 240-S, Arlington VA 22201-2107, www.jedec.org.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of a conflict between the requirements of this specification and the specification sheet, and unless otherwise specified (see exception below), the latter shall govern. (If a specific requirement specified herein is not required for an item, it shall be so indicated on the specification sheet; for example, "Shock - N/A").

Exception - Specification sheets containing acceptable quality levels (AQL) and lot total percent defective (LTPD) shall not take precedence over the requirements contained herein. AQLs and LTPDs are no longer acceptable and shall be replaced with the appropriate c = o sampling plans as specified in appendix E.

3.1.1 Implementation date. The qualifying activity has the authority to extend the implementation date on all MIL-PRF-19500 documents. It is the supplier's responsibility to request and to justify any implementation date extension. In addition, the qualifying activity has the authority to disqualify suppliers from any 5961 document which fails to meet the implementation date. Re-qualification and re-audit may be required to determine the root cause of this non-conformance. The qualifying activity will determine the implementation date whenever the specification fails to specify one; however, all suppliers shall be granted a minimum of 90 days (from the date of the specification or standard) to implement all requirements.

3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3). The general requirements of referenced documents shall also apply. Only devices or die listed or approved for listing on the QML which meet all the performance requirements of the applicable specification sheets for the applicable quality level shall be marked and delivered. Any DoD specification or standard referred to in this specification may be replaced by an equivalent commercial standard as determined by the preparing activity and the qualifying activity. If a manufacturer opts to use an equivalent system, it is the responsibility of the manufacturer to demonstrate to the qualifying activity equivalence to the applicable requirements. Conversion from the standard flow(Figure E-1) to alternate flow(Figure E-2) (or alternate to standard flow) requires approval from the qualifying activity.

3.2.1 Certification options. MIL-PRF-19500 will offer two options of JAN (except for JANS) certification (in addition to the diminishing manufacturing sources (DMS) provisions): The traditional QPL qualified products approach (see appendix D and E) and the QML qualified processes approach (see appendix C and E). JANS may not be qualified in accordance with appendix C). Certification to JANTX or JANTXV shall precede JANS certification unless justified and approved by the qualifying activity. They will be clearly identified on the device listings document, QML 19500. Methods of certification are further explained in 6.4.1 and the DLA Land and Maritime-VQE-19500 certification and qualification document. The QML format is explained in 6.4.2.

3.2.2 Allowable alternate design, materials, and construction. Multiple device designs may be approved by the qualifying activity on a case by case basis for JAN, JANHC, JANKC, JANTX, and JANTXV for each manufacturer (for each corporation, for companies within multiple divisions), provided appropriate identification is provided to ensure traceability and a valid need is demonstrated. For JANS devices, only one design is allowed for each manufacturer (for each corporation, for companies within multiple divisions), and it may differ from the design of other quality levels. For radiation hardened devices of all levels, only one design is allowed, and it may differ from the design of non-radiation hardened devices. With the approval of the qualifying activity (see D.3.4.2), design, material, and construction alternatives for qualified device types may be allowed only for a limited time. An alternate design submitted for approval shall be definitive of all pertinent construction features. Particular design features are not interchangeable between approved designs. A single inspection lot, or sublot, in the case of lots made up of sublots of structurally identical devices, shall contain only one approved design, material, and construction so that homogeneity is preserved within a given lot identification code and device type. The qualifying activity shall be notified of the first lot incorporating the change and the last lot of the present existing design and effective date codes for each. If the existing design is to be maintained, the manufacturer shall justify the retention, subject to approval by the qualifying activity. The qualifying activity may periodically identify specific alternate designs by device type and specification sheet and request justification for continued retention of that specific alternate design.

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3.3 Performance requirements for JAN, JANTX, JANTXV, and JANS devices, and JANHC and JANKC die. It shall be demonstrated that parts delivered to this specification are capable of passing the tests and inspections specified in appendix E or appendix G, as required, for the applicable quality level.

3.4 Reference to specification sheets. For purposes of this specification, when the term "specified" is used without references to a specific document, the intended reference is to the specification sheets.

3.5 Certification. The qualifying activity shall verify that the manufacturer's quality system and device verification system meet 4.1, 4.2, and, if applicable, 4.4, and that the manufacturer is producing devices which meet 3.3. Wafer fabrication and assembly operations shall be performed in a facility or facilities certified by the qualifying activity for the applicable technology to the applicable level. The two levels of certification available are (JAN, JANTX, JANTXV) certification and (JANS) certification. There are two options for obtaining certification from the qualifying activity, either Quality Management Program of appendix C or to the Quality System of appendix D. Each manufacturer's certification status is listed in the QPDSIS-19500. Re-audits are required to maintain certification and qualification. The standard re-audit frequency is two years. This certification period may be extended or reduced by the qualifying activity depending on past audit performance and other quality issues. A manufacturer seeking certification should refer to the qualifying activity's' certification and qualification information for manufactures' handbook. Initial and continued certification of the basic plant or any certified facility will depend on past audit performance and other quality and specification issues (i.e. alerts, redesigns, staffing, process, audit history, corrective action history, and design control). The qualifying activity may perform drop-in audits and problem audits as necessary without prior notification.

3.5.1 Transitional certification to appendix C. Manufacturers may be granted transitional certification to appendix C once the manufacturer has completed an audit to appendix C, and has submitted a plan to the qualifying activity and preparing activity for achieving full appendix C implementation. The plan will include self-audit results and milestones identifying the tasks necessary for appendix C compliance. As a minimum, the manufacturer will have implemented C.3.1. The transitional certification will be based on a commitment by the manufacturer to become appendix C certified and if the commitment is not met, the qualifying activity reserves the right to remove the transitional certification.

3.5.2 DMS approval. This provision was created to provide sources of supply for difficult procurement situations. A DMS source has a verification system that qualifies devices for listing based on MIL-STD-750 laboratory suitability and a limited approval of all the quality system requirements that pertain to laboratory suitability, as follows:

a. Conversion of specification sheet requirements (see D.3.3).

b. Document control (see D.3.5) including production travelers.

c. Product identification and traceability (see D.3.8).

d. Inspection and test (see D.3.10).

e. Control of measuring and test equipment (see D.3.11).

f. Inspection and test status (see D.3.12).

g. Handling, storage, packaging, and delivery (see D.3.15).

h. Quality records (see D.3.16).

i. Internal quality audits (see D.3.17).

j. Training (see D.3.18).

In addition, the following shall be addressed as they apply to inspection and test:

k. Management responsibility (see D.3.1.1, D.3.1.2, D.3.1.3.3, D.3.1.3.4, D.3.1.4, D.3.1.5, D.3.1.6).

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l. Quality system (see D.3.2.1, D.3.2.2, D.3.2.3).

m. Control of non-conforming product (see D.3.13).

n. Corrective and preventative action (see D.3.14).

3.5.3 DMS listing. DMS allows qualification of JAN products without certification based on an acceptable qualification report, MIL-STD-750 laboratory suitability, and DMS approval (see 3.10.10 for part marking). DMS approval will be granted on a case-by-case basis by the qualifying activity. The DMS listings may be superseded by any multiple listings of the same part number by certified manufacturers. The qualifying activity will determine when it is appropriate to add or delete a DMS listing from the QML. These listings may be an interim step to achieving a full qualification listing, or a manufacturer's listing may remain as a DMS listing indefinitely.

3.5.4 MIL-STD-750 laboratory suitability. All testing of JAN devices shall be performed at a facility with MIL-STD-750 laboratory suitability, as granted by the qualifying activity, for the applicable test methods (see appendix D).

3.6 Traceability. All devices delivered to this specification shall be identified (see 3.7 and 3.10.1) such that they shall be traceable through the lot identification code and inspection lot records. JAN, JANTX, JANTXV devices shall have a lot control system from wafer processing through conformance inspection which provides date of operation, operator(s) identification, and quantity. In addition, JANS devices shall have a lot control system from wafer processing through conformance inspection which provides wafer lot identification, operation (machine), date of operation, operator(s) identification, quantity, and serial numbers of devices processed.

3.7 Certification of conformance and acquisition traceability. Manufacturers and distributors who offer the products as described in this specification shall provide written certification signed by the company or corporate official who has management responsibility for the production of the products, (1) that the product being supplied has been manufactured and shall be capable of passing the tests in accordance with this specification and conforms to all of the requirements as specified herein, and (2) that all products are as described on the certificate which accompanies the shipment. The responsible official may, by documented authorization, designate other responsible individuals to sign the certificate, but the responsibility for conformance to the facts shall rest with the responsible official. The certification shall be confirmed by documentation to the Government or to users with Government contracts or subcontracts, regardless of whether the products are acquired directly from the manufacturer or from another source such as a distributor. When other sources are involved, their acquisition certification shall be in addition to the certificates of conformance and acquisition traceability provided by the manufacturer and previous distributors. The contract number shall be included on the certificate of the supplier offering the product to the Government. In no case shall the manufacturer's certificate be altered or show signs of alteration. The certificate shall include the following information:

a. Manufacturer documentation:

(1) Manufacturer's name and address.

(2) Customer's or distributor's name and address.

(3) Device type, quality level, and specification sheet number.

(4) Only one lot identification code for each C of C (including plant code).

(5) Conformance inspection acceptance date.

(6) Quantity of devices in shipment from manufacturer.

(7) Statement certifying product conformance and traceability (see 3.6).

(8) Signature and date of transaction.

(9) Solderability inspection date and re-inspection date (only when performed, see 3.11).

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(10) DMS marking in accordance with 3.5.2 and 3.10.10.f herein if applicable.

(11) ESD classification, when not marked on the device.

(12) Statement certifying lead finish is in accordance with 3.14 and appendix H herein.

b. Distributor documentation for each distributor:

(1) Distributor's name and address.

(2) Name and address of customer.

(3) Quantity of devices in shipment.

(4) Certification that this shipment is a part of the shipment covered by the manufacturer's documentation, and an attached copy of the manufacturer's original certification.

(5) Signature and date of transaction.

(6) Certification that authorized dealers and distributors have handled the products in accordance with the requirements of JEDEC Standard JESD-31 and JESD-625.

(7) Device type, quality level, and lot identification code.

(8) Manufacturer’s name.

3.8 Critical interface and materials. Critical interface and materials for devices furnished under this specification shall be such that the devices meet the performance requirements of 3.3 (see appendix H).

3.9 Lead and terminal finish. The lead and terminal finish shall be in accordance with appendix H herein. The lead finishes applicable to each specification sheet are listed on the Supplemental Information Sheet for Electronic QML-19500 (QPDSIS-19500).

3.10 Marking.

3.10.1 Marking on each device. The following marking shall be placed on each encapsulated semiconductor device and shall be legible. The following marking categories shall be visibly separate and distinct from all other markings on the package. Devices having inadequate marking area for all applicable markings shall have as many of the following as possible, in the following order of precedence (with "a." being most important). Unencapsulated semiconductor devices (die) do not require individual device marking. JAN marking shall be performed by the basic plant. Marking may be performed in any wholly owned certified facility with the approval of the qualifying activity. Marking shall not be performed by any contracted facility. The manufacturer shall define and document the orientation of part marking for consistency.

a. Polarity marking, when applicable (see 3.10.5). Pin 1 shall be indicated on devices with more than 2 leads.

b. PIN (see 3.10.6).

c. Manufacturer's name, trademark, or identification (see 3.10.12) or manufacturer's designating symbol (see 3.10.7).

d. Lot identification code and code for plants (see 3.10.8 and 3.10.8.2).

e. Serial number, if applicable (see 3.10.9).

f. DMS marking, if applicable (See 3.10.10).

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g. Special marking (see 3.10.3). Beryllium oxide identifier (see 3.10.3.2). The “P” suffix marking for PIND testing (see 1.3.7) may follow the type number or be marked any place on the device within the marking area.

h. Electrostatic discharge sensitivity (ESDS) identifier (optional see 3.10.3.1).

i. Country of origin (optional see 3.10.11).

3.10.2 Marking on initial container (unit package). All device marking including an ESD label, except for polarity and serial numbers, shall also appear on the unit package used as the initial protection for delivery. See appendix G, section G.6.2 for marking requirements of die unit packaging.

3.10.3 Special marking. If any special marking is used, it shall in no way interfere with, or obscure, the marking required in 3.10.1.

3.10.3.1 ESDS identifier. ESDS testing shall be done in accordance with test method 1020 of MIL-STD-750. The testing procedure of JESD22-A114 may be used in lieu of method 1020 provided the manufacturer is able to demonstrate correlation between the two methods. Unless otherwise specified, tests shall be performed for initial qualification and product redesign. When a device's ESDS class is determined by the ESDS classification test (see E.4.2.1), the devices represented by the test may, at the option of the manufacturer, be marked as follows:

Class 0	Less than 250 V	∆0 - single equilateral triangle (outline or solid) +0 (still acceptable as pin one designator).
Class 1A	250 V to 499 V	∆A - single equilateral triangle (outline or solid) +A (still acceptable as pin one designator).
Class 1B	500 V to 999 V	∆B - single equilateral triangle (outline or solid) +B (still acceptable as pin one designator).
Class 1C	1,000 V to 1,999 V	∆C - single equilateral triangle (outline or solid) +C (still acceptable as pin one designator).
Class 2	2,000 V to 3,999 V	∆∆ - double equilateral triangle (outline or solid) (still acceptable as pin one designator).
Class 3A	4,000 V - 7,999 V	∆∆∆A - triple equilateral triangle (outline or solid) +A (still acceptable as pin one designator).
Class 3B	8,000 V - 15,999V	∆∆∆B - triple equilateral triangle (outline or solid) +B (still acceptable as pin one designator).
Non-sensitive	Above 15,999 V	No designator.

3.10.3.2 Beryllium oxide package identifier. If a semiconductor package contains beryllium oxide (beryllia), the device shall be marked with the designation 'BeO'.

3.10.4 Marking legibility. Marking shall remain legible. Marking damage caused by mechanical handling shall not be cause for lot rejection. Devices having damaged markings shall be remarked prior to shipment to insure legibility.

3.10.5 Polarity marking of unidirectional diodes and thyristors. The polarity shall be indicated by one of the following methods.

3.10.5.1 Diodes.

a. A diode graphic symbol or arrow with the arrow pointing toward the cathode terminal for forward bias.

b. A single contrasting color band or a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

c. An ESD identifier may be used to indicate polarity for sensitive devices.

NOTE: U, US, or UR diodes shall have a cathode band or three dots indicating cathode end.

3.10.5.2 Thyristors. A graphic symbol for a thyristor with the arrow pointing toward the cathode terminal (for stud-mounted thyristors only).

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3.10.6 PIN. Each semiconductor device shall be marked with the PIN which shall be formulated as shown in 1.3. If the device size does not allow the complete PIN to be marked on the device, the component designation portion (see 1.3.5) shall be omitted first, followed by the quality level. RHA designators will be included in the abbreviated PIN as appropriate.

3.10.6.1 JAN and J marking. The United States Government has adopted and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of the specification. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The “JAN“ or “J” certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 2,577,735 for the certification mark "J".

3.10.6.1.1 JAN branded prefix. The type number of all semiconductor devices acquired to and meeting the requirements of this specification, and the applicable specification sheet, shall bear the applicable prefix. In the case of small size semiconductor devices, the abbreviated prefix J (for JAN), JX (for JANTX), JV (for JANTXV), or JS (for JANS) may be used. Failed devices or lots shall have the JAN brand obliterated or removed within 30 days of failure.

3.10.7 Manufacturer's designating symbol. The manufacturer's designating symbol shall be as listed in the QML and assigned by the qualifying activity. The symbol shall be used only by the manufacturer to whom it has been assigned and only on those devices manufactured at that manufacturer's plant. The manufacturer's designating symbol may be abbreviated by omitting the first "C" in the series of letters (see 3.10.12).

3.10.8 Lot identification code. Semiconductor devices shall be marked by a code indicating the last week of sealing for the inspection lot accumulation period. The first two numbers in the code shall be the last two digits of the number of the year. The third and fourth numbers shall be two digits indicating the calendar week of the year. When the number of the week is a single digit, it shall be preceded by a zero. Reading from left to right or top to bottom, the code number shall designate the year and week. For axial diodes, the lot identification code is required for DO-35 package sizes and larger.

3.10.8.1 Lot identification code suffix letter. When more than one inspection lot of a device type sealed within the same lot accumulation period is submitted for conformance inspection, a lot identification code suffix letter, consisting of a single capital letter, shall appear on each semiconductor device immediately following the date code. This letter shall be chosen by the manufacturer so that each inspection lot is uniquely identified by the lot identification code and by the lot identification suffix letter, if one is required.

3.10.8.2 Code for plants. If the devices are assembled or the wafers are fabricated at a plant other than the basic plant, the lot identification code shall include a single letter which uniquely identifies the plant. This plant designator shall appear immediately preceding and adjacent to the date code. The plant designator will be listed with the addresses in the QML. The code for plants may be omitted if adequate marking area is not available.

3.10.8.3 Shipping containers. Mixing of date codes in intermediate shipping containers are not allowed.

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3.10.9 Serialization. JANS devices and, when specified, other device levels shall be marked with a unique serial number assigned consecutively within the inspection lot. JANS devices shall be marked with the serial number in screening, and inspection lot records shall be maintained to provide traceability from the serial number to the specific wafer lot from which the devices originated. For small devices with insufficient area for serialization, lead or body tags may be used. JANS devices are allowed to be shipped without serialization tags as long as it is requested by the customer in writing. In some cases the lead and body tags are not wanted. In these cases, it is better to have the manufacturer remove the tags than it is to have the original equipment manufacturer (OEM) remove them and possibly damage the devices. Once the tags are removed these devices cannot be resold as JANS devices on the open market to anyone other than the original purchaser who has agreed to the removal of the serialization tags.

3.10.10 DMS marking. The DMS mark is not part of the part number, will not be included in the PIN, and shall be distinctly separate from all other marking. DMS may be abbreviated as “D”.

3.10.11 Country of origin. The manufacturer shall indicate the country of origin of the device if other than USA. At the option of the manufacturer the country of origin marking may be omitted from the body of the device but shall be retained on the initial container.

3.10.12 Manufacturer's name, abbreviation, or trademark. At the manufacturer's option, devices supplied to this specification may be marked with the device manufacturer's name, abbreviation, or trademark in place of the designating symbol described in 3.10.7, except for a manufacturer with multiple facility locations. The name or trademark of only the original manufacturer shall appear on the device or initial container. Alteration of name or trademark shall not be permitted.

3.10.13 Marking option. Except for serialization, the manufacturer has the option of marking the entire lot or only the sample devices prior to CI. If the manufacturer exercises the option to mark only the sample devices, the procedure shall be as follows:

a. The sample devices shall be marked prior to performance of conformance or qualification inspection.

b. At the completion of inspection, the marking of the sample devices shall be inspected for conformance with the requirements of 3.10.1 and 3.10.4.

c. The inspection lot represented by a conforming inspection sample shall then be marked and any specified visual and mechanical inspection performed.

d. The marking materials and processing applied to the inspection lot shall be to the same specification as those used for the inspection sample.

3.11 Solderability. All parts shall be capable of passing the solderability test in accordance with method 2026 of MIL-STD-750 on delivery. Re-inspection of solderability is not required by this specification. However, when solderability re-inspection is performed (either at the option of the manufacturer, supplier, or by customer contract), the re-inspection date shall appear on all subsequent certificates of conformance and traceability (see 3.7).

3.12 ESD control. Semiconductors classified as class 0, 1A, 1B, 1C, 2, 3A, or 3B (see E.4.2.1) shall be handled in accordance with JESD-625, or other industry standard practice, to safeguard against discharge damage, as applicable. A minimum humidity level of 30 percent shall be maintained unless approved by the qualifying activity. Work stoppage shall occur when relative humidity drops below the approved minimum. An effective corrective and preventative action(s) shall be implemented prior to work resumption.

3.13 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, environmentally preferable or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.14 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.8)

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3.15 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see appendix E).

b. Screening (see appendix E).

c. Conformance inspection (see appendix E).

d. Element evaluation (see appendix G).

4.2 Quality system. The manufacturer shall implement and maintain a quality system. This system shall assist the manufacturer in producing devices, which meet section 3. See appendix D for the quality system to be maintained by qualified suppliers to this specification. See appendix C for the optional QM program.

4.3 Device verification. The manufacturer shall implement and maintain a verification program which demonstrates that devices meet section 3. Appendix E is the standard verification flow for qualified products, appendix G is the standard verification flow for unencapsulated devices.

4.4 Test modification, reduction, or elimination. Tests and inspections may be modified, reduced, or eliminated with the approval of the preparing activity and the qualifying activity. When a manufacturer modifies, reduces, or eliminates a test or inspection, the manufacturer is only relieved of the responsibility of performing that test or inspection. The manufacturer is still responsible for providing the product, which meets the requirements section 3 herein.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service’s system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the general requirements for semiconductor devices used in electronic equipment. The semiconductor devices covered by this specification are unique due to the fact that these devices must be able to operate satisfactorily in systems under demanding conditions such as: 20 g's vibration, 100 g's of shock, salt atmosphere, wide temperature range (e.g. -55°C to +150°C). In addition, these requirements are verified under a qualification system. Commercial components are not designed to withstand these environmental conditions.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. PIN (see 1.3).

d. Number of the applicable specification sheets (see 3.1). For additional information on PINs, contact DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or https://landandmaritimeapps.dla.mil/programs/milspec/DocSearch.aspx.

e. Lead formation, length, finish, if other than that specified, or when a choice is required by the device application.

f. Data requirements, when applicable.

g. Specify point of shipment.

h. PIND screening when required.

i. Date code and specification revision letter should not be restricted by the acquisition documents (see 3.10.8).

6.2.1 Order requirements. The date code and specification revision letter should not be restricted by the acquisition documents (see 3.10.8). The JAN brand or the abbreviations (see 3.10.6.1) must not be used on any semiconductor device acquired under contracts which permit or require any changes to this specification or the applicable specification sheet, except for:

a. Lead length.

b. Lead finish.

c. PIND testing. PIND screening to requirements beyond those specified herein may be performed only when imposed by the acquisition document upon the component manufacturer. A new PIN will be created in accordance with 1.3 and 3.10.3. Devices which pass such screens may be JAN branded or may retain the JAN brand if already marked. All devices failing such screens must not be JAN branded or, if already marked, must have the JAN brand removed or the device destroyed. PIND screening will follow the flow outlined in Figure E-4 unless otherwise specified by the customer.

d. Lead forming (see E.5.3.4 and E.5.4.2). The forming of leads must not be performed except for specific customer orders where the lot is controlled throughout processing by specific lot travelers. The bend must not be closer than .050 inch (1.27 mm) to the lead to package seal.

e. Additional marking.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of a contract, qualified for inclusion in the Qualified Manufacturers List (QML), whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. See the "Certification and Qualification Information for Manufacturers" booklet (DLA Land and Maritime-VQE-19500) for details on the certification and qualification program. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 Supersession information. Devices covered by specification sheets are substitutable for the manufacturer's and user's PIN's. This information in no way implies that manufacturer's PINs are suitable as a substitute for the military PIN. The table below is an example of the type of cross reference data that should be available in the specification sheets.

Supersession and cross reference data.

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN

6.4.1 Methods of qualification. There are two methods to supply JAN semiconductors to MIL-PRF-19500. Appendix D and appendix E defines the historical screening, sampling, and quality system approach formerly listed on the QPL. Appendix C defines the QML approach. It includes the following important features:

a. Technical Review Board (TRB).

b. Optimization.

c. Quality planning.

d. Statistical techniques.

e. On-going reliability program (fit rate prediction).

f. QM plan.

g. Product characterization and capability evaluation.

h. Failure mode and effects analysis (FMEA).

One method is not preferred over the other. When large volume device production is typical, the qualified manufacturer approach, which focuses on qualifying the process and production line (see appendix C), offers significant advantages to the supplier. Small companies with modest volumes often find the historical qualified products approach more cost effective. Suppliers should choose their method based on their business environment. Leveraging off high volume commercial lines has proven to be effective in the Microcircuit QML. However, the majority of MIL-PRF-19500 suppliers do not have similar high volume commercial production to leverage from.

6.4.2 QML format. QPDSIS-19500 will be divided into two sections. The first section will include all device listings qualified by the manufacturer. This section lists part number, electrostatic discharge (ESD) class, manufacturer's designating symbol, qualification reference, specification sheets, qualified products approval status (appendix D certification or DMS approval) status, and manufacturer's plants. The second section alphabetically lists manufacturers with the following information as a minimum:

a. Manufacturers' name, manufacturers' management representative, address, fax, e-mail, and phone number.

b. Location of wafer fabrication, assembly, and test operations.

c. Contracted operations (if applicable).

d. Commercial and Government Entity (CAGE) code and manufacturing designating symbol and logo.

e. Qualified products approval status (appendix D certification, or DMS approval).

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6.5 Subject term (key word) listing:

Conformance inspection
Double plug construction
Electrostatic discharge sensitivity
Failure analysis
Metallurgical bond
Qualification
Qualified Manufacturers Listing
Radiation hardness assurance
Statistical process control
Technical review board
Test optimization
Thermal match
Thermal response

6.6 PIN. This specification requires a PIN that describes codification and/or classification and appropriate references to associated documents (see 1.3 and 3.1 herein).

6.7 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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DEFINITIONS

A.1 SCOPE

A.1.1 Scope. This appendix contains the definitions of terms used with semiconductor devices. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.2 APPLICABLE DOCUMENTS. This section not applicable to this appendix.

A.3 SEMICONDUCTOR COMMON DEFINITIONS

A.3.1 Absolute maximum ratings. The values specified for "ratings", "maximum ratings", or "absolute maximum ratings" are based on the "absolute system" and unless otherwise required for a specific test method, are not to be exceeded under any service or test conditions. These ratings are limiting values beyond which the serviceability of any individual semiconductor device may be impaired. Unless otherwise specified, the voltage, current, and power ratings are based on continuous dc power conditions at free air ambient temperature of +25°C. For pulsed or other conditions, or operation of similar nature; the current, voltage, and power dissipation ratings are a function of time and duty cycle. In order not to exceed absolute ratings, the equipment designer has the responsibility of determining an average design value, for each rating, below the absolute value of that rating by a safety factor, so that the absolute values will never be exceeded under any usual conditions of supply-voltage variation, load variation, or manufacturing variation in the equipment itself.

A.3.2 Ambient temperature. Ambient temperature is the air temperature measured below a semiconductor device, in an environment of substantially uniform temperature, cooled only by natural air convection, and not materially affected by reflective and radiant surfaces.

A.3.3 Anode. The electrode from which the forward current flows within the device.

A.3.4 Assembly lot. A collection of devices manufactured on the same production line by means of the same production technique, controls, and design.

A.3.4.1 Sublot. A sublot shall be a single (non JANS) device type (part number) contained on a single specification sheet manufactured on the same production line(s) through final seal and to the same device design with the same material requirements and within the same 6-week lot accumulation period unless a different lot accumulation period is specified. A sublot is a single device type(part number) in an inspection lot. May contain parts from multiple assembly lots.

A.3.5 Specification sheet. Since this specification covers the general requirements and tests for semiconductor devices, the details of performance of the semiconductor device shall be specified in the specification sheet. Items listed below should be covered in the specification sheet:

a. PIN (see 1.3).

b. Generic design, construction, and material (see appendix H).

c. The marking to be omitted, if any (see 3.10). The order of precedence for marking is as listed in 3.10.1.

d. Classification of inspection.

e. Examination and tests to be performed under qualification inspection.

f. Examination and tests to be performed under CI.

g. Examination and tests to be performed under screening inspection.

h. End-point measurements to be made for group A, B, C, D, and E inspections (see appendix E).

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i. Quality levels covered.

j. Device level and screening procedure if other than table E-IV.

k. Sequence of test, test method, test condition, limit, cycles, temperature, or axis; when not specified, or if other than specified herein.

l. Interim (pre- and post-burn-in) electrical parameters.

m. Burn-in test condition and burn-in test circuit.

n. Delta parameter measurements or provisions for percent defective allowed (PDA) including procedures for traceability, where applicable.

o. Final electrical measurements.

p. Requirements for data recording and reporting, where applicable.

A.3.5.1 MIL-STD-750 details. In addition to the items as specified in A.3.5, the applicable details required by MIL-STD-750 should be listed in the specification sheet.

A.3.6 Blocking. A term describing the state of a semiconductor device or junction which eventually prevents the flow of current.

A.3.7 Breakdown voltage. The breakdown voltage is the maximum instantaneous voltage, including repetitive and nonrepetitive transients, which can be applied across a junction in the reverse direction without an external means (circuit) of limiting the current. It is also the instantaneous value of reverse voltage at which a transition commences from a region of high small-signal impedance to a region of substantially lower small-signal impedance.

A.3.8 Case mount. A type of package (outline) which provides a method of readily attaching one surface of the semiconductor device to a heat dissipater to achieve thermal management of the case temperature (example: TO-3, DO-4).

A.3.9 Case outline. The embodiment of the external geometric characteristics of an electronic device including dimensions and dimensional tolerances.

A.3.10 Case temperature. Case temperature is that temperature measured at a specified point on the case of a semiconductor device.

A.3.11 Cathode. The electrode to which the forward current flows within the device.

A.3.12 Characteristic. An inherent and measurable property of a device. Such a property may be electrical, mechanical, thermal, hydraulic, electromagnetic, or nuclear, and can be expressed as a value for stated or recognized conditions. A characteristic may also be a set of related values usually shown in graphical form.

A.3.13 Control plans. Control plans are written descriptions of the system for controlling parts and processes. They are written by suppliers to address the importance characteristics and engineering requirements of the product. Each part shall have a control plan, but in many cases “family “ control plans can cover a number of part products using a common process.

A.3.14 Constant current source. A current source shall be considered constant if halving the generator impedance does not produce a change in the parameter being measured that is greater than the required precision of the measurement.

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A.3.15 Constant voltage source. A voltage source shall be considered constant if doubling the generator impedance does not produce a change in the parameter being measured that is greater than the required precision of the measurement.

A.3.16 Disc type. A type of package (outline) for very high power devices which provides two parallel surfaces for mounting into a specialized heat dissipator capable of applying a specified compressive force to the device.

A.3.17 Engineering evaluation. An engineering evaluation is an in-depth scientific investigation of anomalies or potential problems to determine root cause and provide solutions. An analysis of a possible non-conformance or quality defect will be performed by appropriate engineers or applicable technical experts. An assessment of the damage or non-conformance will be documented and corrective action will be implemented. Conclusions can be based on visual inspection, data, past history, failure analysis, destructive physical analysis (DPA), delidding, step stress testing, electric screening, or any reasonable tool or combinations of tools. A detailed report shall be documented and made available to the qualifying activity.

A.3.18 Expanded metallization. Expanded metallization is metallization that increases in area (example, metal line to bond pad area) (see figure A-1).

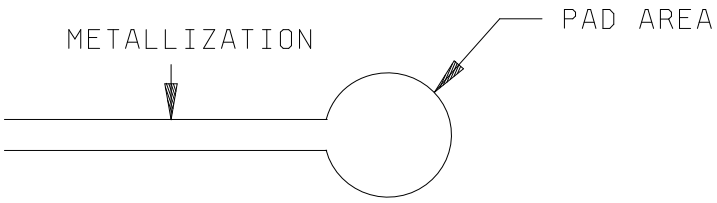


FIGURE A-1. Example of expanded metallization.

A.3.19 Failure analysis. Failure analysis (FA) is the discipline incorporating all techniques of physical, chemical and electro-magnetic (EM) characterization of product and materials to isolate the mode of failure and define the mechanism of occurrence. FA is used as an integral tool in root cause determination and process-segment targeting in the corrective action procedure.

A.3.20 Forward bias. The bias which tends to produce current flow in the forward direction (p-type semiconductor region at a positive potential relative to n-type region).

A.3.21 Failure mode and effects analysis (FMEA). A FMEA is an analytical technique used as a means to assure potential failure modes and their associated causes/mechanisms have been considered and addressed. The process and design FMEAs should identify process and design failure modes, and identify process and design variables on which to focus controls for detection and occurrence reduction. It develops a list of potential failure modes and establishes a priority system for corrective action considerations. Creation of the FMEA should start with the process flow. A potential failure mode is defined as the manner in which the process could potentially fail to meet the process requirements or design intent. FMEAs also include information on potential effects of failure, severity level, potential causes, current process controls, as well as other information.

A.3.22 Hermetically sealed package A hermetically sealed package is considered to be airtight and impervious to external influences. A device is considered to be hermetic if the device passes the requirements of test method 1071 of MIL-STD-750.

A.3.23 Impulse waveform. A pulse with a defined virtual front and impulse duration for either a voltage or current amplitude of unidirectional polarity.

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A.3.23.1 Virtual front duration. The pulse time as defined by 1.67 times time for voltage to increase from 30 percent to 90 percent of crest (peak value) or 1.25 times time for current to increase from 10 percent to 90 percent of crest.

A.3.23.2 Impulse duration. The time required for an impulse waveform to decay to 50 percent of the peak value measured from the start of the virtual front duration of zero crossover.

A.3.24 Internal Free Cavity Volume. The internal volume of a device less the area/volume of the die and any conformal coating.

A.3.25 Line. A collection of similar wafer fabrication flows, or similar package assembly flows used to manufacture semiconductors in accordance with a specified process flow.

A.3.26 Metallurgical bond, diode construction, and thermal matching. Metallurgical bonds as used in JAN-brand semiconductor devices will be identified by one of the following categories. The listing of the three types of metallurgical bonds is for clarification and may not necessarily be listed in order of merit.

A.3.26.1 Double plug construction. Double plug construction is one where the terminal plugs have equal nominal diameters, except for a Category I metallurgical bond device which may utilize an asymmetrical plug design. Plug contact with the semiconductor die may be achieved either through direct contact with the die metallization materials or via brazing or solder preform metallization. The use of a point contact whisker or other wire conductors is not allowed.

A.3.26.2 Dash-one construction. Dash-one diodes shall be of double plug construction utilizing high temperature metallurgical bonding between both sides of the silicon die and attach preform or terminal pins.

A.3.26.3 Category I metallurgical bond. A category I metallurgical bond is formed when the bond between the semiconductor element (such as silicon or germanium) and the package consists of a phase which melts during the bonding process and which includes in the solidified melt both a portion of the semiconductor element and a portion of the metallization layer which is on the package mounting surface. Category I bonds between adjacent semiconductor elements (as in stacks) shall include portions of both semiconductor elements in the solidified melt. Unless otherwise specified in the specification sheet, category I metallurgical bonds are typically required for all axial leaded diodes, equal to and greater than 1 watt or 1 amp.

A.3.26.4 Category II metallurgical bond. A category II metallurgical bond is formed utilizing a brazing or soldering alloy which melts during the bonding process and bonds to a metallization layer on each of the surfaces being joined. Dissolution of the semiconductor element or any of the wetted surface layers is not required.

A.3.26.5 Category III metallurgical bond. A category III metallurgical bond is formed when the surfaces to be bonded are brought together under conditions of temperature and pressure such that a diffusion bond is formed between the outermost metallization layer of the elements being joined. This bond is characterized by having species from both sides of the original interface diffused across the interface without any molten phase having been present.

A.3.26.6 Non-cavity double plug diode. Double plug construction where the package glass is in intimate contact with the semiconductor die isolating the anode and cathode regions, and insuring immunity from particle related failures. Voids may be present provided isolation (to prevent arcing) and particle immunity are insured.

A.3.26.7 Thermally matched axial leaded diodes. Diode construction with the coefficients of thermal expansion of the die, plug, and package materials will be thermally matched such that the diodes are immune to intermittent opens caused by thermal stress. Axial-leaded diode critical interfaces which comply with this definition shall utilize tungsten or molybdenum plugs.

A.3.26.8 Metallurgically bonded-thermally matched-noncavity double plug construction. Diodes meeting this definition shall be of double plug construction utilizing tungsten or molybdenum plugs. Both sides of the diode chip (die) shall be bonded to the corresponding plug by a category I metallurgical bond. The package shall be thermally

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matched, non-cavity construction (see A.3.23.1 through A.3.23.3, A.3.23.6, and A.3.23.7). The plated silver button contact design is not permitted.

A.3.26.9 Soft glass diode construction.A Diode construction with the following characteristics - Axial lead and surface mount (round end-cap), double plug, opaque or transparent body, Dumet plug. Category II or III.

A.3.26.10 Hard glass diode construction. A Diode construction with the following characteristics - Axial lead and surface mount (square end-cap), double plug, opaque or transparent body, tungsten or molybdenum plug. Category I or III.

A.3.27 Noise figure. At a selected input frequency, the noise figure is the ratio of the total noise power per unit bandwidth (at a corresponding output frequency) delivered to the output termination, to the portion thereof contributed at the input frequency by the input termination, whose noise temperature is standard (293°K) at all frequencies.

A.3.28 Open circuit. A circuit shall be considered as open circuited if halving the magnitude of the terminating impedance does not produce a change in the parameter being measured greater than the specified accuracy of the measurement.

A.3.29 Package type. A package type is a package which has the same case outline (varying only in size), configuration, materials (including bonding, wire, or ribbon and die attach), piece parts (excluding preforms which differ only in size), and assembly processes.

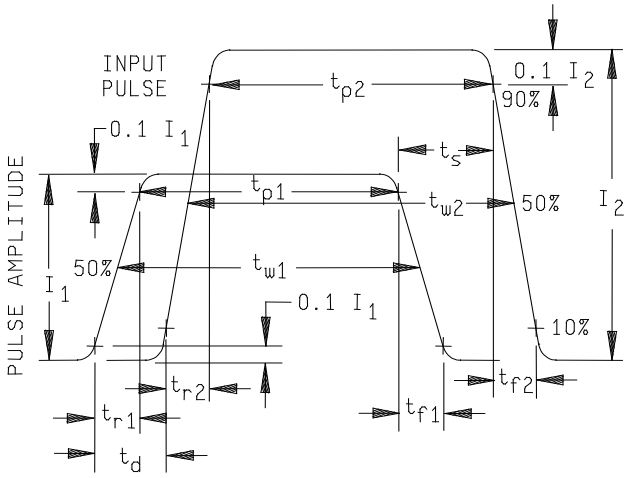
A.3.29.1 Case isolated package. A package in which the active die connection terminals are electrically isolated from conductive surfaces of the case including the stud, header, sidewalls, or lid.

A.3.29.2 Metal Electrode Leadless Face (MELF).. A leadless cylindrical surface mount diode that is metallized at its ends.

A.3.30 Power device.. A diode with a power rating of 1 watt or higher or current rating of 1 Amp or larger, transistors with a power rating of 4 watts or larger, and MOSFETs with a power rating of 1 watt or larger.

A.3.31 Pulse. A pulse is a flow of electrical energy of short duration. See figure A-2 for illustrations of the characteristics defined in A.3.29 through A.3.34, inclusive.

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t_r = Pulse rise time.

t_d = Pulse delay time.

t_p = Pulse time.

t_s = Pulse storage time.

t_f = Pulse fall time.

t_w = Pulse average time.

I = Pulse amplitude.

NOTES:
1. Subscript 2 denotes output.
2. Subscript 1 denotes input.

FIGURE A-2. Pulse measurements.

A.3.32 Pulse average time. The average pulse time of a pulse is the time duration from a point on the leading edge which is 50 percent of the maximum amplitude to a point on the trailing edge which is 50 percent of the maximum amplitude.

A.3.33 Pulse delay time. The delay time of a pulse is the time interval from a point at which the leading edge of the input pulse has risen to 10 percent of its maximum amplitude to a point at which the leading edge of the output pulse has risen to 10 percent of its maximum amplitude.

A.3.34 Pulse fall time. The fall time of a pulse is that time duration during which the amplitude of its trailing edge is decreasing from 90 to 10 percent of the maximum amplitude.

A.3.35 Pulse rise time. The rise time of a pulse is that time duration during which the amplitude of its leading edge is increasing from 10 to 90 percent of the maximum amplitude.

A.3.36 Pulse storage time. The storage time of a pulse is the time interval from a point 10 percent down from the maximum amplitude on the trailing edge of the input pulse to a point 10 percent down from the maximum amplitude on the trailing edge of the output pulse.

A.3.37 Pulse time. The pulse time of a pulse is the time interval from the point on the leading edge which is 90 percent of the maximum amplitude, to the point on the trailing edge which is 90 percent of the maximum amplitude.

A.3.38 Radiation failures. A radiation failure is defined at the lowest radiation level when either any device parameter exceeds its specified post irradiation parameter limits (PIPL) or the device fails any functional test in accordance with stated test conditions.

A.3.39 Radiation hardness assurance (RHA). That portion of performance verification testing that assures that parts meet the radiation response characteristics as specified in this specification and the specification sheet.

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A.3.40 Rating. The nominal value of any electrical, thermal, mechanical, or environmental quantity assigned to define the operating conditions under which a component, machine, apparatus, or electronic device is expected to give satisfactory service.

A.3.40.1 Targeted parameter. Parameters that consist of a min/max or range for the limit, specified in the specification sheet.

A.3.41 Reverse bias. The bias which tends to produce current flow in the reverse direction (n-type semiconductor region at a positive potential relative to the p-type region).

A.3.42 Semiconductor devices. Electronic device in which the characteristic distinguishing electronic conduction takes place within a semiconductor.

A.3.42.1 Die design A semiconductor die manufactured using the same physical size, materials, topology, mask set, process flow, on a single fabrication line.

A.3.43 Semiconductor diode. A semiconductor device having two terminals and exhibiting a nonlinear voltage-current characteristic.

A.3.44 Semiconductor junction. A region of transition between semiconductor regions of different electrical properties (e.g., n-n+, p-n, p-p+ semiconductors) or between a metal and a semiconductor.

A.3.45 Short circuit. A circuit shall be considered short-circuited if doubling the magnitude of the terminating impedance does not produce a change in the parameter being measured that is greater than the specified accuracy of the measurement.

A.3.46 Small signal. A signal shall be considered small if doubling its magnitude does not produce a change in the parameter being measured that is greater than the specified accuracy of the measurement.

A.3.47 Stack junction rectifier diode. A semiconductor device having two terminals and exhibiting a nonlinear voltage-current characteristic with two or more active and/or passive die that are metallurgically bonded together in a series configuration to gain the additive effect of the reverse blocking capabilities of each individual active die.

A.3.48 Storage temperature. Storage temperature is a temperature at which the device may be stored without any power being applied.

A.3.49 Temperature coefficient. The ratio of the change in a parameter to the change in temperature.

A.3.50 Thermal compression bond. A bond achieved when pressure and temperature are present regardless of how the temperature rise was achieved except without ultrasonic assist.

A.3.51 Thermal equilibrium. Thermal equilibrium is reached when doubling the test time interval does not produce a change, due to thermal effects, in the parameter being measured that is greater than the specified accuracy of the measurement.

A.3.52 Thermal resistance. Thermal resistance is the temperature rise, per unit power dissipation, of a junction above the temperature of a stated external reference point under conditions of thermal equilibrium.

A.3.53 Thyristor. A bi-stable semiconductor device that comprises three or more junctions and can be switched from the off-state or on-state to the opposite state.

A.3.54 Transistor. An active semiconductor device capable of providing power amplification and having three or more terminals.

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A.4 TRANSISTOR DEFINITIONS

A.4.1 Junction transistors, multijunction types.

A.4.1.1 Base. A region which lies between an emitter and collector of a transistor and into which minority carriers are injected.

A.4.1.2 Collector. A region through which a primary flow of charge carriers leaves the base.

A.4.1.3 Cutoff current. The cutoff current is the measured value of dc current when a transistor is reverse biased by a voltage less than the breakdown voltage.

A.4.1.4 Emitter. A region from which charge carriers that are minority carriers in the base are injected into the base.

A.4.1.5 Junction, collector. A semiconductor junction, normally biased in the reverse direction, the current through which can be controlled by the introduction of minority carriers into the base.

A.4.1.6 Junction, emitter. A semiconductor junction normally biased in the forward direction to inject minority carriers into the base.

A.4.1.7 Saturation. A base current and a collector current condition resulting in a forward-biased collector junction.

A.4.2 Unijunction transistors.

A.4.2.1 Peak point. The point on the emitter current-voltage characteristic corresponding to the lowest current at which the change in emitter base voltage with respect to emitter current equals zero.

A.4.2.2 Unijunction transistor. A three-terminal semiconductor device having one junction and a stable negative-resistance characteristic over a wide temperature range.

A.4.2.3 Valley point. The point on the emitter current-voltage characteristic corresponding to the second lowest current at which the change in emitter base voltage with respect to emitter current equals zero.

A.4.3 Field-effect transistors (FET).

A.4.3.1 Depletion-mode operation. The operation of a FET such that changing the gate to source voltage from zero to a finite value decreases the magnitude of the drain current.

A.4.3.2 Depletion-type FET. A FET having appreciable channel conductivity for zero gate to source voltage. The channel conductivity may be increased or decreased according to the polarity of the applied gate to source voltage.

A.4.3.3 Drain. A region into which majority carriers flow from the channel.

A.4.3.4 Enhancement-mode operation. The operation of a FET such that changing the gate to source voltage from zero to a finite value increases the magnitude of the drain current.

A.4.3.5 Enhancement-mode FET. A FET having substantially zero channel conductivity for zero gate to source voltage. The channel conductivity may be increased by the application of a gate to source voltage of appropriate polarity.

A.4.3.6 FET. A transistor in which the conduction is due entirely to the flow of majority carriers through a conduction channel controlled by an electric field arising from a voltage applied between the gate and source terminals.

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A.4.3.7 Gate. The electrode associated with the region in which the electric field due to the control voltage is effective.

A.4.3.8 Insulated-gate FET. A FET having one or more gate electrodes which are electrically insulated from the channel.

A.4.3.9 Junction-gate FET. A FET that uses one or more gate regions to form p-n junction(s) with the channel.

A.4.3.10 MOSFET. An insulated gate FET in which the insulating layer between each gate electrode and the channel is oxide material.

A.4.3.11 N-channel FET. A FET that has an n-type conduction channel.

A.4.3.12 P-channel FET. A FET that has a p-type conduction channel.

A.4.3.13 Source. A region from which majority carriers flow into the channel.

A.5 DIODE AND RECTIFIER DEFINITIONS

A.5.1 Signal diodes and rectifier diodes.

A.5.1.1 Semiconductor rectifier diode. A device having an asymmetrical voltage-current characteristic used for rectification.

A.5.1.2 Semiconductor signal diode. A device having an asymmetrical voltage-current characteristic and used for signal detection.

A.5.2 Microwave diodes.

A.5.2.1 Detector diode. A device which converts rf energy into dc or video output.

A.5.2.2 Gunn diode. A microwave diode that exhibits negative resistance arising from the bulk negative differential conductivity occurring in several compound semiconductors such as gallium arsenide, and that operates at a frequency determined by the transit time of charge bunches formed by this negative differential conductivity.

A.5.2.3 Impact, avalanche, and transit time diode (IMPATT). A semiconductor microwave diode that, when its junction is biased into avalanche, exhibits a negative resistance over a frequency range determined by the transit time of charge carriers through the depletion region.

A.5.2.4 Limited space-charge accumulation diode (LSA). A microwave diode similar to the Gunn diode except that it achieves higher output power at frequencies, determined by the microwave cavity, that are several times greater than the transit-time frequency by avoiding the formation of charge bunches or domains.

A.5.2.5 Matched pair. A pair of diodes identical in outline dimensions and with matched electrical characteristics. The two diodes may both be forward polarity, or one forward and one reverse polarity, or both reverse polarity.

A.5.2.6 Microwave diode. A two terminal device that is responsive in the microwave region of the electromagnetic spectrum, commonly regarded as extending from 1 GHz to 300 GHz.

A.5.2.7 Mixer diode. A microwave diode that combines rf signals at two frequencies to generate an rf signal at a third frequency.

A.5.2.8 Trapped plasma avalanche transit time diode (TRAPATT). A microwave diode that, when its junction is biased into avalanche, exhibits a negative resistance at frequencies below the transit time frequency range of the diode due to generation and dissipation of trapped electron-hole plasma resulting from the intimate interaction between the diode and a multiresonant microwave cavity.

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A.5.3 Tunnel diodes.

A.5.3.1 Tunnel diodes. A device in which quantum-mechanical tunneling leads to a region of negative slope in the forward direction of the current-voltage characteristic.

A.5.3.2 Backward diode. A device in which quantum-mechanical tunneling leads to a current-voltage characteristic with a reverse current greater than the forward current, for equal and opposite applied voltages.

A.5.4 Voltage-regulator and voltage-reference diodes.

A.5.4.1 Voltage-reference diode. A diode which is normally biased to operate in the breakdown region of its voltage-current characteristic and which develops across its terminals a reference voltage of specified accuracy, when biased to operate throughout a specified current and temperature range.

A.5.4.2 Voltage-regulator diode. A diode which is normally biased to operate in the breakdown region of its voltage-current characteristic and which develops across its terminals an essentially constant voltage throughout a specified current range.

A.5.5 Current-regulator diodes.

A.5.5.1 Current-regulator diode. A diode which limits current to an essentially constant value over a specified voltage range.

A.5.6 Varactor diodes.

A.5.6.1 Varactor diode. A two terminal semiconductor device in which use is made of the property that its capacitance varies with the applied voltage.

A.5.6.2 Tuning diode. A varactor diode used for rf tuning including functions such as automatic frequency control (AFC) and automatic fine tuning (AFT).

A.5.7 Transient voltage suppressors.

A.5.7.1 Varistor. A transient voltage suppressor that is a nonlinear resistor with symmetrical characteristics.

A.5.7.2 Avalanche-junction. A transient voltage suppressor that is a semiconductor diode that can operate in either the forward or reverse direction of its voltage-current characteristic to limit voltage transients.

A.5.7.3 Clamping voltage. The voltage in a region of low differential resistance that serves to limit the transient voltage across the device terminals.

A.5.7.4 Clamping factor. The ratio of clamping voltage to breakdown voltage.

A.5.7.5 Peak impulse current. The peak current for a series of essentially identical impulses.

A.5.7.6 Standby current. The dc current through a transient voltage suppressor at rated standoff voltage.

A.5.7.7 Repetitive peak pulse power. The peak power dissipation resulting from the peak impulse current.

A.5.7.8 Response time. The time interval between the point on the impulse waveform at which the amplitude exceeds the clamping voltage level and the peak of the voltage overshoot.

A.5.7.9 Voltage overshoot. The excess voltage over the clamping voltage that occurs when a current impulse having short virtual front duration is applied.

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A.5.7.10 Forward surge current. The peak current for a single impulse for forward biased diode.

A.5.7.11 Working peak voltage. The peak voltage, excluding all transient voltage, usually referred to as standoff voltage.

A.6 CLASSES OF THYRISTORS DEFINITIONS

A.6.1 Thyristor. A bi-stable semiconductor device that comprises three or more junctions and can be switched between conducting and nonconducting status.

A.6.1.1 Bi-directional diode thyristor. A two terminal thyristor having substantially the same switching behavior in the first and third quadrants of the principal voltage-current characteristic.

A.6.1.2 Bi-directional triode thyristor. An n-gate or p-gate thyristor having substantially the same switching behavior in the first and third quadrants of the principal voltage-current characteristic.

A.6.1.3 N-gate thyristor. A three-terminal thyristor in which the gate terminal is connected to the n-region adjacent to the region to which the anode terminal is connected and that is normally switched to the on-state by applying a negative signal between gate and anode terminals.

A.6.1.4 P-gate thyristor. A three-terminal thyristor in which the gate terminal is connected to the p-region adjacent to the region to which the cathode terminal is connected and that is normally switched to the on-state by applying a positive signal between gate and cathode terminals.

A.6.1.5 Reverse blocking diode thyristor. A two-terminal thyristor that switches only for positive anode to cathode voltages and exhibits a reverse blocking state for negative anode to cathode voltages.

A.6.1.6 Reverse blocking triode thyristor. An n-gate or p-gate thyristor that switches only for positive anode to cathode voltages and exhibits a reverse blocking state for negative anode to cathode voltages.

A.6.1.7 Reverse conducting diode thyristor. A two terminal thyristor that switches only for positive anode to cathode voltages and conducts large currents at negative anode to cathode voltages comparable in magnitude to the on-state voltage.

A.6.1.8 Reverse conducting triode thyristor. An n-gate or p-gate thyristor that switches only for positive anode to cathode voltages and conducts large currents at negative anode to cathode voltages comparable in magnitude to the on-state voltages.

A.6.1.9 Turn off thyristor. A thyristor that can be switched between conducting and nonconducting states by applying control signals of appropriate polarities to the gate terminal, with the ratio of triggering power to triggered power appreciably less than one.

A.6.2 Physical structure terms.

A.6.2.1 Gate. An electrode connected to one of the semiconductor regions for introducing control current.

A.6.2.2 Main terminals. The two terminals through which the principal current flows.

A.6.3 Electrical characteristic and rating terms.

A.6.3.1 Anode to cathode voltage-current characteristic (anode characteristic). A function, usually represented graphically, relating the anode to cathode voltage to the principal current, with gate current where applicable, as a parameter.

A.6.3.2 Breakover point. Any point on the principal voltage-current characteristic for which the differential resistance is zero and where the principal voltage reaches a maximum value.

A.6.3.3 Negative differential resistance region. Any portion of the principal voltage-current characteristic in the switching quadrant within which the differential resistance is negative.

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A.6.3.4 Off impedance. The differential impedance between the terminals through which the principal current flows when the thyristor is in the off state.

A.6.3.5 Off-state. The condition of a thyristor corresponding to the high resistance low current portion of the principal voltage-current characteristic between the origin and the breakover point in the switching quadrant.

A.6.3.6 On impedance. The differential impedance between the terminals through which the principal current flows when the thyristor is in the on state.

A.6.3.7 On-state. The condition of a thyristor corresponding to the low resistance, low voltage portion of the principal voltage-current characteristic in the switching quadrant.

A.6.3.8 Principal current. A generic term for the current through the device excluding gate current.

A.6.3.9 Principal voltage. The voltage between the main terminals.

A.6.3.10 Principal voltage-current characteristic (principal characteristic)). A function, usually represented graphically, relating the principal voltage to the principal current, with gate current where applicable, as a parameter.

A.6.3.11 Reverse blocking impedance. The differential impedance between the two terminals through which the principal current flows when the thyristor is in the reverse blocking state at a stated operating point.

A.6.3.12 Reverse blocking state. The condition of a reverse blocking thyristor corresponding to the portion of the anode to cathode voltage-current characteristic for which the reverse currents are of lower magnitude than the reverse breakdown current.

A.6.3.13 Switching quadrant. A quadrant of the principal voltage-current characteristic in which a device is intended to switch between an off state and an on state.

A.7 OPTOELECTRONIC DEVICE DEFINITIONS

A.7.1 Optoelectronic device. A device that is responsive to or that emits or modifies electromagnetic radiation in the visible, infrared, or ultraviolet spectral regions; or a device that utilizes such electromagnetic radiation for its internal operation.

A.7.1.1 Conversion efficiency. The ratio of maximum available power output resulting from photovoltaic operation to total incident radiant flux.

A.7.1.2 Dark condition. The condition attained when the electrical parameter under consideration approaches a value which cannot be altered by further irradiation shielding.

A.7.1.3 Dark current. The current that flows through a photosensitive device in the dark condition.

A.7.1.4 Light current. The current that flows through a photosensitive device when it is exposed to radiant energy.

A.7.1.5 Photoconductive diode. A photodiode that is intended to be used as a photoconductive transducer.

A.7.1.6 Photocurrent. The difference in magnitude between light current and dark current.

A.7.1.7 Photodiode. A diode that is intended to be responsive to radiant energy.

A.7.1.8 Photodiode, avalanche. A photodiode that is intended to take advantage of avalanche multiplication of photocurrent.

A.7.1.9 Photoemitter. A device that emits electromagnetic radiation in the visible, infrared, or ultraviolet spectral regions.

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A.7.1.10 Photosensitive device. A device that is responsive to electromagnetic radiation in the visible, infrared, or ultraviolet spectral regions.

A.7.1.11 Photothyristor. A thyristor that is intended to be responsive to radiant energy for controlling its operation as a thyristor.

A.7.1.12 Phototransistor. A transistor that is intended to be responsive to radiant energy.

A.7.1.13 Photovoltaic diode. A photodiode that is intended to generate a terminal voltage in response to radiant energy.

A.7.2 Photoemitting devices.

A.7.2.1 Avalanche luminescent diode. A light emitting diode that emits luminous energy when a controlled reverse current in the breakdown region is applied.

A.7.2.2 Infrared emitting diode. A diode capable of emitting radiant energy in the infrared region of the spectrum resulting from the recombination of electrons and holes.

A.7.2.3 Light emitting diode. A diode capable of emitting luminous energy resulting from the recombination of electrons and holes.

A.7.2.4 Radiant efficiency. The ratio of the total radiant flux emitted to the total input power.

A.7.3 Optocouplers.

A.7.3.1 Photodarlington coupler. An opto-coupler in which the photo-sensitive element is a darlington connected phototransistor.

A.7.3.2 Photodiode coupler. An opto-coupler in which the photosensitive element is a photodiode.

A.7.3.3 Photothyristor coupler. An opto-coupler in which the photosensitive element is a photothyristor.

A.7.3.4 Phototransistor coupler. An opto-coupler in which the photosensitive element is a phototransistor.

A.8 ELECTRICAL AND ENVIRONMENTAL STRESS SCREENING DEFINITIONS

A.8.1 Electrical and environmental stress screening.

a. Electrical stressing near maximum rating of semiconductor devices is performed to remove devices within a given lot which are subject to early life failures due to improper processing.

b. Determine if wear-out mechanisms are present in a given lot which will shorten the time to failure (life tests).

A.8.2 Power burn-in. A generic term describing a screening test which operates the device by internally dissipating sufficient power to significantly heat the device junction for a specified time.

A.8.2.1 Rectifying ac power burn-in. Power burn-in whereby junction heating is accomplished through the alternate application every half cycle of forward current and reverse voltage.

A.8.2.2 Steady-state dc power burn-in. Power burn-in whereby junction heating is accomplished through the application of steady-state forward current, reverse current, or forward power for diodes (including rectifiers), Zeners, and transistor respectively.

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A.8.3 High temperature reverse bias. A generic term describing a screening test which applies a blocking voltage and is normally performed at ambient temperature (T_A) = +150°C through the external application of heat.

A.8.3.1 Steady-state dc high temperature reverse bias. High temperature reverse bias which applies steady-state dc blocking voltage.

A.8.3.2 Half-wave high temperature reverse bias. High temperature reverse bias which applies half-wave blocking voltage.

A.8.3.3 Full-wave high temperature blocking bias. High temperature reverse bias which applies full-wave blocking voltage; sometimes applicable to symmetrical thyristors or transient voltage suppressors.

A.8.4 Operating life. A generic term describing a sample test which operates and internally heats a device junction for an extended time to verify lot integrity. This is generally an extension of power burn-in.

A.8.4.1 Rectifying ac operating life. Operating life whereby heating is accomplished through the alternate application of forward current and reverse voltage.

A.8.4.2 Steady-state dc operating life. Operating life whereby heating is accomplished through the application of steady-state forward current, reverse current, or forward power for diodes (including rectifiers), Zeners, and transistors respectively.

A.8.4.3 Intermittent operating life. Operating life whereby junction temperature (T_J) and case temperature (T_C) is cycled through a specified temperature range by a heating current or power and a cooling period, when current or power is removed.

A.8.4.3.1 Rectifying ac intermittent operating life. Intermittent operating life whereby the device is subjected to forward current and reverse voltage, during the heating period.

A.8.4.3.2 DC intermittent operating life. Intermittent operating life whereby the device is subjected to steady-state forward current or equivalent half sine forward current, during the heating period. This test is also known as power cycling or thermal fatigue.

A.8.5 Blocking life. A generic term describing a sample test which applies a blocking voltage and is normally performed at a specified high ambient or case temperature through the external application of heat.

A.8.5.1 Steady-state dc blocking life. Blocking life which applies steady-state dc blocking voltage.

A.8.5.2 Half-wave blocking life. Blocking life which applies half-wave blocking voltage.

A.8.5.3 Full-wave blocking life. Blocking life which applies full-wave blocking voltage.

A.8.6 Temperature cycling (air to air). Temperature cycling at device's case temperature through a specified range by the external heating and cooling of the device in an air to air environment.

A.8.7 Thermal shock (liquid to liquid). Thermal shock cycling at device's case temperature through a specified range by the external heating and cooling of the device in a liquid to liquid environment.

A.8.8 Thermal impedance. Thermal impedance for the purpose of this specification is the application of an electrical stress sufficient to pass heat through the interface of dissimilar materials, primarily to determine the quality of attachment by measuring the electrical characteristics of the temperature sensitive parameter. Thermal impedance measurement is performed by quantifying changes to the heat sensitive parameter caused by current flow and its induced heating. It is used to detect interface voids between die, attachment medium and the header or heat sink.

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A.8.9 Surge. Surge is the application of a high peak current ten times (minimum) the device average current maximum rating applied for a short pulse width appropriate to determine processing defects (e.g., wire bond integrity, micro cracks, and bond voids).

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ABBREVIATIONS AND SYMBOLS

B.1 SCOPE

B.1.1 Scope. This appendix covers the abbreviations and symbols for use with semiconductor devices. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

B.2 APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

B.3 SEMICONDUCTOR ABBREVIATIONS AND SYMBOLS

B.3.1 Abbreviations.

AFC	Automatic frequency control
AFT	Automatic fine tuning
AQL	Acceptable quality level
BIST	Backward instability shock test
CAGE	Commercial and Government Entity
CI	Conformance inspection
DMOS	Diffusion metal oxide semiconductor
DMS	Diminishing manufacturing sources
DPA	Destructive physical analysis
EE	Element evaluation
EM	Electro-magnetic
ESD	Electrostatic discharge
ESDS	Electrostatic discharge sensitive
FA	Failure analysis
FET	Field-effect transistor
FIST	Forward instability shock test
FIT	Failure in time
FMEA	Failure mode and effects analysis
FRED	Fast recovery epitaxial diode
GaAs	Gallium arsenide

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GIDEP ..... Government Industry Data Exchange Program

HTRB ..... High temperature reverse bias

IGBT ..... Insulated gate bipolar transistor

IMPATT ..... Impact, avalanche, and transit time diode

JFET ..... Junction field effect transistor

LTPD ..... Lot total percent defective

LSA ..... Limited space-charge accumulation

MOSFET ..... Metal oxide semiconductor field-effect transistors

NF ..... Noise figure

OEM ..... Original equipment manufacturer

PD ..... Percent defective

PDA ..... Percent defective allowed

PDQ ..... Product design qualification

PIN ..... Part or Identifying number

PIND ..... Particle impact noise detector

PIPL ..... Post-irradiation parameter limits

PMQ ..... Product manufacturing qualification

POC ..... Point of contact

PPM ..... Parts per million

QM ..... Quality management

QML ..... Qualified Manufacturers List

QPL ..... Qualified Products List

RHA ..... Radiation hardness assurance

RMS ..... Root mean square

SEM ..... Scanning electron microscope

SOA ..... Safe operating area

SPC ..... Statistical process control

TQM ..... Total quality management

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TRAPATT ..... Trapped plasma avalanche transit time diode

TRB ..... Technical review board

UHF ..... Ultra high frequency

VMOS ..... V-Groove metal oxide semiconductor

B.3.2 Symbols. Previous symbols are in parentheses.

F ..... Spot noise figure (NF)

F ..... Average NF

$R_{\theta JAM}$ ..... Thermal resistance junction to adhesive mount

$R_{\theta JSP}$ ..... Thermal resistance surface mount junction to solder point

$R_{\theta JX}$ ..... Thermal resistance (θ)

$R_{\theta CA}$ ..... Thermal resistance, case to ambient

$R_{\theta JA}$ ..... Thermal resistance, junction to ambient (θ_{J-A})

$R_{\theta JC}$ ..... Thermal resistance, junction to case (θ_{J-C})

$R_{\theta JEC}$ ..... Thermal resistance, junction to end cap

$R_{\theta JL}$ ..... Thermal resistance, junction to lead

$R_{\theta JR}$ ..... Thermal resistance, junction to reference

T_A ..... Ambient or free air temperature

T_C ..... Case temperature

T_{EC} ..... End cap temperature

T_J ..... Junction temperature

T_L ..... Lead temperature

T_{op} ..... Operating temperature

T_{STG} ..... Storage temperature

t_d ..... Delay time

t_f ..... Fall time

t_{off} ..... Turn off time

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t_{on} ..... Turn on time

t_p ..... Pulse time

t_r ..... Rise time

t_s ..... Storage time

t_w ..... Pulse average time

$V_{(BR)}$ ..... Breakdown voltage (BV)

Z_θ ..... Thermal impedance

B.4 TRANSISTORS SYMBOLS

B.4.1 Junction transistors, multijunction type symbols.

C_{ibo}, C_{ieo} ..... Input capacitance, (common base, common emitter) collector open circuited for ac

C_{ibs}, C_{ies} ..... Input capacitance, (common base, common emitter) collector short-circuited to reference terminal for ac

C_{obo}, C_{oeo} ..... Output capacitance, (common base, common emitter) input open circuited to ac

C_{obs}, C_{oes} ..... Output capacitance, (common base, common emitter) input short-circuited to reference terminal for ac

f_{hfb}, f_{hfc}, f_{hfe} ..... Small signal, short-circuit forward current transfer ratio cutoff frequency (common base, common collector, common emitter)

f_{max} ..... Maximum frequency of oscillation

f_T ..... Extrapolated unity gain frequency

g_{MB}, g_{MC}, g_{ME} ..... Static transconductance (common base, common collector, common emitter)

g_{mb}, g_{mc}, g_{me} ..... Small signal transconductance (common base, common collector, common emitter)

G_{PB}, G_{PC}, G_{PE} ..... Large signal insertion power gain (common base, common collector, common emitter)

G_{pb}, G_{pc}, G_{pe} ..... Small signal insertion power gain (common base, common collector, common emitter)

h_{FB}, h_{FC}, h_{FE} ..... Static forward current transfer ratio (common base, common collector, common emitter)

h_{fb}, h_{fc}, h_{fe} ..... Small signal short circuit forward current transfer ratio (common base, common collector, common emitter)

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$\lvert h_{fe} \rvert$	Magnitude of common emitter small signal short circuit forward current transfer ratio
h_{IB}, h_{IC}, h_{IE}	Static input resistance (common base, common collector, common emitter)
h_{ib}, h_{ic}, h_{ie}	Small signal short circuit input impedance (common base, common collector, common emitter)
h_{ob}, h_{oc}, h_{oe}	Small signal open circuit output admittance (common base, common collector, common emitter)
h_{rb}, h_{rc}, h_{re}	Small signal open circuit reverse voltage transfer ratio (common base, common collector, common emitter)
I_B	Base current (dc)
I_C	Collector current (dc)
I_E	Emitter current (dc)
i_B	Base current (instantaneous total value)
i_C	Collector current (instantaneous total value)
i_E	Emitter current (instantaneous total value)
I_{CBO}	Collector cutoff current (dc) emitter open
I_{CEO}	Collector cutoff current (dc) base open
I_{CER}	Collector cutoff current (dc) with specified resistance between base and emitter
I_{CES}	Collector cutoff current (dc) base short circuited to emitter
I_{CEV}	Collector cutoff current (dc) with specified voltage between base and emitter
I_{CEX}	Collector cutoff current (dc) with specified circuit between base and emitter
I_{EBO}	Emitter cutoff current (dc) collector open
I_{ECS}	Emitter cutoff current (dc) base short-circuited to collector
I_S	Collector efficiency

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P_C	Collector power dissipation
P_T	Total power dissipation, all terminals
R_B	External base resistance
r_b	Base spreading resistance
rb'cc	Collector-base time constant
R_C	External collector resistance
$r_{CE(sat)}$	Collector to emitter saturation resistance
R_E	External emitter resistance
r_{iep}	Small signal short circuit parallel input resistance (common emitter)
t_c	t_{OFF} crossover time (the time interval during which the collector voltage decreases from 10 percent of its peak off-state value and the collector current decreases to 10 percent of its peak on state value)
V_{BB}	Base supply voltage
V_{BE}	Base to emitter voltage (dc)
$V_{BE(sat)}$	Base to emitter saturation voltage
$V_{(BR)CBO}$	Breakdown voltage collector to base, emitter open
$V_{(BR)CEO}$	Breakdown voltage collector to emitter, base open
$V_{(BR)CER}$	Breakdown voltage collector to emitter, with specified resistance between base and emitter
$V_{(BR)CES}$	Breakdown voltage collector to emitter, with base short-circuited to emitter
$V_{(BR)CEX}$	Breakdown voltage collector to emitter, with specified circuit between base and emitter
$V_{(BR)EBO}$	Breakdown voltage emitter to base, collector open
V_{CB}	Collector to base voltage (dc)
V_{CBF}	DC open circuit voltage (floating potential) between the collector and base, with the emitter biased in the reverse direction with respect to the base

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V_{CBO} ..... Collector to base voltage (static), emitter open

V_{CC} ..... Collector supply voltage

V_{CE} ..... Collector to emitter voltage (dc)

V_{ce} ..... Collector to emitter voltage (rms)

v_{ce} ..... Collector to emitter voltage (instantaneous)

$V_{CE(sat)}$ ..... Collector to emitter saturation voltage

V_{CEO} ..... Collector to emitter voltage (static) base open

$V_{CEO(sus)}$ ..... Breakdown voltage, collector to emitter, sustained

V_{CER} ..... Collector to emitter voltage (dc), with specified resistance between base and emitter

V_{CES} ..... Collector to emitter voltage (dc), base short-circuited to emitter

V_{EB} ..... Emitter to base voltage (dc)

V_{eb} ..... Emitter to base voltage (rms)

v_{eb} ..... Emitter to base voltage (instantaneous)

V_{EBF} ..... DC open circuit voltage (floating potential) between the emitter and base, with the collector biased in the reverse direction with respect to the base

$V_{(BR)CEV}$ ..... Breakdown voltage collector to emitter, with specified voltage between base and emitter

V_{EBO} ..... Emitter to base voltage (static) collector open

V_{EC} ..... Emitter to collector voltage (dc)

V_{ECF} ..... DC open circuit voltage (floating potential) between the emitter and collector, with the base biased in the reverse direction with respect to the collector

V_{EE} ..... Emitter supply voltage

V_{RT} ..... Reach through voltage

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B.4.2 FET symbols.

b_{is}	Small-signal, common-source, short-circuit, input susceptance
b_{os}	Small-signal, common-source, short-circuit, output susceptance
b_{fs}	Small-signal, common-source, short-circuit, forward transfer susceptance
b_{rs}	Small-signal, common-source, short-circuit, reverse transfer susceptance
C_{ds}	Small-signal drain to source capacitance
C_{du}	Small-signal drain to substrate capacitance
C_{iss}	Small-signal, common-source, short-circuit, input capacitance
C_{oss}	Small-signal, common-source, short-circuit, output capacitance
C_{rss}	Small-signal, common-source, short-circuit, reverse transfer capacitance
D, d	Drain terminal
E_{AR}	Repetitive avalanche energy capability
E_{AS}	Single pulse avalanche energy capability
G, g	Gate terminal
g_{fs}	Small-signal, common-source, short-circuit, forward transfer conductance
g_{is}	Small-signal, common-source, short-circuit, input conductance
g_{os}	Small-signal, common-source, short-circuit, output conductance
G_{pg}	Small-signal, common-gate insertion power gain
G_{ps}	Small-signal, common-source insertion power gain
g_{rs}	Small-signal, common-source, short-circuit, reverse transfer conductance
G_{tg}	Small-signal, common-gate, transducer power gain
G_{ts}	Small-signal, common-source transducer power gain

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I_D ..... Drain current

I_{AR} ..... Rated avalanche current (repetitive and nonrepetitive)

$I_{D(on)}$ ..... On-state drain current

$I_{D(off)}$ ..... Drain cutoff current

I_{DSS} ..... Zero-gate-voltage drain current

I_G ..... Gate current

I_{GF} ..... Forward gate current

I_{GR} ..... Reverse gate current

I_{GSS} ..... Reverse gate current with all other terminals short-circuited to source (junction-gate)

I_{GSSF} ..... Forward gate current with all other terminals short-circuited to source (insulated-gate)

I_{GSSR} ..... Reverse gate current with all other terminals short-circuited to source (insulated-gate)

I_S ..... Source current through drain diode (forward biased V_{SD})

$I_{S(off)}$ ..... Source cutoff current

I_{SDS} ..... Zero-gate-voltage source current

$I_{(ISO)}$ ..... Source pin to case isolation current

$Q_g(th)$ ..... Gate charge that shall be supplied to reach minimum specified gate-source threshold voltage

$Q_{g(on)}$ ..... Gate charge that shall be supplied to reach the gate-source voltage specified for the device $r_{DS(on)}$ measurement

$Q_{gm(on)}$ ..... Gate charge that shall be supplied to the device to reach the maximum rated gate-source voltage

Q_{gs} ..... Charge required by C_{GS} to reach a specified I_D

Q_{gd} ..... Charge supplied to the drain from the gate to change the drain voltage under constant drain current conditions

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$r_{ds(on)}$ ..... Small-signal, drain to source on-state resistance

$r_{DS(on)}$ ..... Static drain to source on-state resistance

S, s ..... Source terminal

$t_{d(off)}$ ..... Turn-off delay time

$t_{d(on)}$ ..... Turn-on delay time

U, u ..... Substrate (bulk) (terminal, when substrate is externally terminated)

$V_{(BR)GSS}$ ..... Gate to source breakdown voltage, all other terminals short-circuited to source (junction-gate)

$V_{(BR)DSS}$ ..... Drain to source breakdown voltage, all other terminals short-circuited to source (junction-gate)

$V_{(BR)GSSF}$ ..... Forward gate to source breakdown voltage

$V_{(BR)GSSR}$ ..... Reverse gate to source breakdown voltage

V_{DD} ..... Drain supply voltage

V_{DG} ..... Drain to gate voltage

V_{DS} ..... Drain to source voltage

$V_{DS(on)}$ ..... On-state drain to source voltage

V_{DU} ..... Drain to substrate voltage

V_{GG} ..... Gate supply voltage

V_{GP} ..... Gate plateau voltage

V_{GS} ..... Gate to source voltage

V_{GSF} ..... Forward gate to source voltage

V_{GSR} ..... Reverse gate to source voltage

$V_{GS(off)}$ ..... Gate to source cutoff voltage

$V_{GS(th)}$ ..... Gate to source threshold voltage

V_{GU} ..... Gate to substrate voltage

V_{ISO} ..... Source pin to case isolation voltage

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V_{SS} ..... Source supply voltage

V_{SU} ..... Source to substrate voltage

y_{fs} ..... Magnitude of small-signal common-source short-circuit forward transfer admittance

y_{is} ..... Magnitude of small-signal common-source short-circuit input admittance

y_{rs} ..... Magnitude of small-signal common-source short-circuit reverse transfer admittance

B.4.3 Unijunction transistor symbols.

$I_{B2(mod)}$ ..... Interbase modulated current

I_{EB20} ..... Emitter reverse current

I_P ..... Peak point current

I_V ..... Valley point current

r_{BB} ..... Interbase resistance

V_{B2B1} ..... Interbase voltage

$V_{EB1(sat)}$ ..... Emitter saturation voltage

V_{OB1} ..... Base - 1 peak voltage

V_p ..... Peak point voltage

V_V ..... Valley point voltage

η ..... Intrinsic standoff ratio

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B.5 DIODES AND RECTIFIERS SYMBOLS

B.5.1 Diodes and rectifier symbols. (See table B-I.)

C_J ..... Junction capacitance

$I_{F(RMS)}$, I_f, I_F, $I_{F(AV)}$, i_F, I_{FM} ..... Forward current

I_{FSM} ..... Forward current, surge peak

$I_{FM(OV)}$ ..... Forward current, overload

I_O ..... Average forward current, over a full cycle 180 degrees conduction angle, 60 (or 50) Hz, half sine wave equal to I_{FM}/3.14 (Pi).

$I_{R(RMS)}$, I_r, I_R, $I_{R(AV)}$, i_R, I_{RM} ..... Reverse current

$i_{R(REC)}$, $I_{RM(REC)}$ ..... Reverse recovery current

I_{RRM} ..... Reverse current, repetitive peak

I_{RSM} ..... Reverse current, surge peak

P_F, $P_{F(AV)}$, P_F, P_{FM} ..... Forward power dissipation

P_R, $P_{R(AV)}$, P_R, P_{RM} ..... Reverse power dissipation

Q_S ..... Stored charge

t_{fr} ..... Forward recovery time

t_{rr} ..... Reverse recovery time

$V_{(BR)}$, $v_{(BR)}$ ..... Breakdown voltage (dc, instantaneous total value)

$V_{F(RMS)}$, V_f, V_F, $V_{F(AV)}$, v_F, V_{FM} ..... Forward voltage

$V_{R(RMS)}$, V_r, V_R, $V_{r(AV)}$, v_R, V_{RM} ..... Reverse voltage

V_{RWM} ..... Working peak reverse voltage

V_{RRM} ..... Repetitive peak reverse voltage

V_{RSM} ..... Nonrepetitive peak reverse voltage

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TABLE B-I. Letter symbol for diodes and rectifiers.

	Total RMS value	RMS value of alternating component	DC value, no alternating component	DC value with alternating component	Instantaneous total value	Maximum (peak) total value
Forward current	$I_{F(RMS)}$	I_f	I_F	$I_{F(AV)}$	i_F	I_{FM}
Forward current average, over a full cycle at least 150° minimum to 180° conduction angle 60 (or 50) Hz, half sine wave				I_O		
Forward current repetitive peak						I_{FRM}
Forward current, surge peak						I_{FSM}
Forward current, overload						$I_{F(OV)}$
Reverse current	$I_{R(RMS)}$	I_r	I_R	$I_{R(AV)}$	i_R	I_{RM}
Reverse recovery current					$i_{R(REC)}$	$I_{RM(REC)}$
Forward power dissipation			P_F	$P_{F(AV)}$	p_F	P_{FM}
Reverse power dissipation			P_R	$P_{R(AV)}$	p_R	P_{RM}
Forward voltage	$V_{F(RMS)}$	V_f	V_F	$V_{F(AV)}$	v_F	V_{FM}
Reverse voltage	$V_{R(RMS)}$	V_r	V_R	$V_{R(AV)}$	v_R	V_{RM}
Reverse voltage working peak						V_{RWM}
Reverse voltage repetitive peak						V_{RRM}
Reverse voltage nonrepetitive peak						V_{RSM}
Breakdown voltage			$V_{(BR)}$		$v_{(BR)}$	

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B.5.2 Microwave diode symbols.

$\overline{F}_o$ ..... Overall average noise figure (of a mixer diode)

$\overline{F}_{os}$ ..... Standard overall average noise figure (of a mixer diode)

L_c ..... Conversion loss

M ..... Figure of merit (of a detector diode)

N_r ..... Output noise ratio

TSS ..... Tangential signal sensitivity

VSWR ..... Voltage standing wave ratio

z_{if} ..... Impedance, intermediate-frequency

z_{rf} ..... Impedance, radio-frequency

z_m ..... Impedance, modulator-frequency load

z_v ..... Video impedance

B.5.3 Tunnel diodes and backward diode symbols.

I_I ..... Inflection point current

I_p ..... Peak point current

I_V ..... Valley point current

r_i ..... Dynamic resistance at inflection point

V_{PP} ..... Projected peak point voltage

V_I ..... Inflection point voltage

V_P ..... Peak point voltage

V_V ..... Valley point voltage

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B.5.4 Voltage regulator and voltage-reference diode symbols.

I_F ..... Forward current, dc

I_R ..... Reverse current, dc

I_Z, I_{ZK}, I_{ZM}, I_{ZSM} ..... Regulator current, reference current (dc, dc near breakdown knee, dc maximum rated current, dc maximum rated surge current)

V_F ..... Forward voltage, dc

V_R ..... Reverse voltage, dc

V_Z, V_{ZM} ..... Regulator voltage, reference voltage (dc, dc at maximum rated current)

z_z, z_{zk}, z_{zm} ..... Regulator impedance, reference impedance (small-signal, at I_Z, at I_{ZK}, at I_{ZM})

α_{VZ} ..... Temperature coefficient of regulator voltage, temperature coefficient of reference voltage

B.5.5 Current regulator diode symbols.

I_L ..... Limiting current

I_S ..... Regulator current, (I_P may be used as an alternate symbol)

V_K ..... Knee voltage

V_L ..... Limiting voltage

ΔI_S ..... Regulator current variation, (ΔI_P may be used as an alternate symbol)

V_S ..... Regulator voltage

z_k ..... Knee impedance

z_S ..... Regulator impedance

α_{IS} ..... Temperature coefficient of regulator current, (α_{IP} may be used as an alternate symbol)

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B.5.6 Varactor diode symbols.

C_C ..... Case capacitance

C_j ..... Junction capacitance

C_t ..... Total capacitance

$\frac{C_{t1}}{C_{t2}}$ ..... Capacitance ratio

f_{co} ..... Cut-off frequency

L_S ..... Series inductance

Q ..... Figure of merit

r_S ..... Series resistance, small-signal

α_C ..... Temperature coefficient of capacitance

η ..... Efficiency

B.5.7 Transient voltage suppressor symbols.

CF ..... Clamping factor, the ratio of Vc to $V_{(BR)}$

I_D ..... Standby current

I_{FS} ..... Forward surge current

I_{FSM} ..... Rated forward surge current

I_{PP} ..... Peak impulse current

I_{PPM} ..... Rated peak impulse current

I_S ..... Surge peak transient current

I_{SM} ..... Rated surge peak transient current

$P_{(AV)}$ ..... Average power dissipation

$P_{M(AV)}$ ..... Rated average power dissipation

P_{PP} ..... Repetitive peak pulse power dissipation

P_{PPM} ..... Rated repetitive peak pulse power dissipation

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t_{os} ..... Overshoot duration

t_{res} ..... Response time

$V_{(BR)}$ ..... Breakdown voltage

V_C ..... Clamping voltage

V_W ..... Working peak voltage, also standoff voltage

$V_{WM(RMS)}$ ..... Working rms voltage

V_{WM} ..... Rated working peak voltage

V_{OS} ..... Voltage overshoot

$\alpha_{V(BR)}$ ..... Temperature coefficient of breakdown voltage

B.6 THYRISTORS SYMBOLS

B.6.1 Thyristor symbols. (See table B-II.)

dv/dt ..... Critical rate of rise of off-state voltage

$I_{(BO)}$, $i_{(BO)}$ ..... Breakover current

$I_{(BR)}$, $i_{(BR)}$ ..... Reverse breakdown current (of a reverse-blocking thyristor)

$I_{D(RMS)}$, I_D, $I_{D(A)}$, i_D, I_{DM} ..... Off-state current

I_{DRM} ..... Repetitive peak off-state current

I_G, $I_{G(A)}$, i_G, I_{GM} ..... Gate current

I_{GD}, i_{GD}, I_{GDM} ..... Gate nontrigger current

I_{GQ}, i_{GQ}, I_{GQM} ..... Gate turn-off current (of a turn-off thyristor)

I_{GT}, i_{GT}, I_{GTM} ..... Gate trigger current

I_H, i_H ..... Holding current

I_L, i_L ..... Latching current

$I_{R(RSM)}$, I_R, $I_{R(A)}$, i_R, I_{RM} ..... Reverse current (of a reverse-blocking or reverse-conducting thyristor)

I_{RRM} ..... Repetitive peak reverse current (of a reverse-blocking thyristor)

I_{RSM} ..... Nonrepetitive peak reverse current (of a reverse-blocking thyristor)

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$I_{T(RMS)}$, I_T, $I_{T(A)}$, i_T, I_{TM} ..... On-state current

I_{TRM} ..... Repetitive peak on-state current

I_{TSM} ..... Nonrepetitive peak on-state current

P_G, $P_{G(A)}$, P_G, P_{GM} ..... Gate power dissipation

P_R, $P_{R(A)}$, P_R, P_{RM} ..... Reverse power dissipation

t_{gd} ..... Gate-controlled delay time

t_{gq} ..... Gate-controlled turn-off time (of a turn-off thyristor)

t_{gt} ..... Gate-controlled turn-on time

t_q ..... Circuit-commutated turn-off time

$V_{(BO)}$, $v_{(BO)}$ ..... Breakover voltage

$V_{(BR)}$, $v_{(BR)}$ ..... Reverse breakdown voltage (of a reverse-blocking thyristor)

$V_{D(RMS)}$, V_D, $V_{D(A)}$, v_D, V_{DM} ..... Off-state voltage

V_{DRM} ..... Repetitive peak off-state voltage

V_{DSM} ..... Nonrepetitive peak off-state voltage

V_{DWM} ..... Working peak off-state voltage

V_G, $V_{G(A)}$, v_G, V_{GM} ..... Gate voltage

V_{GD}, v_{GD}, V_{GDM} ..... Gate nontrigger voltage

V_{GQ}, v_{GQ}, V_{GQM} ..... Gate turn-off voltage (of a turn-off thyristor)

V_{GT}, v_{GT}, V_{GTM} ..... Gate trigger voltage (of a reverse-blocking thyristor

V_{RRM} ..... Repetitive peak reverse voltage (of a reverse-blocking thyristor)

V_{RSM} ..... Nonrepetitive peak reverse voltage (of a reverse-blocking thyristor)

V_{RWM} ..... Working peak reverse voltage (of a reverse-blocking thyristor)

$V_{T(RMS)}$, V_T, $V_{T(A)}$, v_T, V_{TM} ..... On-state voltage

$V_{T(MIN)}$ ..... Minimum on-state voltage

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TABLE B-II. Letter symbol for thyristors.

	Total RMS value	DC value, no alternating component	DC value with alternating component	Instantaneous total value	Maximum (peak) total value
On-state current	$I_{T(RMS)}$	I_T	$I_{T(A)}$	i_T	I_{TM}
Repetitive peak, on-state current					I_{TRM}
Surge (nonrepetitive) on-state current				I_{TSM}	
Overload on-state current					$I_{T(OV)}$
Breakover current		$I_{(BO)}$		$i_{(BO)}$	
Off-state current	$I_{D(RMS)}$	I_D	$I_{D(A)}$	i_D	I_{DM}
Repetitive peak, off-state current					I_{DRM}
Reverse current	$I_{R(RMS)}$	I_R	$I_{R(A)}$	i_R	I_{RM}
Repetitive peak, reverse current					I_{RRM}
Reverse breakdown current		$I_{(BR)R}$		$i_{(BR)R}$	
On-state voltage	$V_{T(RMS)}$	V_T	$V_{T(A)}$	v_T	V_{TM}
Breakover voltage		$V_{(BO)}$		$v_{(BO)}$	
Off-state voltage	$V_{D(RMS)}$	V_D	$V_{D(A)}$	v_D	V_{DM}
Minimum on-state voltage		$V_{T(MIN)}$			
Working peak, off-state voltage					V_{DWM}
Repetitive peak off-state voltage					V_{DRM}

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TABLE B-II. Letter symbol for thyristors - Continued.

	Total RMS value	DC value, no alternating component	DC value with alternating component	Instantaneous total value	Maximum (peak) total value
Nonrepetitive off-state voltage					V_{DSM}
Reverse voltage	$V_{R(RMS)}$	V_R	$V_{R(A)}$	v_R	V_{RM}
Working peak reverse voltage					V_{RWM}
Repetitive peak, reverse voltage					V_{RRM}
Nonrepetitive peak, reverse voltage					V_{RSM}
Reverse breakdown voltage		$V_{(BR)R}$		$v_{(BR)R}$	
Holding current		I_H		i_H	
Latching current		I_L		i_L	
Gate current		I_G	$I_{G(A)}$	i_G	I_{GM}
Gate trigger current		I_{GT}		i_{GT}	I_{GTM}
Gate nontrigger current		I_{GD}		i_{GD}	I_{GDM}
Gate turn-off current		I_{GQ}		i_{GQ}	I_{GQM}
Gate voltage		V_G	$V_{G(A)}$	v_G	V_{GM}
Gate trigger voltage		V_{GT}		v_{GT}	V_{GTM}
Gate nontrigger current		V_{GD}		v_{GD}	V_{GDM}
Gate turn-off voltage		V_{GQ}		v_{GQ}	V_{GQM}
Gate power dissipation		P_G	$P_{G(A)}$	P_G	P_{GM}

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B.7 OPTOELECTRONIC DEVICES SYMBOLS

B.7.1 Optoelectronic device symbols.

Q, (Q_e) ..... Radiant energy

Q, (Q_v) ..... Luminous energy

t_d ..... Delay time

t_f ..... Fall time

t_{off} ..... Turn-off time

t_{on} ..... Turn-on time

t_r ..... Rise time

t_s ..... Storage time

τ ..... Time constant

ϕ ..... Luminous flux, radiant flux

B.7.2 Photosensitive device symbols.

A_D ..... Area, detector

E ..... Illuminance (illumination); irradiance

f_{mod} ..... Modulation frequency

I_n ..... Detector noise

I_S, I_s ..... Detector signal current (dc; rms value of ac component)

P_n ..... Noise equivalent power

V_n ..... Detector noise voltage

V_S, V_s ..... Detector signal voltage (dc; rms value of ac component)

μf ..... Noise equivalent bandwidth

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B.7.3 Photoemitting device symbols.

I	Luminous intensity; radiant intensity
L	Luminance; radiance
t_f	Radiant-pulse fall time
t_r	Radiant-pulse rise time
w	Luminous density; radiant density
$\Delta\lambda$	Spectral bandwidth
λ_p	Peak wavelength

B.7.4 Optocoupler (photocoupler and opto-isolator) symbols.

C_{io}	Input-to-output internal capacitance; transcapacitance
h_F	Current transfer ratio
I_{IO}	DC input-to-output current; isolation current
r_{IO}	Isolation resistance
V_{IO}	DC input-to-output voltage; isolation voltage

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QUALITY MANAGEMENT PROGRAM

C.1 SCOPE

C.1.1 Scope. This appendix outlines a Quality Management (QM) program and has been prepared to allow manufacturers to produce high quality, reliable military discrete semiconductors under one quality system utilizing best commercial practices. However, manufacturers who choose to participate shall be fully complaint to appendix D and shall implement a Technical Review Board (TRB), and conduct product quality planning. Manufacturers with a certified QM program will be approved by the qualifying activity to pursue process and test optimization. This level gives unique flexibility to applicable suppliers. A qualified manufacturer is a supplier who is compliant to this appendix and who exercises a system which focuses on the manufacturing process and technology capability. The DLA Land and Maritime audit is process oriented. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 APPLICABLE DOCUMENTS

C.2.1 General. The documents listed in this section are specified in sections C.3, C.4, or C.8 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they shall meet all specified requirements of documents cited in sections C.3, C.4, or C.8 of this specification, whether or not they are listed.

C.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

JEDEC - SOLID STATE TECHNOLOGY ASSOCIATION (JEDEC)

JEP-121 - Guidelines for MIL-STD-883 Screening and QCI Optimization.

(Application for copies should be addressed to JEDEC, 3103 North 10th Street, Suite 240-S, Arlington, VA 22201-2107 www.jedec.org.)

SAE International

EIA-557 - Statistical Process Control Systems.

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096 www.sae.org.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

C.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

C.3. QUALITY MANAGEMENT (QM) PROGRAM

C.3.1 QM approach. A program that embodies the principles of total quality management (TQM) will be developed and implemented by the manufacturer and documented in the QM plan. This is a management approach that is applied over the life cycle of the product to provide visibility and control of the functional and physical characteristics of devices marked with the JAN designator.

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C.3.1.1 QM plan. The QM plan documents the QM program and consists of the following three major elements: 1) The quality manual, in accordance with the quality system specified in appendix D; 2) The TRB information (see C.4.1); and 3) Product quality planning information (see C.5.1). Product quality planning is expected to be comprised of individual groups of documents that are specific to the technology being offered. Each technology that is offered will have its own product quality planning information. The QM plan, itself, may be an index of references to other documents which include the information required. In addition to these three major elements, the QM plan will include the following as a minimum.

C.3.1.1.1 Quality improvement plan. The quality improvement plan documents are the specific procedures to be followed by the manufacturer to assure continuous improvement (defined by C.8.4) in quality and reliability of the process and the product being produced.

C.3.1.2 Failure analysis. This establishes the procedures that a manufacturer self-imposes to test and analyze a sufficient quantity of failed parts to determine each failure category from all stages of manufacturing and the field. This should also identify corrective actions or specify the use of a corrective action plan based on the findings of the failure analysis.

C.3.1.3 Statistical process control (SPC) plan. A specific plan defining the manufacturer's SPC program within the manufacturing process to the requirements of EIA-557.

C.3.1.4 Certification and qualification plan. The certification and qualification plan will be defined (see C.6 and C.8).

C.3.1.5 Conversion/review of customer requirements. A procedure for reviewing customer’s requirements, as expressed in specifications and orders to determine if the device will meet expectations. This includes determining if certification and QML coverage does not exist, and it may include converting the customer's requirements into in-house requirements. This procedure should be documented (in addition to D.3.3).

C.4 TECHNICAL REVIEW BOARD

C.4.1 TRB responsibilities. The manufacturer defines its' own TRB structure. The purpose of the TRB is to assess the impact of proposed product/process changes for reliability and form, fit, and function. The TRB is responsible for product quality planning, covered by the QM program for the introduction of new products and maintains the QML lines. Written procedures which will govern the operation of the TRB shall be maintained and updated as defined in the procedures.

C.4.1.1 Organizational structure. The manufacturer’s TRB will ensure communication is established and maintained among representatives from device design, technology development, wafer fabrication, assembly, test, production, and quality assurance. The TRB is intended to be a cross-functional technical group. The manufacturer will submit the name(s) and telephone numbers of their TRB contact person(s) to the qualifying activity. The members of the TRB shall have the responsibility and authority to make decisions and the resources to implement these decisions. Records of the TRB deliberations and decisions will be maintained and made available to the qualifying activity.

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C.4.2 TRB duties. The TRB duties are defined as follows:

a. The TRB will keep the qualifying activity updated on the status of QML technologies and products.

b. The TRB will have a methodology in place for assessing and monitoring the quality and reliability of its' products.

c. The TRB shall set measurable quality objectives and monitor their progress towards meeting those objectives.

d. The TRB is responsible for control of all certified technology families.

e. The technology families will be defined by the TRB.

f. A manufacturer can have more than one TRB if appropriate.

g. The TRB controls all aspects of military product design changes, material changes, and process changes. The TRB monitors performance of all contract services.

h. The TRB will approve all process flow charts, failure mode effects analysis (FMEA), and control plans for each individual process. The TRB determines what test or verifications are needed to prove out proposed design and construction changes (See table E-IV of appendix E).

i. The TRB decides what devices (i.e., worst case) shall be used to qualify a new technology family.

j. The TRB will also address the impact of key personnel changes on the QM system.

k. The TRB shall maintain records and make them available for qualifying activity review.

l. The TRB is required to report periodically to the qualifying activity on the status of the QML technology and products (see C.4.5).

C.4.3 TRB input The TRB shall be provided input in order to monitor the quality of products. Examples of needed input are: Self-audit results, SPC data, production yields, customer returns, and corrective/preventive action initiatives. Other TRB responsibilities include, but are not limited to:

a. Quality improvement initiatives (see C.8.4).

b. Reliability program results (see.C.8.1.1.1, C.8.1.2, and C.8.4.1).

c. Qualification test data and process validation review.

d. Design and process change test data review (see C.7.1).

e. Reviewing results of failure analysis.

f. Monitor performance of any contract services.

g. Monitor customer returns and complaints.

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C.4.4 TRB authority. After the manufacturers QM program has been certified by the qualifying activity, the manufacturer's TRB will have the authority to review and approve the following without prior approval of the qualifying activity:

a. Laboratory suitability for new test methods.

b. Qualification and design change test plans.

c. Qualification, design, and construction test data.

d. Ship ahead of group B or group C completion.

e. Submit plans to the qualifying activity to modify, replace, reduce, and eliminate test.

In addition, when errors and omissions are identified in a specification sheet, the manufacturer's TRB may make the appropriate corrections provided the qualifying activity and preparing activity are notified prior to implementation.

C.4.5 TRB status reporting. The manufacturer's TRB will submit a status report to the qualifying activity describing the health of the QML technology families including all changes and the criticality of the changes with respect to quality, reliability, performance, and interchangeability. This report will be submitted semi-annually unless modified by the qualifying activity. Supporting test data shall be retained by the manufacturer where applicable. The following areas shall be addressed in each report as a minimum:

a. TRB meeting minutes and decisions.

b. Production summaries on all QML products shall be available.

c. Customer returns and corrective actions.

d. Changes, additions, and improvements in design, fabrication, assembly, or test process.

e. Changes to the facility and new process equipment.

f. Qualification planning (proposed new qualifications).

g. How well the TRB is doing at meeting their quality objectives.

h. Changes to FMEAs and control plans.

i. Contracted services.

j. Reliability program results (data).

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C.5 QUALITY PLANNING

C.5.1 Product quality planning. Quality planning is a controlled method of assuring that a product meets the customer's performance needs. This is accomplished through characterization of all applicable technology families and associated processes. Formal technology process flowcharts, FMEA's, and control plans shall be generated and maintained under the manufacturer's document control system. All contracted processes shall be identified and FMEA'S shall be performed for assembly and test.

C.5.1.1 Technology process flowchart. Flowcharts shall be generated for each certified technology. The flow chart consists of a diagram showing the sequence of material processing from incoming material through final shipment, including any production testing.

C.5.2 FMEA. The TRB is responsible for assuring that all actions recommended have been implemented or adequately addressed in the FMEA's.

C.5.3 Control plans. The control plan is prepared to summarize the process control planning for each technology. Product characteristics that should be included in a control plan are control item characteristics and significant characteristics. The evaluation method, sampling plans, data analysis method, and out of control plans are part of the control plan. Control plan updating should be tied to the corresponding FMEA updates.

C.6 QM CERTIFICATION

C.6.1 QM program certification process. The effectiveness of the manufacturer's QM program can only be determined over time. Manufacturers shall have a working TRB in place prior to a QM program certification audit. A comprehensive self-audit shall be performed by the manufacturer after a sufficient implementation period to get a realistic snapshot of the QM program and TRB effectiveness. All correspondence sent to the qualifying activity shall be first reviewed by the TRB. Results of such reviews are required for qualification reports and design changes. These reviews will be used in part to gauge the effectiveness of the TRB. The following items are examples of pre-audit submission:

a. Quality planning information.

b. Manufacturers QM plan (see C.3.1.1).

c. Quality policy.

d. TRB procedure (see C.4.1) and meeting minutes.

e. List of methods for lab suitability.

f. Self-audit results.

g. Appropriate documents to demonstrate compliance to MIL-PRF-19500 requirements.

The qualifying activity will review the pre-audit submissions for compliance to MIL-PRF-19500. Any initial concerns will be addressed at this time. Additional information, such as detailed procedures may be requested. The audit will be scheduled once all the pre-audit information is approved. The validation will focus on the TRB, QM plan, and process controls for each technology family. Once corrective actions have been taken and approved by the qualifying activity, certification will be issued.

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C.7 CHANGE PROCEDURES

C.7.1 Design, construction and process change control procedures. Any changes to the manufacturer's certified process flow or design and construction are to be governed by the manufacturer's TRB. All changes should be documented as to the reason of the change with supporting data taken to justify the change, as appropriate. All changes shall be classified by the TRB as major or minor. For any change that merits consideration for requalification, the TRB will decide if requalification is needed. The effect on any test optimization shall be evaluated prior to the implementation of a design change. Notification of major changes should be made concurrently to the qualifying activity.

C.7.1.1 Change notification. Any changes that will affect the design and construction information are considered major changes (see DLA Land and Maritime Form 36D). The qualifying activity shall be provided the latest design and construction information to keep DLA Land and Maritime files current.

C.7.2 Design and process methodology change. Changes in the design and process methodology to be evaluated by the TRB will include, but not be limited to, changes in the following areas:

a. Technology flow chart (see C.5.1.1).

b. FMEA (see C.5.2).

c. Control plan (C.5.3).

d. Bill of materials.

e. Any changes that affect form, fit, or function.

C.8 QUALIFICATION, VALIDATION, OPTIMIZATION, AND CONTINUOUS IMPROVEMENT

C.8.1 QML processing. The QML manufacturer is expected to use the following items within each technology family.

a. Design qualification.

b. Process optimization.

c. Process validation.

d. Continuous improvement.

e. Qualification.

C.8.1.1 Design qualification. The qualification activity will commence as soon as a "design freeze" is agreed upon that can meet the targeted performance criteria and the product target specification (mechanical/electrical/ reliability/thermal). Prior to the start of the product design qualification (PDQ), a meeting will be called by the TRB to determine the tests that need to be performed. The general principle of testing product, under worst case conditions, in accordance with the target specification; should be adhered to, to determine product capability at published operating levels.

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C.8.1.1.1 General elements of a design qualification program. A design qualification program is a qualification of the design and the manufacturing process, more commonly referred to as PDQ. These tests are usually performed as part of the quality design process for determining the long-term reliability of the design and the manufacturing process. Once performed and acceptable, these tests are not repeated unless a major design change is made which would require a new PDQ.

a. Product failures: Perform step stress testing in order to generate failures, not produce the absence of failures.

b. Failure analysis: Once failure occurs, perform failure analysis (if needed) to ascertain the mechanisms and failure rates or time to wearout.

c. Engineering evaluations: Evaluate to constantly improve the final product quality and reliability.

d. Failure mechanism identification: Identify and focus on failure mechanisms typical for the technology (assembly, packages, and die-level failures).

C.8.1.2 Detailed elements of a design qualification program. Defining a device family: In order to provide acceptable coverage during representative reliability assessment testing, all devices shall first be grouped into their respective device families. Device families may be selected based on characteristics from the following three categories:

a. Package grouping based on: Package profile, volume, complexity of package construction, and number of pins or wire bonds.

b. Overall construction grouping based on: Die attach method, interconnect construction techniques (e.g., spring contact), or category of bond.

c. Die grouping based on: Overall dimensions, aspect ratio, thickness, number of bonds, voltage, frequency, or power rating.

When selecting device families, consideration shall also be given to differences due to the use of different production facilities, variations in fabrication procedures, and differing design techniques. Differences or variations, which may influence device reliability, imply that a separate device family should be established. Ignoring these factors in creating device families shall be justified to the qualifying activity.

Identification of worst case devices: Once device families have been established, worst case device types from within those families shall be selected which will provide coverage for the reliability assessment of all other devices within the family. Worst case parameters (or combinations of parameters) from within the three categories listed above (i.e., packaging, overall construction, and die) shall be identified in order to establish which devices are worst case.

C.8.2 Process optimization. Process optimization is the mechanism to provide for continuous improvement as equipment and materials technology changes. Following process characterization, it is necessary to optimize the process parameters. Optimization is accomplished through identification of optimal targets and continual reduction of variation around those targets. Process characterization should be conducted on the entire process (i.e., design through delivery). A matrix should be developed that identifies failure mechanisms and their controls (see C.5.2 FMEA). The ability to maintain control of key process points will determine process capability.

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C.8.2.1 Test optimization. The manufacturer may use statistical techniques and historical data to reduce, modify, move, or eliminate tests. Manufacturer specific optimization requires the approval of the qualifying activity and preparing activity. The manufacturer shall have a test optimization program in place that meets JEP121 or an equivalent system, which justifies the proposed changes. Process monitoring and controls, statistical techniques (i.e., process capability (Cpk), failure in time (FIT)), and accumulated data may be used to prove the validity of test optimization. All optimizations shall be monitored for compliance to initial qualification baseline requirements.

C.8.2.2 Methods of test optimization. Test optimization includes the alternative methods listed below.

a. Reduction of testing/sampling. This includes reduction in the number of samples or test frequency, whether it is within a lot or periodic sampling.

b. Modification of tests. This includes using alternative test procedures, or equipment.

c. Moved tests. This pertains to performing tests in process instead of after all processing is complete.

d. Elimination of tests. The tests are no longer performed.

e. Any combination of the above.

C.8.3 Process validation: On-going reliability or conformance inspection. The process needs to be validated on a regular basis to confirm continuing conformance to the originally qualified design parameters. The two methods of validation are either conformance inspection (appendix E) or on-going reliability. Conformance inspection may only be used as a validation for this appendix if a continuous improvement feedback loop is documented that ensures conformance failures are analyzed for root cause and corrective actions are taken to modify both the FMEA and the control plan, thereby eliminating the failure mechanism.

C.8.3.1 On-going reliability guidance criteria. With the approval of the qualifying activity and preparing activity, this program, or an equivalent program, may be used as an alternative to the conformance inspection.

C.8.3.1.1 General elements of an on-going reliability program. An on-going reliability program is designed to create a closed loop feedback system, which will generate continuous improvement. Existing production lines are constantly sampled and tested. This proves the critical interfaces and manufacturing process are stable and repeatable, as well as providing a mechanism for failures to be evaluated and the process improved. The on-going reliability program shall include, as a minimum, the following:

a. Program will formalize the routine reliability testing and tie failures to a specific manufacturing process.

b. Program will integrate FMEA with reliability surveillance.

c. Restructure surveillance program by "root technologies", rather than by individual part numbers.

d. All on-going reliability testing will continue routinely.

e. Samples will be tested as specified in a specified frequency in key environmental test, e.g., High temperature reverse bias (HTRB), solder dip, and thermal shock.

f. Failures will be reviewed and tracked to a specific manufacturing process.

g. The on-going reliability program will record type of failure, manufacturing process, and root cause/corrective action.

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C.8.4 Continuous improvement. A comprehensive continuous improvement philosophy shall be fully deployed throughout the company's organization. Manufacturers shall continuously improve in quality, service, and reliability for all customers. This requirement does not replace the need for innovative improvements. Manufacturers shall develop specific action plans for continuous improvement in processes that are most important to the customer once those processes have demonstrated stability and acceptable capability. Continuous improvement means optimizing the characteristics and parameters at a target value and reducing variation around that value.

C.8.4.1 Quality and productivity improvement. The manufacturer shall identify opportunities for quality and productivity improvement and implement appropriate action plans. Some examples are:

a. Excessive variation.

b. Manufacturing yields.

c. Scrap, rework, and repair.

d. Optimization of tests.

e. Excessive cost of non-quality.

f. Machine down-time.

g. Machine set-up.

h. Cycle time.

i. Customer dissatisfaction, e.g., complaints, repairs, returns, mis-shipments, incomplete orders, customer plant concerns, and warranty.

C.8.4.2 Measures and methodologies. The manufacturer shall demonstrate knowledge of the following measures and methodologies and shall use those that are appropriate:

a. Capability indices.

b. Control charts.

c. Design of experiments.

d. Cost of quality.

e. Parts per million (PPM) analysis.

f. Problem solving.

g. Benchmarking.

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C.8.5 Qualification. The manufacturer shall have a qualification procedure. Initial qualifications for any product shall demonstrate compliance to the specification sheet. It may be an extension of the PDQ or a modification to the Product Manufacturing Qualification (PMQ). The qualification plan should include the following elements:

a. Multiple samples.

b. Hour or cycle tests.

c. Pre-conditioning, if warranted.

d. Mechanical test.

e. Electrical test.

f. Environmental stress test.

g. Reliability program results (data).

Existing data can be submitted for evaluation by the qualifying activity if the data is from the same product family and the designs are similar. The TRB defines the qualification test plan and monitors lot formation and inspection results. This plan may be based on previous data recorded for other purposes. Equivalence of all tests and inspections shall be determined by the TRB. The TRB shall be able to demonstrate to the preparing and qualifying activities that the product is capable of meeting the performance requirements outlined in MIL-PRF-19500 and the specification sheets. The TRB shall perform a comprehensive analysis of the qualification data. The data and the results of the TRB review shall be available to the qualifying activity upon request. The TRB then notifies the qualifying activity of part numbers and quality assurance levels for inclusion on the QPDSIS-19500. Design and construction information shall also be submitted with qualification notification.

C.9 Maintaining QM program certification. Certification and Lab suitability will be maintained through DLA Land and Maritime re-audits based on the previous audit findings, status reports, and other correspondence. As the confidence level and relationship between the manufacturer and the qualifying activity increases, the need for re-audits decreases.

C.10 Removal of TRB authority. The success of the TRB depends on the manufacturer. Continuing participation is conditioned upon compliance with the QML requirements. Responsibility for removing TRB authority is a function of the qualifying activity.

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QUALITY SYSTEM

D.1 SCOPE

D.1.1 Scope. This appendix contains details of the quality system for MIL-PRF-19500 semiconductor devices. The quality system will address the verification system of appendix E which measures and evaluates the manufacturer's manufacturing process against a baseline for that process. This baseline can include innovative and improved processes that result in an equivalent or higher quality product, provided that the process used to evaluate and document these changes has been reviewed and approved by the qualifying activity (see 4.4). Changes to the process baseline can be made by the manufacturer after achieving approval with documented reliability and quality data. The approach outlined in this appendix is a proven baseline, which contains details of a quality system including best commercial practices. Manufacturers shall demonstrate to the qualifying activity a quality system that achieves at least the same level of quality as could be achieved by complying with this appendix. Certification is provided by the qualifying activity upon approval by the preparing activity (for equivalent quality systems) and qualifying activity. Reduction and alternate test will be approved on a case-by-case basis. A qualified products supplier is compliant to this appendix. A qualified products supplier is one who exercises a system, which focuses on product and inspections. The DLA Land and Maritime audit is product, system, and process oriented. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

D.2 APPLICABLE DOCUMENTS

D.2.1 General. The documents listed in this section are specified in sections D.3 or D.4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they shall meet all specified requirements of documents cited in sections D.3 or D.4 of this specification, whether or not they are listed.

D.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract

NCSL INTERNATIONAL (NCSL)

NCSL Z540.3	-	Calibration Laboratories and Measuring and Test Equipment - General Requirements.

(Copies of these documents are available online at http://www.ncsli.org or from NCSL International 2995 Wilderness Place, Suite 107 Boulder, CO 80301-5404.)

JEDEC - SOLID STATE TECHNOLOGY ASSOCIATION (JEDEC)

JEDEC Publication 114	-	Guidelines for Particle Impact Noise Detection (PIND) Testing, Operator Training and Certification

(Application for copies should be addressed to the JEDEC, 3103 North 10th Street, Suite 240-S, Arlington, VA 22201-2107, www.jedec.org.)

SAE International

EIA-557	-	Statistical Process Control Systems.

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096, www.sae.org.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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D.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

D.3 REQUIREMENTS

D.3.1 Responsibility and authority.

D.3.1.1 Basic and contracted plant requirements. Devices furnished under this specification shall comply with the performance requirements of section 3 herein, and shall be prepared for delivery in accordance with section 5 herein. The verification system shall assure that the design, processing, assembly, inspection, and testing of semiconductor devices comply with this specification and the specification sheet. The quality system shall incorporate the applicable requirements of appendix E, appendix G, and appendix H.

D.3.1.1.1 Definition of a manufacturer. The manufacturer is the basic plant. For guidance on specific business scenarios, details of implementation, financial commitment to this program, or interpretation of criteria herein, contact the qualifying activity.

D.3.1.1.1.1 Basic plant. The basic plant is where wafer fabrication or device assembly is performed except the corporate office may be listed as the basic plant whenever any wholly owned wafer fabrication or device assembly plant is utilized for production. The basic plant has the responsibility for device critical interfaces, performance, quality, and reliability. The manufacturer's designating symbol of the basic plant shall appear on the finished product. Total ownership of the basic plant is required for listing and JAN branding. The basic plant shall be certified and qualified by the qualifying activity. The basic plant shall ensure their subcontractors meet the requirements of MIL-PRF-19500. The basic plant shall demonstrate control of all contracted plants to the qualifying activity.

D.3.1.1.1.2 Contracted plant. The basic plant may contract assembly or wafer fabrication provided control of the contracted plant is demonstrated to the qualifying activity. A written agreement between a basic plant and a contracted plant shall be required for all contracting scenarios. This agreement shall establish who is the basic plant and the contracted plant. For any device type, a basic plant may not contract both assembly and wafer fabrication. The contracted plant shall be certified by the qualifying activity. Qualification of assembled devices or wafers will be the responsibility of the basic plant. Manufacturers may offer their certified line services to each other through the contracted plant provisions.

D.3.1.1.1.2.1 Contracted assembly plant. The basic plant may contract assembly when control of the contracted plant is demonstrated by the basic plant to the qualifying activity. The basic plant shall have a system to address the applicable requirements of this appendix as determined by and approved by the qualifying activity. All contracted processes shall be identified and FMEAs and control plans shall be implemented for assembly.

D.3.1.1.1.2.2 Contracted wafer fabrication. Contracted wafer fabrication facilities shall have wafer lot identification, wafer lot process documentation, and wafer lot inspection. The wafer lot inspection shall ensure critical interface requirements meet the semiconductor die properties as stated on DLA Land and Maritime Form 36D. A JANS certified assembly facility may contract any JANS certified wafer fabrication facility for the purpose of qualifying JANS devices.

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D.3.1.1.1.3 Wafer fabrication . The wafer fabrication plant (basic wafer fabrication plant or contracted wafer fabrication plant) may contract any or all of the following processes to specialty laboratories:

a. Epitaxial deposition.

b. Ion implantation.

c. Irradiation (for carrier lifetime suppression).

d. Back grinding or polishing.

The applicable flow chart and traveler shall identify any contracted process. Back metallization, dicing the wafer, and loading into trays shall be performed at a plant certified by the qualifying activity. The basic plant or the contracted plant shall maintain dicing capability unless otherwise approved by the qualifying activity. Except for die/chips covered by appendix G of this specification, wafer probe is an optional screen that may be contracted to a certified facility with the approval of the qualifying activity.

D.3.1.2 Responsibility of the manufacturer. The manufacturer is responsible for the performance of all inspection requirements as specified herein, and in the specification sheet. The manufacturer may use their own or other suitable facilities which have been approved and granted laboratory suitability by the qualifying activity for the performance of MIL-STD-750 inspection requirements specified herein. The Government or acquiring activity reserves the right to witness or perform any of these inspections set forth herein or in the specification sheet and to audit the data resulting from the manufacturer's performance of these specifications.

D.3.1.3. Management responsibility. Management shall provide sufficient resources, personnel, and authority necessary for performing all applicable verification and manufacturing certification requirements specified herein. The qualifying activity may require the supplier to contract or utilize an expert of applicable subject matter to resolve ongoing technical issues.

D.3.1.3.1 Personnel. Personnel performing quality functions shall have sufficient well-defined responsibility, authority, and the organizational freedom to identify and evaluate quality problems and to initiate, recommend, or provide solutions.

D.3.1.3.2 Resources. The manufacturer shall identify requirements and provide adequate resources for management, performance of work, verification activities, and internal audits. For the purpose of this document, resources shall include, but not be limited to, materials, equipment, training, and personnel.

D.3.1.4 Organization. A functional block organization chart shall be included in the quality system program (see D.3.2) which shows the lines of responsibility and authority for organization, approval, and implementation of all aspects of the quality program. A management representative shall be identified who has defined responsibility and authority to ensure that the requirements of this specification are met.

D.3.1.4.1 Management representative. The basic plant and contracted plant shall appoint a member of management who, has the authority and responsibility for development and implementation of the quality system, this specification, and associated specification sheets. The management representative is also responsible for coordination with the preparing activity on all specification issues. This individual shall insure that the quality system is maintained and shall periodically report to the manufacturer’s management on the performance of the system and opportunities to improve it. The qualifying activity shall be informed whenever there is a change in the management representative. In addition, the plant shall provide a POC (point of contact) with the authority and responsibility to coordinate specifications. All suppliers shall be responsive to all applicable specification issues. The duties and responsibilities of the DLA Land and Maritime POC shall be documented and shall be made available to DLA Land and Maritime.

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D.3.1.5 Management review. The manufacturers management shall maintain an ongoing cognizance of the status of the quality system (D.3.2) and the design control system (D.3.4) to ensure continuous suitability and effectiveness in satisfying the requirements herein with periodic cross-functional reviews of these systems through a Qualified Products Technical Review Board (QPTRB).

D.3.1.5.5 Qualified Products Technical Review Board (QPTRB) .

D.3.1.5.5.1 Organizational structure. The QPTRB is intended to be a cross-functional management and technical group to ensure communication is established and maintained among representatives from: Management, device design, technology development, wafer fabrication, assembly, test, production, and quality assurance. The members of the QPTRB shall have the responsibility and authority to make decisions and the resources to implement these decisions. Written procedures which will govern the operation of the QPTRB shall be maintained and updated. Records of the QPTRB deliberations and decisions will be maintained and made available to the qualifying activity.

D.3.1.5.5.2 QPTRB duties. The QPTRB shall review the quality system at planned intervals to ensure continuing stability, adequacy, and effectiveness. This review shall include assessing opportunities for improvement and the need for changes to the quality system including quality policy and objectives.

D.3.1.5.5.2.1 QPTRB input:

a. Results of internal and external audits

b. Production yields.

c. Customer returns/feedback.

d. Corrective/preventive actions.

e. Failure analysis.

f. Design control / qualification initiatives.

g. Contract services.

h. Follow-up actions from previous management reviews.

i. Changes that could affect the quality system.

j. Recommendations for improvement.

D.3.1.5.5.2.2 QPTRB output:

a. The QPTRB shall document and communicate, when applicable, the results of the review.

b. The QPTRB shall set measurable quality objectives and monitor their progress towards meeting those objectives.

c. The QPTRB shall review changes to all governing specifications and comment back to the preparing activity: MIL-PRF-19500, MIL-STD-750, and performance specification sheets.

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D.3.2 Quality system program.

D.3.2.1 Quality system requirements. A quality system shall be established by each manufacturer which implements all requirements of this appendix. This system shall be documented in a quality manual. The current revision of the quality manual shall serve to demonstrate to the qualifying activity that the manufacturer's system is adequate to assure compliance with the applicable specifications and quality standards. Proprietary documents shall be clearly identified by category in the system.

D.3.2.2 Process flows. The system includes process flows for each distinct product family. As a minimum, flows shall include wafer fabrication, assembly, screening, and conformance inspection (CI), and shall indicate which CI option has been selected.

D.3.2.3 Quality system procedures. The manufacturer shall implement the approved system. The range and detail of the procedures shall ensure compliance with this system

D.3.2.4 Quality planning. The manufacturer shall define how the requirements for quality will be met. Quality planning shall be consistent with all other requirements of the manufacturer's quality system. The manufacturer shall give consideration to the following activities, as appropriate, in meeting the specified requirements for products.

a. The preparation of quality plans.

b. The identification and acquisition of any controls, processes, equipment (including inspection and test equipment), fixtures, resources, and skills that may be needed to achieve the required quality.

c. Ensuring the compatibility of the design, production process, inspection, and test procedures.

d. The updating, as necessary, of quality control, inspection, and testing techniques, including the development of new instrumentation.

e. The identification of any measurement requirement involving capability that exceeds the known state of the art, in sufficient time for the needed capability to be developed.

f. The identification of suitable verification at appropriate stages in the realization of product.

g. The clarification of standards of acceptability for all features and requirements, including those which contain a subjective element.

h. The identification of all contracted processes. Unless approved by the qualifying activity, all devices assembled in a contracted plant shall be tested (screening and CI) in the basic plant.

D.3.2.4.1 FMEA. Changes in design, materials, or processes for any device shall be reviewed in accordance with established design control procedures. Tools can include process flow charts, control plans, SPC, DOEs, or FMEAs.

D.3.2.4.2 Control plans. The control plan is prepared to summarize the process control planning for each technology. Product characteristics that should be included in a control plan are control item characteristics and significant characteristics. The evaluation method, sampling plans, data analysis method, and out of control plans are part of the control plan.

D.3.2.4.3 Plant moves. For all plant moves, all die banking inventory information shall be made available to the qualifying activity. All die not suitable for JAN assembly shall be controlled and identified. Die shall not be transferred from one facility to another facility without the approval of the qualifying activity.

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D.3.2.4.4 Business interruption control plans. Each supplier shall create a detailed business interruption control plan (e.g. using a Program Evaluation Review Technique (PERT) approach) and it shall be available upon request. Manufacturers shall notify the qualifying activity when implementing their business interruption control plan within five business days. The plan shall address some or all of the following as applicable:

a. Finished goods inventory.

b. Die inventory.

c. Multi-qualified wafer fabs including contract wafer fabs (mirror image qualified product).

d. Multi-qualified assembly plants including contracted assembly plants (mirror image qualified product).

e. Training new personnel.

f. New equipment.

g. New design.

h. New processes.

i. New materials.

j. Independent consultants/subject matter experts.

k. Top management understanding and commitment of resources.

l. Impact on quality.

m. Market analysis (customer needs).

n. Sole source and high volume product special attention.

o. Containment.

p. Off site storage of die and finished goods.

q. Security and safety officer (early detection).

r. Deliberate malice.

D.3.2.4.5 Incoming, in-process, and outgoing inventory control. The manufacturer shall employ procedures to control storage and handling of incoming materials, work in-process, warehoused, and outgoing product in order to 1) achieve such factors as age control of limited-life materials; and 2) prevent inadvertent mixing of conforming and nonconforming materials, work in process, finished product, resubmitted lots, or customer returns.

D.3.3 Conversion of specification requirements. The manufacturer shall have a system to convert specification sheets, orders, and contract requirements to in-house procedures, methods, travelers, and specifications. The system shall include revision and distribution controls to assure that product meets current requirements. The manufacturer's system ensures review of purchase requirements (and modifications) received from their customer. The review shall insure that the order does not violate this specification for JAN product of any level and that any special instructions are accounted for. The system shall provide objective evidence of this review.

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D.3.4 Critical interface control.

D.3.4.1 Design, materials, and processing documentation. Each product design shall be fully specified such that all aspects of design, verification testing, processing, and materials are described. In addition, each manufacturer shall have a flow chart, traveler, and DLA Land and Maritime Form 36D for each certified product line (see qualifying activity).

D.3.4.2 Design changes. After qualification, the manufacturer shall notify the qualifying activity prior to the release and shipment (for JANS prior to line implementation, except for evaluation samples) of product which undergoes any change in the product or verification program which affects performance, quality, appearance, reliability, or interchangeability (see appendix E-III, table E-III). Details of any change affecting DLA form 36D shall be sent to the qualifying activity. The changes shall be approved by the qualifying activity prior to release and shipment of product. Changes in design, materials, or processes for any device shall be processed in accordance with established change control procedures for all affected documents (see D.3.6.1 and D.3.7). Such notification shall include a thorough description of the proposed change and a test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability, or interchangeability and that the changed product will continue to meet all specification requirements. The completed test results shall be approved by the qualifying activity. After approval of the design change, all product inventory of the old design shall be submitted to CI within 6 months, unless authorization is extended by the qualifying activity (see H.3.1). Unless the design change has been required to correct or eliminate a verified design defect, finished devices manufactured to the previously approved design which are in inventory or in process of testing (i.e., inspection lot identification code assigned) will remain qualified only until that inventory is depleted.

D.3.5 Document control.

D.3.5.1 Document control procedures. The manufacturer shall establish and maintain procedures to control all documents that relate to the requirements of this specification. This includes, to the extent applicable, military and industry specifications and standards and their revisions, amendments, and notices. The manufacturer's procedures are required to be in a language that is understandable by the operator performing the given operation.

D.3.5.2 Document approval and issue. Documents shall be reviewed and approved for adequacy by authorized personnel prior to issue. A procedure shall be established to ensure that the current issue of appropriate documents are available at all required locations, that invalid or obsolete documents are promptly removed from all points of issue, and that any obsolete documents which are retained are suitably identified.

D.3.5.3 Document changes. Unless designated otherwise, changes to documents shall be reviewed and approved by the same functions/organizations that performed the original review and approval.

D.3.6 Purchasing.

D.3.6.1 Acquisition documentation. A system for the acquisition of supplies and approval of suppliers shall be established.

D.3.7 Control of customer supplied material. The manufacturer shall establish and maintain procedures for the control, verification, maintenance and storage, as applicable, of customer supplied materials and equipment. Any material which is lost or becomes unsuitable for use shall be recorded and reported to the customer.

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D.3.8 Product identification and traceability. All devices delivered to this specification shall be traceable through the lot identification code and inspection lot records, and identified as in D.3.9 through D.3.12, inclusive. In addition, JANS devices shall have a lot control system from wafer processing through screening which provides wafer lot identification; operation (machine), date of operation, operator(s) identification, quantity, and serial numbers (after step 8 of appendix E, table E-IV) of devices processed.

D.3.9 Process control. The manufacturer shall assure that all manufacturing operations are carried out to insure continued process capability. Operations shall be controlled as to the manner of production, requirements for monitoring and control, and output product characteristics. Where necessary, due to the complexity or sensitivity of operation parameters, the process shall consider working environment, workmanship criteria, equipment set-up, and the need for special operator certification or continuous monitoring of critical parameters.

D.3.9.1 Statistical process control (SPC). Where SPC is used to control a process, the control shall be established in accordance with EIA-557. If a sample exhibits an out-of-control condition, the product represented by the sample shall be subjected to evaluation and disposition. The evaluation process, results, and disposition shall be documented and traceable.

D.3.9.2 Environmental controls. The relative humidity, temperature, and particle count for each applicable critical process step (e.g., wafer fabrication, assembly) shall be specified, controlled, and recorded. The procedures and techniques for measuring these environmental parameters and limits shall be documented. The procedures shall contain corrective actions for out-of-tolerance environmental conditions. Unsealed parts shall be handled in such a way as to minimize the introduction of foreign material into the cavity. In addition, spittle protection is required in applicable critical areas. See test method 5010 of MIL-STD-750.

D.3.9.3 Chemical controls. The purity of chemicals, including water, shall be specified and controlled. The purity of process water shall be measured and recorded in terms of resistivity at +25°C (resistivity cells and meters shall be calibrated), total solids, organic impurities, and bacteria count.

D.3.9.4 Wafer lot process and inspection. All suppliers shall develop wafer lot acceptance procedures and criteria. These procedures and criteria shall address probe testing and lot acceptance testing, element evaluation or equivalent for each wafer, or wafer lot as applicable unless otherwise approved by the qualifying activity. The methods and procedures, which are utilized within wafer production for the following functions, as a minimum, shall be documented:

D.3.9.4.1 Process.

a. Sample sizes.

b. Wafer cleaning and handling operations.

c. Junction and surface preparations.

d. Alloying processes.

D.3.9.4.2 Inspection.

a. Control of wafer thickness. (For JANS, deviation shall be within 20 percent of the approved design nominal.)

b. Metal deposition and thickness. (For JANS, maximum thickness deviation shall be within 25 percent for the approved design nominal of > 35K Angstroms, and within 30 percent for ≤ 35K Angstroms. For JANS, the minimum front metallization thickness shall be 8,000 Å.)

c. Glassivation and passivation thickness. (For JANS, deviation shall be within 30 percent of the approved design nominal. For overlay structures or expanded metallization see H.3.7 for the minimum thickness that is required.)

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d. Gold backing thickness, when applicable. (For JANS, deviation shall be within 30 percent of the approved design nominal.)

e. SEM, see E.3.1.2.1 (for JANS).

Where limits are based on tolerances about a "approved design nominal", the nominal shall be stated in DLA Land and Maritime Form 36D, and shall either be identified on the JANS fab traveler or on a JANS inspection record submitted for approval by the qualifying activity.

D.3.9.5 Process monitor programs. Process monitor programs shall be established as referenced below, for processes performed by the manufacturer. A fully implemented and approved SPC program (in accordance with D.3.20.1) may replace all or portions of the process monitor programs with the approval of the qualifying activity. These programs shall be documented and made available to the certification team for review. The implementing procedures shall define sampled population frequency, sample size, reject criteria, allowable rework, and disposition of failed batch/product. Investigative and corrective actions shall be established which address noted deficiencies. Unless otherwise specified, a failure in the process monitor does not count against the PDA requirements. A procedure is required for the traceability, recovery, and disposition of all units monitored since the last successful test. All monitors shall provide for continual process improvement. Records of these monitors shall be available to any (Government or military user) audit team for review. As a minimum, the process monitors shall include the following, or equivalent, as approved by the qualifying activity:

a. Die attachment: The type of die mounting method, die material, die mounting material, package material, and mounting configuration shall be documented. The time, temperature, pressure, scrubbing, cleanliness, and environment shall also be specified. The manufacturer shall monitor the die attach integrity for all silicon transistors and case mounted diodes in accordance with test method 2017, 2076, acoustical techniques, or thermal response test methods 3101, 3161, 3131, 3103, and 3104 of MIL-STD-750 using the manufacturer's documented procedure. In addition, any die attached to a package or substrate in a cavity device shall be monitored for die attach integrity. This procedure shall be performed at each equipment set up for JANTX and JANTXV and, considered for other related factors (such as change of operator, etc.). This test shall be conducted on a minimum of two devices from each die attach station at the start and finish of operator change, package type change, die size change, and after every two hours of production for JANS. In the event that the die shear is less than the value shown in test method 2017, or the test leaves less than 75 percent silicon remaining of die-to-header bond surface, the output of the die attach station shall not be used until tests show that satisfactory operation has been re-established. A procedure for the traceability, recovery, and disposition of all units bonded since the last successful die attach integrity test is required. This procedure shall provide for sample size, reject criteria, and disposition of failed lots. This test may be conducted on the same samples used for the wire bond strength test.

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b. Wire bonding or interconnection: The bonding techniques, type of bond wire, and lead material used in connecting the die to the package leads shall be documented and comply with this specification and the applicable specification sheets. The temperature, pressure, dwell time, control of condition of capillary or electrode, ultrasonic power, composition of metals, lead dress, thickness to width ratio of bond, and environment shall be specified. The manufacturer shall monitor the wire bond strength in accordance with test method 2037 of MIL-STD-750 using the manufacturer's documented procedure. Any bi-metallic (e.g. gold-aluminum) interface at the die shall be baked for 1 hour at +300°C in either an air or inert atmosphere prior to the performance of the first wire bond strength testing with a minimum sample size of 11 wires, c = 0. See H.3.6.1 for bi-metallic bond restrictions. This procedure shall be performed at each equipment setup for JANTX and JANTXV and change of operators, lot, shift start and stop, and other related factors known to affect wire bond integrity shall also be considered. At manufacturer's discretion, wire bonding may continue up to the start of the sealing operation prior to the successful completion of wire bond process monitor testing. However, if any process monitors samples fail, then all devices wire bonded since the previous successful process monitor sample test shall be rejected. As a minimum for JANS, each operator/wire bonding station shall have two device samples taken at the start and end of each shift, after each two hours of production, and after changing operators, spools, shifts, packages, wire size, and maintenance of equipment. When more than one lot is processed for JANS in a two-hour period, samples shall be tested from each lot. Pull strength data shall be read and recorded and shall be control charted and maintained in accordance with the specified requirements. Data shall include the force, in grams, required for failure, the physical location of the point of failure, and the nature of the failure. In event that any bond strength is less than the pre-seal value shown in test method 2037 of MIL-STD-750, the bonder shall be inactivated immediately and not returned to production until tests show that satisfactory operation has been re-established. When the system at the die surface is bimetallic and the lead wires are less than 5 mils in diameter, the lead shall be pulled to destruction and if the chip bond lifts before the wire breaks, the lot shall be rejected. A procedure for the traceability, recovery, and disposition of all units bonded since the last successful bond strength test is required. This procedure shall provide for sample size, number of bonds, and device to be tested, reject criteria, and disposition of failed lots.

c. Glass-to-lead seals on clear glass diodes: Visual inspection procedures shall specify inspection criteria, the use of visual aids, and shall comply with the requirements of test methods 2069, 2070, 2072, or 2073 (die visual), and 2074 of MIL-STD-750. In addition, the manufacturer shall monitor the lead-to-glass seal following final plating operation for all JANTX, JANTXV, and JANS clear glass diodes constructed with borated seal using the manufacturer's documented procedure. The procedures shall specify criteria and visual aids and shall be capable of detecting significant loss of glass-to-lead seal.

d. Internal gas content: The moisture content of the sealing environment shall be controlled. The internal moisture content of ceramic and metal packages shall not exceed a value of 5,000 PPM for the sample when tested in accordance with test method 1018 of MIL-STD-750 at +100°C. All manufacturers shall exercise package internal moisture content monitors at key locations in the manufacturing flow. The internal water content vapor content test shall be performed on three devices for each active package design bi-monthly (twice a month) until sufficient data determines which packages are susceptible to elevated internal moisture and oxygen levels. A special internal gas content monitor schedule shall be determined for the packages until all problems are resolved. The problematic packages could be determined by pareto analysis or other statistical means. Any gas analysis with the presence of bombing gases that are indicative of non-hermeticity, O_2 to Ar ratio indicative of room air, dissimilar concentration of internally sealed gases (e.g. nitrogen, helium) than originally sealed in the device package, or the presence of leak test fluid (i.e. fluorocarbon) shall fail IGA even when the moisture content of the device has not exceeded 5,000 PPM.

When the internal water vapor content test is performed, the samples shall have been subjected to screen 3a of table E-IV or equivalent; not required for devices which are inactive for new design. All devices not utilizing a eutectic or epoxy die attach shall have the internal oxygen content of the sealing environment controlled. For devices not utilizing a eutectic or epoxy die attach the internal oxygen content of the device package shall not exceed the limits listed in test method 1018, of MIL-STD-750. All gases detected should be reviewed and recorded as this information can help the manufacturer determine at which process step the gases were introduced. Control limits for the following gasses shall be developed

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by each supplier; He, H_2, O_2, fluorocarbons, argon and CO_2. Devices shall be marked or otherwise traceable before being sent to the laboratory for testing. All devices not utilizing a eutectic or epoxy die attach shall not exceed 2,000 PPM of O_2 at +100°C. When the internal oxygen content test is performed, the samples shall have been subjected to screen 3 of table IV or equivalent; not required for devices which are inactive for new design. As part of the cap weld device setup, a procedure shall be used to verify that the cap welding electrodes provide uniform pressure during the cap weld process.

e. Particle contamination: The manufacturer shall establish a particle detection monitoring program which assesses the source of particle contamination of sealed cavity devices on an individual manufacturing line basis. The monitor shall have provisions for testing in accordance with method 2052, condition A, of MIL-STD-750. JEDEC Publication 114, may be used as a guideline. Unless otherwise exempt herein, this monitor is applicable to all metal or ceramic cavity package outlines. Particle Impact Noise Detection(PIND) monitor must be performed, as a minimum, at the start of the sealing lot, and at the end of the sealing lot. A minimum sample size of 10 devices per PIND sampling event is required and passing devices may be placed back into the lot. This monitor shall be performed before the lot is moved to the next operation, to ensure the most effective feedback loop for corrective action. As an example, the monitor could track by package type the sample or set-up Particle Impact Noise Detection (PIND) yields using an np chart. After historical data has been established for each package type, and the upper limits are set, any sample that goes out of control should have an assignable cause established by established techniques such as particle entrapment and analysis. If a JAN, JANTX, or JANTXV lot is outside the upper control limits or monitor criteria, it shall be tested 100 percent.

f. The manufacturer shall monitor thickness, composition (including Lead (Pb) content), and adhesion of final lead finish: (see appendix H). Test method 2100 of MIL-STD-750 may be used for thickness and composition measurements.

g. Wafer fabrication: The manufacturer shall establish the necessary controls (or equivalent, subject to the approval of the qualifying activity) for D.3.9.4 above. For JANS, compliance to tolerance about the “approved design nominal” shall be demonstrated.

h. Lot norm testing:

(1) For all semiconductors the manufacturer shall have a documented program to characterize thermal impedance, leakage current(s), and all other critical parameters, as applicable, to identify atypical devices. If a correlation of die attach voiding is approved by the qualifying activity, thermal impedance for lot norm testing is not required. Devices which exhibit atypical behavior shall be rejected and removed from the lot even when they pass the group A, subgroup 2 or thermal curve limits. The documented procedure shall discuss sample selection and how limits will be determined.

(2) The manufacturer shall have procedures in place to monitor and control lot-to-lot variations and central tendencies.

Lot norm limit: When there is no previous device lot history, or in the event of a major design change affecting the applicable critical parameters, the first five manufacturing lots will use their lot statistical averages as calculated in D.3.9.5.h.1 and charted using SPC techniques. At no time shall the lot statistical limits exceed those limits used in D.3.9.5.h.1. Atypical (maverick) lots shall be excluded. Lot norm testing as described in D.3.9.5.h.1 shall remain in effect for all lots processed thereafter.

Lot norm requirements for JANS:

(1) The manufacturer shall perform lot norm after burn-in for JANS. An additional lot norm may be performed prior to burn-in at the option of the manufacturer. Mean and standard deviation may be calculated from the full lot or from the Group A, subgroup 2, n = 116 sample. If calculated from a sample, the limits from the sample shall be applied to the full lot. The probability distribution may be represented in either normal or log-normal.

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If lot norm is applied pre-burn-in with 3-sigma limits, then 4-sigma limits may be applied after burn-in. Tests done at hot or cold temperature may be evaluated with 6-sigma limits, otherwise 3-sigma limits shall be applied. Devices outside of the limits shall be rejected from the lot, otherwise engineering comment shall be provided, if parts are considered for use. Devices which exhibit atypical behavior shall be rejected and removed from the lot even when they pass the group A, subgroup 2 limits, unless engineering comment is provided. If an unusual yield is observed, an engineering comment is required. Comparison to specification limits is not a rationale unless other information is provided. Alternative methods may be used with qualifying activity approval.

(2) The manufacturer shall evaluate lot norm data to ensure unimodal distribution within a single lot. If multimodal distribution is detected an engineering comment is required.

(3)The manufacturer shall ensure that the lot's mean and standard deviation are consistent with previous lots. After initial qualification a minimum of 5 lots must be statistically monitored for spread and location. Significant changes between lots for average and standard deviation shall be addressed by engineering comment. Lot-to-lot variation control in spread and location shall be performed for each critical parameter. Control shall be based on a moving average of at least 5 lots, see D.3.9.1. This control is for comparing lots only; evaluation of individual parts in (1) above shall be performed using the single lot mean and standard deviation. The requirements of paragraph 3 do not apply to parameters considered targeted parameters, such as h_{FE} on small signal transistors.

(4) These requirements may be waived for parts where production lot size is consistently less than 20 pieces with Qualifying Activity approval.

JANS lot norm parameters and test temperature per device type are listed below. For technology types not listed, monitored parameters shall be approved by the Qualifying Activity. Unless otherwise specified, all testing shall be performed at room temp. (e.g.…leakage at hot, Vf at cold)

Bipolar Transistors		MOSFET		Rectifier		Zener	
Parameter	Additional test temp	Parameter	Additional test temp	Parameter	Additional test temp	Parameter	Additional test temp
H_{FE} (for at least one condition)		V_{th} / $V_{GS(th)}$	Hot	I_R(Post HTRB) 1/	Hot	I_R(Post HTRB) 1/	Hot
I_{CBO}	Hot	IGSS	Hot	V_F(not applicable for devices above 1 watt)	Cold	$Z_{\theta JC}$(Post HTRB) 1/	
I_{EBO}		IDSS	Hot	$Z_{\theta JC}$ (post HTRB) 1/		V_F (post burn-in)	Cold
I_{CES} / I_{CEO}		RDSON	Hot				
$Z_{\theta JC}$		$Z_{\theta JC}$					

1/ Post HTRB hot and cold testing and lot norm analysis may be performed as part of screen 13b. All specification sheet post HTRB testing must be within specified test windows, or if not specified be within 16 hours after removal of applied voltage in HTRB.

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i. Internal visual: The manufacturer shall develop and document an internal visual monitor for JAN/JANTX levels.

j. Glass evaluation: The manufacturer shall monitor all hard glass diodes, tungsten or-molybdenum plug, in accordance with method 1057 of MIL-STD-750, condition B. Monitor may be performed at incoming inspection as sample evaluation of each glass lot in accordance with method 1057. In addition, the manufacturer shall establish a monitoring program to examine glass stress, strain, or excessive forces leading to potential glass cracking post assembly.

k. Nickel particles: All manufacturers using nickel caps in the encapsulation process shall implement a cap cleaning process. An in-line monitor shall be implemented to ensure the cleanliness of the nickel caps.

l. Lead integrity: Manufacturers shall have procedures in place to monitor and ensure the integrity of the lead. For axial devices this will apply to lead tension and fatigue in accordance with method 2036 of MIL-STD-750. For US devices it will apply to terminal tension only in accordance with method 2038, condition B, of MIL-STD-750.

m. Wafer saw monitor: For all semiconductors, the manufacturer shall establish an effective monitoring program for wafer saw operations. Saw monitoring shall be effective in preventing saw inflicted damage.

n. Laser Marking: The manufacturer shall establish a process monitor for laser marking legibility in accordance with paragraph 3.10, and laser marking depth to ensure corrosion prone base metal is not exposed in accordance with paragraph H.4.1. The manufacturer may use methodologies such as: non-contact depth measuring microscopy or laser depth measuring instrumentation to measure laser marking depth or any other comparable methodology.

D.3.10 Inspection and testing.

D.3.10.1 Inspection of purchased materials. The manufacturer is responsible for assuring that all supplies and services procured from suppliers conform to the contract. The manufacturer shall have a system which controls inspection, storage, and handling of incoming materials, and periodic evaluation of material received. The system may include verification of chemical, physical, and functional characteristics required by manufacturer drawings and specifications. The manufacturer may utilize a certified supplier program to reduce or eliminate receiving inspection.

D.3.10.1.1 Certified supplier program. The capability of supplied material may be validated through a supplier certification system. This program selects and monitors suppliers in order to guarantee that the supplied material will meet and maintain required capability levels (e.g., Cpk, PPM).

D.3.10.2 In-process quality control. The manufacturer shall perform in-process inspections to the extent necessary to assure product conformance to the end item specification. This shall include inspections and tests which assure that the manufacturing processes are in a state of control as indicated in D.3.9 herein.

D.3.10.2.1 Manufacturer imposed tests. Any manufacturer imposed test(s) (e.g., gross and fine leak) conducted prior to any qualification or conformance inspection, are to be performed on all subsequent lots until re-qualified. If any manufacturer imposed tests detect a problem, the manufacturer shall submit all devices in the lot to those tests to eliminate rejects and shall take steps to determine and eliminate the cause of failure (e.g., rough handling which has produced gross leaks).

D.3.10.3 Testing and screening operations. Testing and screening operations may only be performed in a facility which has received laboratory suitability approval from the qualifying activity for each MIL-STD-750 test method to be employed. Each facility lists their approved methods and equipment on DLA Land and Maritime Form 36 (Test Equipment List) or equivalent. The manufacturer shall document and update all burn-in socket capacity by plant location on the applicable DLA Land and Maritime Form 36(s). The environmental conditions where testing and screening are performed shall be maintained to assure compliance with the requirements of MIL-STD-750.

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D.3.10.4 Final inspection and testing.

D.3.10.4.1 Qualification and conformance inspection. Qualification and conformance inspection shall be performed as required by the specification sheet and herein at a facility which has received laboratory suitability approval from the qualifying activity. The results of testing shall be reviewed and approved.

D.3.10.4.2 Visual inspection. Shall be performed on outgoing product prior to shipping. Visual examination shall be in accordance with test method 2071 of MIL-STD-750. Sampling plans and acceptance are to be determined by the supplier and approved by the qualifying activity.

D.3.10.5 Inspection and test verification. The manufacturer shall verify that all inspection and testing required by this specification and their quality plan (see D.3.2.4) has been accomplished as specified. Verification shall clearly show whether the product passed or failed any requirement and shall contain sufficient detail to allow traceability to a specific operator, test date, equipment, and program.

D.3.11 Control of inspection measuring and test equipment.

D.3.11.1 Test programs and setups. The test programs and setups used to sort, classify, and test for MIL-PRF-19500 requirements are required to be traceable through each document control system to insure the correct revision was used when testing MIL-PRF-19500 devices.

D.3.11.2 Test equipment maintenance and calibration system. Maintenance and calibration systems and the frequency of scheduled actions, for gauges and test equipment, shall be established. The system may use NCSL Z540.3, or equivalent, as a guideline.

D.3.11.3 Conditions and methods of test. The general requirements, conditions and methods of test shall be in accordance with MIL-STD-750.

D.3.11.4 Electrical test equipment verification. The manufacturer shall define and utilize a method (e.g., correlation samples, diagnosis routines) to verify the measurement and operation characteristics of the test equipment when in use. In the event of verification of failure, the manufacturer shall utilize a procedure which will determine the requirements for traceability, recovery, and when re-testing is required of all units tested since the last successful verification.

D.3.12 Inspection and test status.

D.3.12.1 Inspection and test status. A system shall be maintained to insure that products are identified by their test or inspection status. The system shall insure the separation of products that have been tested or inspected from product that has not.

D.3.13 Control of nonconforming product. The manufacturer shall maintain a system which will prevent the shipment or use of nonconforming product. The system shall provide for identification, segregation, and evaluation of product, which does not conform to specified requirements at any point in the manufacturing and screening flow. Devices or lots, which fail any specification or internal tests, procedures, and requirements above and beyond the specification, except electrical selections, are not suitable for any JAN level. Lots failing electrical selections for any JAN level which, 1) are above and beyond this specification and, 2) are intended to select a portion of a lot having greater performance or reliability than the overall average performance or reliability of the lot other than through statistical methods are not suitable for any JAN level.

D.3.13.1 Reworked product. Reworked or repaired product shall be re-inspected in accordance with the appropriate procedures. All rework performed shall be documented and shall be attached to the original lot travelers and be traceable and may be part of management review.

D.3.13.2 Rework provisions. All allowable rework on devices manufactured under this specification shall be accomplished in accordance with the manufacturer's established procedures. Lots shall not exceed one rework for any process except additional or rediffusion, photo resist strip, and recoat.

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D.3.13.2.1 Wafer rework. For wafers, rework is limited to the following:

a. Additional etch to complete the process.

b. Photo resist strip and recoat.

c. Strip and redeposit of non-junction passivation. For JANS, strip and redeposition of any oxide, or passivation layer is not allowed.

d. Frontside and backside metal prior to sintering with the following stipulations.

(1) Visual examination of the front side of the wafer shall not reveal any residue or “ghost” footprint of the previous metallization. It shall essentially be impossible to differentiate a reworked stripped metallization wafer from a wafer that has never been metalized.

(2) Rework on front side metal is only permitted on junction devices (Zeners, diodes, rectifiers, bipolar transistors, SCR’s, JFET’s) or Schottkys. MOSFETs are not permitted.

(3) It is not permitted on any device for which the metal is patterned by an evaporation or sputtering metal mask rather than using photoresist and etching.

(4) It is permitted on backside metal for all types provided suitable masking or protection is used for the front side metal, unless the rework is purposely being done to rework both the front and the back metals that are of identical construction.

e. Additional or re-diffusion provided a photolithography step has not occurred between diffusions.

D.3.13.2.2 Assembly rework. Devices shall not be reworked during assembly. Devices shall not be subjected to temperatures or processes that will reflow the die attach material. Wire bonded devices shall not be subjected to rebonding. Devices shall not be resealed under any circumstances.

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D.3.13.2.3 Rework of assembled devices. No delidding or package opening shall be permitted except for disc packages. Unless otherwise specified, rework of sealed packages is limited to recleaning, rebranding to correct defective marking, and lead straightening, replating, or resolder dipping of the leads.

D.3.13.2.3.1 Rebranding and resolder dipping. Rebranding and resolder dipping may be performed as needed and are not subject to the one rework limit provided that the package body is not exposed to the abrasive effect of stripping solution. In cases that the removal of original plating or solder layer requires a stripping solution that etches out the layer, and exposes the package body to the same solution a sample (n=11, c=0), shall undergo test method 1057, condition B, of MIL-STD-750 for glass body diodes, and test method 1041 of MIL-STD-750 for metal hermetic packages in addition to the following standard tests. After any replating or resolder dipping, all JANS shall pass as a 100-percent screen, the requirements of visual and mechanical inspection criteria of test method 2071 of MIL-STD-750, group A, subgroup 2 of the specification sheet and the hermetic seal requirements of screen 14 of table E-IV herein. For JANS clear glass diodes only, the parts shall pass the hermetic seal requirements with a sample size of 116 pieces with no failures. All JANTXV and JANTX shall pass the requirements of group A, subgroup 2 of the specification sheet and the hermetic seal requirements of screen 14 of table E-IV herein with a sample size of 116 pieces with no failures allowed. Solder dip and plating rework shall be in accordance with appendix H.

D.3.13.2.3.2 Irradiation of assembled devices. Irradiation is allowed when part of the qualified flow. If any portion of a lot requires re-irradiation, a new inspection lot shall be formed, identified, and CI performed as required.

D.3.13.3 Rejected lots. Lots with an unscreenable failure mode shall be rejected (see E.6.1.1).

D.3.14 Corrective and preventive action.

D.3.14.1 Corrective action. The system identifies the necessity for corrective action as a result of failure, defect analysis, inappropriate performance of inspections/procedures, product trends, product returns, inappropriate records and audits (both customer and internal). The corrective action system shall establish a classification system to identify the severity of the nonconformance. The corrective action system shall provide for the use of a standard format for documenting corrective actions. This standard format shall provide evidence of: Documents created and /or revised to correct the problem, training, and follow-up to verify implementation and effectiveness. Failure to initiate corrective action may result in removal of products from the qualified product list. The corrective action system shall identify when a failure analysis is to be employed (i.e., field use, user analysis, Government Industry Data Exchange Program (GIDEP), competitive analysis).

D.3.14.2 Preventive action. Appropriate sources of information such as processes and work operations which affect product quality, audit results, quality records, and customer complaints shall be used to detect, analyze, and eliminate potential causes of non-conformities. The methodology for the initiation, implementation, and follow up of preventive actions shall be documented by the manufacturer.

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D.3.15 Handling, storage, packaging, and delivery.

D.3.15.1 Security of completed devices. Marked devices which have passed all screening and conformance requirements shall be retained in a secure area prior to shipment delivery. Device inventory shall be controlled by device type, lot identification code, quantity, product assurance level, and transaction date. This requirement applies to both the manufacturer and the distributor(s).

D.3.15.2 Electrostatic discharge sensitive (ESDS) program. The manufacturer of ESDS class 0, 1A, 1B, 1C, and class 2 devices shall institute an ESDS program commensurate with the product classification. The product classification shall be as indicated in appendix E. The requirements of JESD-625 apply but may be tailored for establishing an ESDS program. Justification for the tailoring shall be made available to the qualifying activity for approval upon request.

D.3.16 Quality records. A system shall be in place to track all quality records, including but not limited to the results of all qualification, screening, conformance tests (attributes and variable data as applicable) and inspections, any required failure analysis, subcontractor records, internal quality audit results, training, and management reviews. The retention period for each type of quality records shall be a minimum of 10 years based on lot date code. Quality records shall be available for review and maintained in an organized manner. Quality records may be retained using paper hard copy or suitable electronic medium. If electronic records are used, the manufacturer shall establish appropriate procedures to ensure the security of such records from loss or alteration. Corrections to “hard copy” quality records shall be made by "lining out" the incorrect entry with a single line (maintaining the legibility of original data) and inserting the correct entry immediately adjacent to the "lined out" entry. The operator making the change shall initial by the "lined out" entry. Erasures, mark overs, and "white out" are not permitted on any quality record. For electronic records, the manufacturer shall establish procedures for making corrections to quality records such that the original record, the date of the change, and the person making the change are clearly identifiable. Quality records of any type and data of any type shall be sent to the qualifying activity or preparing activity upon request.

D.3.17 Internal quality audits.

D.3.17.1 Internal audit program. The manufacturer shall establish an independent internal audit program which shall be included in the quality system. The internal audit program shall assess compliance with all MIL-PRF-19500 requirements (processes and quality systems) and shall identify key review areas, their frequency of audit, and corrective actions shall be resolved through the corrective action system (see D.3.16).

D.3.17.1.1 Internal audit checklist. The internal audit checklist shall assure that the quality system is adequate and followed by all personnel in each area.

D.3.17.1.2 Subcontractor Audits. All subcontractors are to be audited by the certified manufacturer (or by DLA Land and Maritime) annually, to have an available checklist of the audit, a list of any concerns, and corrective actions available to the qualifying activity for review. This does not apply to facilities with DLA Land and Maritime lab suitability. All certified manufacturers shall demonstrate effective control of relevant contracted services as approved by the qualifying activity.

D.3.17.2 Audit schedules and frequencies. The audit frequency shall in no case exceed 1 year for each area unless authorized by the qualifying activity. An internal audit shall be conducted and corrective actions completed prior to the initial qualifying activity audit. The qualifying activity may modify the frequency of the internal audit(s) or require additional inspection based on the effectiveness of the manufacturer's internal audit program, and assessment of the internal audit findings.

D.3.17.3 Auditors. The designated auditors shall be independent from the area being audited. If the use of an independent auditor is not practical, a second individual should be assigned to participate in the audit or review the results. Auditors shall be trained in the area to be audited, in the applicable military specification requirements and provided with an appropriate checklist for annotating deficiencies. Prior to the audit, the assigned auditor(s) shall review the previous internal audit results to assure corrective actions have been implemented and are effective.

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D.3.17.4 Audit deficiencies. All audit deficiencies shall be conveyed to the responsible individual for corrective action(s). All corrective actions shall be agreed to by the management representative. A procedure shall be established to follow up on all audit deficiencies to assure that the corrective actions have been implemented in a timely manner. If recurrent deficiencies are found, additional corrective action shall be taken to assure correction of the problem and the qualifying activity shall be notified. The internal audit team shall perform a follow-up within 6 months for verification of corrective actions and to assure that they are adequate.

D.3.18 Training.

D.3.18.1 Personnel performing quality operations. All personnel performing an operation affecting quality shall be trained in, and familiar with, that part of the operation relevant to their function. They shall have sufficient responsibility and authority to inspect and accept or reject the product according to the applicable specifications.

D.3.18.2 Training requirements. Work training practices shall be established, using formal or on-the-job methods, and utilized in acquiring and maintaining job skills as required in critical work areas. Training shall cover operator certification for critical processes, periodic proficiency reviews, testing for certification, expiration of cert/training status, and identification of minimum quality system training requirements, as applicable.

D.3.18.3 Personnel training records. Records shall cover appropriate documented information as evidence of competence for jobs performed. Records are required only for product-related training and testing as distinguished from safety, first aid, etc.

D.3.19 Servicing. The manufacturer's system shall describe the methods used to meet customer expectations. This includes but is not limited to: Applications support, customer returns, stock rotation, GIDEP issuance, warranty issues, and customer complaints.

D.3.20 Statistical techniques.

D.3.20.1 SPC program. The method used for process control may or may not utilize SPC, but will use a method that is appropriate for the process being controlled. Compliance with EIA-557 is a requirement for JANS certification.

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STANDARD VERIFICATION SYSTEM FOR QUALIFED PRODUCTS

E.1 SCOPE

E.1.1 Scope. This appendix contains the standard verification system which, when performed in its entirety, assures that the product will meet the performance requirements. This program also measures and evaluates the manufacturer's manufacturing process against a baseline for that process. This baseline can include innovative and improved processes that result in an equivalent or higher quality product, provided that the process used to evaluate and document these changes have been reviewed and approved by the qualifying activity. Changes to the process baseline can be made by the manufacturer after achieving approval with documented reliability and quality data. The approach outlined in this appendix is a proven baseline, which contains details of the screening and conformance inspection (CI) procedures. (See tables E-I through E-X.) However, manufacturers shall demonstrate to the qualifying activity, a system that achieves at least the same level of quality as could be achieved by complying with this appendix. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

E.2 APPLICABLE DOCUMENTS

E.2.1 General. The documents listed in this section are specified in sections E.3 and E.6 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they shall meet all specified requirements of documents cited in sections E.3 and E.6 of this specification, whether or not they are listed.

E.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

NCSL INTERNATIONAL (NCSL)

NCSL Z540.3	-	Calibration Laboratories and Measuring and Test Equipment - General Requirements.

(Copies of these documents are available online at http://www.ncsli.org or from NCSL International 2995 Wilderness Place, Suite 107 Boulder, CO 80301–5404.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

E.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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E.3 GENERAL TEST AND INSPECTION INFORMATION

E.3.1 Wafer lots. Wafer lots consist of semiconductor wafers formed into lots at the start of wafer fabrication for processing as a group. Each lot is assigned a unique identifier to provide traceability and maintain lot integrity throughout the fabrication process. The wafer lot shall be traceable to the silicon lot(s) or portion thereof. The maximum number of wafers in the wafer lot size shall be defined by the manufacturer. Wafer lot processing as a group is accomplished by any of the following procedures, providing process schedules and controls are sufficiently maintained to assure identical processing in accordance with process instructions of all wafers in the lot:

a. Batch processing of all wafers in the wafer lot through the same machine process steps simultaneously.

b. Continuous and sequential processing (wafer by wafer or batch portions of wafer lot) of all wafers through the same machine(s) or process steps. No other product, lots, equipment maintenance, repair, or calibration shall interrupt continuous or sequential processing. Any deviation from this processing requires the written approval of the qualifying activity or a new wafer lot identifier will be assigned.

E.3.1.1 Formation of inspection lots. JAN, JANTX, and JANTXV devices shall be assembled into an identifiable inspection lot or collection of inspection sublots. JANS devices shall be assembled into an identifiable inspection lot. Each inspection lot shall be identified by a unique lot identification code (see 3.10.8).

E.3.1.1.1 JAN, JANTX, and JANTXV inspection lot. The total number of devices that the manufacturer submits at any one time for qualification or conformance inspection and which conforms to the following criteria shall constitute an inspection lot.

a. The maximum small inspection lot size shall be 2,500 devices.

b. The lot accumulation period shall not exceed 6 consecutive calendar weeks of device seal operations. The inspection lot is submitted to determine compliance with the requirements of the specification sheets.

c. Each inspection lot shall consist of devices of a single device type or consist of a collection of sublots of structurally identical device types contained on one or more specifications sheets (see E.3.1.1.1.1 and E.3.2).

d. Assembly lot identification shall be maintained from the time the lot is kitted.

e. The entire inspection lot shall be accumulated prior to initiating conformance inspection. Samples shall be randomly selected from the entire inspection lot.

E.3.1.1.1.1 Inspection sublot. An inspection sublot shall be a single device type (part number) contained on a single specification sheet manufactured on the same production line(s) through final seal and to the same device design with the same material requirements and within the same 6-week lot accumulation period.

E.3.1.2 JANS lots.

E.3.1.2.1 Scanning electron microscope (SEM) inspection. If any wafer lot is of a selected die design (such as overlay structure devices and devices with metallization path to bond pad crossing any junction covered by passivation or glassivation where the bonding pad is not on the same active area of the device), SEM inspection is required prior to acceptance of the wafer lot (generally this is applicable only on particular small signal bipolar transistors and MOSFETs). This inspection shall be performed in accordance with, test method 2077 of MIL-STD-750, for each of these lots. If a contracted laboratory is used for the actual SEM inspection, the manufacturer shall document and define the responsibilities of the manufacturer and contractor regarding steps within test method 2077 that may be performed by either party, such as: Sample selection and sample preparation.

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E.3.1.2.2 JANS inspection lots. A JANS inspection lot shall consist of the total number of devices that the manufacturer submits at one time and which conforms to the following criteria:

a. Small lots shall not exceed 1,000 pieces. Sampling inspection for large or small lots shall be in accordance with table E-VIA, and E-VII.

b. All devices shall be of a single device type.

c. All devices shall be from a single wafer lot.

d. All devices shall be assembled on the same production line with the same technique from die attach through final seal, within 21 working days, not to exceed 31 calendar days.

E.3.2 Structurally identical device types. Structurally identical device types are devices manufactured on the same production line(s) within the same plant through final seal using the same fabrication processes within the same generic package outlines and construction materials and differ only electrically. The packages can vary in size but the overall construction shall use the same process. The number and size of wires, or size of pins, can vary to handle the power rating but the assembly process shall be identical with identical materials used. The processes used to attach pins to tubes, slugs to pins, wires to chips and pins, sealing technique, and die attach shall be identical. For glass diodes, surface mounts and axial leads may be grouped together. Examples of such structurally identical device types are as follows:

a. Rectifiers, signal diodes, or thyristors grouped into different voltage ratings. Standard recovery, fast recovery, and ultra fast recovery rectifiers are not typically considered structural identical. Rectifiers and diodes with identical design rules (same doping technique, passivation, and device structure) which differ only in die size and package size are considered structurally identical. Initially, group B and C shall be performed on the highest voltage device, intended for qualification, for each device construction. On subsequent lots, the die sizes/package styles, which receive group B and C inspection, shall be rotated on every lot thus assuring that all die sizes/package styles receive groups B and C inspection.

b. Transistor groupings. Transistors with the same die structures that vary only in die size and package size are considered structurally identical if the following criteria are met.

   Die shall have the same generic design rules and vary only in size. Channel stop, voltage enhancement, and emitter ballasting techniques, epi-base, diffused base, expanded contacts, and metal interconnects over oxide steps shall be the same. The process sequence in the diffusion and photolithographic areas shall be the same. Transistors shall have similar peak frequency responses and V ratings that do not vary more than three to one (e.g., 3 MHz to 9 MHz, 60 VDC to 180 VDC). Darlington transistors cannot be grouped with standard transistors.

   Group C conformance inspection for small die flow products (as defined by the performance specification sheet) does not involve any die related tests, therefore, group C coverage is based only on case outline. Group C for non-RHA product shall not be used to cover RHA product and vice-versa, unless same design is used for both.

c. Power MOSFETs with the same voltage ratings. For qualification, power MOSFETs of the same voltage types with identical design rules (field termination and cell density) and which differ only in die size are considered structural identical. Group C conformance inspection for MOSFET's does not involve any die related tests, therefore, group C coverage is based only on case outline. Group C for non-RHA product shall not be used to cover RHA product and vice-versa, unless same design is used for both.

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d. Zener diodes, reference diodes, and transient voltage suppressors; grouped into different Zener voltage ratings with identical design rules (same passivation and device structure) which differ only in starting silicon, crystal orientation, epitaxial resistivity, or die size or package type shall be considered a family. For purposes of conformance inspection, structurally identical device groupings do not apply for Group B JANS device types but they may be used for qualification purposes. For qualification, for each JAN, JANTX, and JANTXV, group B shall be performed on the highest and lowest Zener voltages in the structurally identical group. On subsequent lots, of JAN, JANTX, and JANTXV levels, the die sizes/package styles which receive group B inspection shall be representative of all the structurally identical family devices contained within the inspection lot, rotated on every lot assuring that all die/package styles receive group B inspection periodically.

e. Stack junction rectifiers and diodes; grouped into different voltage ratings with identical design rules (same doping technique, passivation and device structure) which differ only in active die count and die/package size (X, Z and Y axis 1/) are considered structurally identical and shall be considered a family. For purposes of conformance inspection, structurally identical device groupings do not apply for group B JANS device types but they may be used for qualification purposes. For qualification, for each JAN, JANTX, and JANTXV, group B shall be performed on the highest and lowest voltages in the structurally identical group. On subsequent lots of JAN, JANTX and JANTXV levels, the die sizes/package styles which receive group B inspection shall be representative of all the structurally identical family devices contained within the inspection lot, rotated on every lot assuring that all die/package styles receive group B inspection periodically.

1/ Axis orientation defined in MIL-STD-750.

E.3.3 Disposal of samples. Devices subjected to destructive tests or which fail any test shall not be shipped. Sample devices from lots which have passed CI and which have been subjected to mechanical or environmental tests specified in groups B and C inspection and not classified as destructive, may be shipped provided each of the devices subsequently passes group A, subgroup 2 inspection.

E.3.4 Destructive tests. Unless otherwise demonstrated, the following MIL-STD-750 tests are classified as destructive:

Method number	Test
1017	Neutron irradiation.
1018	Internal gas analysis.
1019	Steady-state total dose irradiation procedure.
1020	Electrostatic discharge sensitivity (ESDS) classification.
1021	Moisture resistance.
1036, 1037	Intermittent operation life.
1041	Salt atmosphere corrosion.
1042 (condition D)	Burn-in and life test for power MOSFETs or insulated gate bipolar transistors (IGBT).
1046	Salt spray (corrosion).
1056	Thermal shock (liquid to liquid).
1057	Resistance to glass cracking.
1080	Single-event burn-out and single-event gate rupture
2017	Die attach integrity.
2031	Soldering heat.
2036	Terminal strength.
2038	Surface mount end cap bond integrity
2037	Bond strength (destructive bond pull test).
2075	Decap internal visual design verification.
2077	Scanning electron microscope (SEM) inspection of metallization.
2101	Destructive physical analysis (DPA) procedures for diodes.
2102	Destructive physical analysis (DPA) procedures for transistors.
3478	Power transistor electrical dose rate.

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All other mechanical or environmental tests (other than those listed in E.3.5) shall be considered destructive initially, but may subsequently be considered nondestructive upon accumulation of sufficient evidence to indicate that the test is nondestructive. The accumulation of data from five repetitions of the specified test on the same sample of product, without significant evidence of cumulative degradation in any device in the sample, is considered sufficient evidence that the test is nondestructive for the device of that manufacturer. Any test specified as a 100-percent screen shall be considered nondestructive for the stress level and duration or number of cycles applied as a screen.

E.3.5 Nondestructive tests. Unless otherwise demonstrated, the following MIL-STD-750 tests are classified as nondestructive:

Method number	Test
1001	Barometric pressure (reduced).
1022	Resistance to solvents.
1026, 1027	Steady-state operation life.
1031, 1032	High temperature life (non-operating).
1038, 1039, 1040	Burn-in screen.
1042 (conditions A, B, and C)	Burn-in and life test for power MOSFETs or insulated gate bipolar transistors (IGBT).
1048	Blocking life.
1051 (100 cycles or less)	Temperature cycling (air to air).
1071	Hermetic seal.
1081	Dielectric withstanding voltage
2006	Constant acceleration.
2016	Shock.
2026	Solderability. (if the original lead finish is unchanged and if the maximum allowable number of reworks is not exceeded).
2052	Particle impact noise detection (PIND) test.
2056	Vibration, variable frequency.
2066	Physical dimensions.
2069, 2070, 2072, 2073, 2074	Internal visual (pre-cap).
2071	Visual and mechanical examination.
2076	Radiographic.
2081	Forward instability shock (FIST).
2082	Backward instability, vibration (BIST).
3051, 3052, 3053, 3474	Safe operating area (SOA) (condition A for test method 3053). (with limited supply voltage)
3101	Thermal impedance (response) testing of diodes.
3103	Thermal impedance measurements for insulated gate bipolar transistors (IGBT) (delta gate-emitter on voltage method).
3104	Thermal impedance measurements of GaAs MOSFET (constant current forward-biased gate voltage method).
3131	Thermal impedance measurements for bipolar transistors (delta base-emitter voltage method).
3161	Thermal impedance measurements for vertical power MOSFET's (delta source-drain voltage method).
3181	Thermal resistance for thyristors.
4066	Surge current.
4081	Thermal resistance of lead mounted diodes (forward voltage, switching method).

When the junction temperature exceeds the device maximum rated junction temperature for any operation or test (including electrical stress test), these tests shall be considered destructive except under transient surge or nonrepetitive fault conditions or approved accelerated screening when it may be desirable to allow the junction temperature to exceed the rated junction temperature. The feasibility shall be determined on a part by part basis and, in the case where it is allowed adequate sample testing, shall be performed to provide the proper reliability safeguards.

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E.3.6 Resubmitted lots.

E.3.6.1 Resubmitted lots of JAN, JANTX, JANTXV, and JANS. Resubmitted lots shall be kept separate from new lots and shall be clearly identified. Any failed lot for groups A, B, or group C may be resubmitted one time only, for the failed subgroup, at double the original lot sample size with zero additional failures; if it is determined by analysis of all the failed devices that the failure mechanism is due to a defect that can be effectively removed by rescreening the entire lot, and that rescreening has been performed. A failed lot which is rescreened (resubmittal to inadvertently missed process steps or tests is not considered a rescreen) may not be used to satisfy the periodic group C coverage. The lot may be resubmitted only to the failed subgroup to determine its' own acceptance. Failure analysis shall be performed on all failed lots. Lots which fail group B, bond strength, decap internal visual, and SEM (when applicable), shall not be resubmitted. For subgroups A-2, A-3, and A-4, see footnote 7 of table E-V).

E.3.6.2 Resubmitted lots of RHA devices. A JANTXV lot which fails group D tests may be resubmitted once, in accordance with footnote 4 or 7, as applicable, of table E-VIII. A JANS lot which fails group D, subgroups 1 or 3 tests, may be re submitted once, in accordance with footnote 3 of table E-VIII. JANS wafers which fail group D, subgroup 2 tests shall not be resubmitted. In lieu of resubmission, a lot or wafer which fails group D testing may be used as a non-RHA device or may be certified at a lower RHA level if the group D data indicates the lot or wafer meets the lower level requirements.

E.3.7 Conditions and methods of test. Conditions and methods of test shall be in accordance with MIL-STD-750. The general requirements of MIL-STD-750 apply as specified. A system for control and calibration of test equipment shall be established. NCSL Z540.3 may be used as guidance for the calibration system.

E.3.7.1 Alternative and equivalent test methods and requirements. Other test methods (flows, criteria, conditions) or circuits may be substituted for those specified in MIL-STD-750 provided equivalency is demonstrated to, and approved by, the qualifying activity. Such a substitution will in no way relax the performance requirements of this specification. The schematic wiring diagram of the test equipment or test methods (including, as applicable, bias conditions, time, temperature, mounting conditions, etc.) shall be made available for review by the qualifying activity. The manufacturing site's performance history will be essential criteria for approval. Any design change (in accordance with table E-III) to a product, using an approved alternate test flow, requires approval by the qualifying activity.

E.3.7.1.1 In-line conformance inspection. Supplier specific (custom) in-line conformance inspection is permitted in lieu of group B conformance with the approval of the qualifying activity on continuous production lines.

E.3.7.2 Procedure in case of test equipment failure or human error. If it is determined through a detailed engineering evaluation that a failed device is the result of test equipment failure or human error, a replacement device from the same inspection lot may be added to the sample. The replacement device shall be subjected to all those tests to which the discarded device was subjected prior to its failure and to any remaining specified tests to which the discarded device was not subjected prior to its failure. Failures occurring as a result of operator error, prior to the start of testing, may be replaced by the manufacturer but shall be noted on the lot history. Any electrostatic discharge (ESD) failures shall be counted as rejects and not be attributed to equipment/operator error for qualification, conformance inspection, screening, group A, and end-point electrical tests of MIL-STD-750. Immediate corrective action is required whenever failures are attributed to test equipment failure or human error. In all cases, a detailed report shall be documented and submitted to the qualifying activity within 10 working days. Each occurrence shall be recorded on the applicable travelers.

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E.3.7.2.1 Particle impact noise detector (PIND) test equipment failure or error. If it is determined through an engineering evaluation and traceable through the manufacturer's quality system, that PIND rejects are the result of equipment failure or human error, then all devices (passed and rejected) subsequent to the equipment failure or human error, may be restarted. The devices shall be restarted at the run in which it was determined the equipment failed or the human error occurred.

E.3.7.3 Standard mixer diodes and holders. The manufacturer of UHF and microwave mixer diodes shall establish and maintain standard mixer diodes and standard mixer holders for use in qualification and quality conformance testing of UHF and microwave mixer diodes. These standards shall be calibrated at least once in each successive 12-month period or prior to use if over 12 months, at a laboratory acceptable to the Government.

E.3.8 JANS electrical test data. Unless otherwise specified in the specification sheet, all electrical measurements performed on devices after screen 8 of table E-IV shall be recorded and traceable to the serial number. This includes all manufacturer imposed tests.

E.3.8.1 Summary of parts fallout. A summary of the JANS parts fallout during screening tests shall be prepared by the manufacturer in accordance with the requirements of the qualifying activity.

E.3.9 JAN, JANTX, JANTXV electrical test data. Unless otherwise specified in the specification sheet, conformance inspection electrical measurements shall be recorded. This includes all manufacturer imposed tests.

E.3.10 Preservation of lot identity. During all screening, inspection, and marking operations, each lot and sublot shall be kept segregated, secure, and traceable.

E.4 VERIFICATION

E.4.1 Qualification inspection. Qualification inspection shall be performed at a facility approved by the qualifying activity and shall be conducted in accordance with the procedures described herein for groups A, B, C, D, and E inspection and by the qualifying activity. Qualification of a particular device type to a given quality level may be extended by the qualifying activity to any other quality level provided all the groups A, B, C, D, and E requirements of the other level have been met and provided that suitable approved screening facilities are available for the other tests and stress levels. In addition, the requirements of appendix H shall be met. Small lot sampling shall not be used for qualification inspection. Variables data is required for qualification. Manufacturers shall notify the qualifying activity when a qualification lot is removed from consideration as a qualification lot. The manufacturer shall notify the qualifying activity in writing within 35 days that the lot is being removed from consideration.

Group D is required for each inspection lot of RHA types as specified in the specification sheet. Qualification for RHA shall be for a specific semiconductor die.

An alternate qualification procedure for RHA devices for levels M, L, and D only, are available for devices with demonstrated RHA. These devices shall be submitted for qualification inspection and conformance inspection, if process or design changes affecting RHA are made.

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E.4.2 Inspection routine. All samples subjected to groups B, C, D, and E shall have been chosen from a lot which has passed the requirements for group A except as modified in E.6.5. The following conditions apply:

a. The required sample plan for series of devices shall consist of group A inspection for the highest and the lowest voltage types or as the qualifying activity requires.

b. A sample from one sublot shall be tested for each group B subgroup. One device representing each process variation within the series to be qualified shall be submitted to the design verification examination. If devices within the series differ only by electrical selection, then only one type needs to be submitted to design verification.

c. A sample from one sublot shall be tested for each group C subgroup. At the option of the manufacturer, devices from group B, tables E-VIA and E-VIB, may be continued on in group C, subgroup 6, to achieve 1,000 hours or 6,000 cycles total, or separate samples may be used.

d. When group D (RHA) qualification extension is granted, the radiation facility shall be approved by the qualifying activity. A sample from a sublot of each device type shall be tested for each group D subgroup.

e. Devices which are constructed using braided leads may be processed through table E-IV screening and qualification high temperature testing prior to the addition of leads. Qualification testing requiring load current conduction will require that leads be attached.

E.4.2.1 Qualification to electrostatic discharge sensitivity (ESDS) classes. Initial qualification to an ESDS class or requalification after redesign shall consist of qualification to the appropriate quality and reliability level plus ESDS classification in accordance with test method 1020 of MIL-STD-750.

a. Although little variation due to case outline is expected, if a device type is available in more than one package type or case outline, ESDS testing and classification shall be applied to at least that one package type shown by experience to be worst case for ESDS. ESDS classification test results shall be submitted to the qualifying activity for all specification sheets for listing. Specifications using structurally identical die designs may be classified with data from previously classified types. Any dissimilar designs within a specification sheet shall have ESDS classifications for each structurally identical grouping.

b. All Zeners, (voltage reference and voltage regulators) and transient suppressors are considered non-sensitive by design.

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ESDS classification levels are defined as follows:

ESDS class designator 1/	Prior designation category	Individual part marking	Package marking	Electrostatic voltage	Device ESD failure threshold classification
0		Δ0	Δ0	<250 V	Fails after exposure to any ESD pulse less than 250 V.
1A		ΔA	ΔA	250 - 499V	Any part that passes after exposure to an ESD pulse of 250 V but fails after exposure to an ESD pulse of 500 V.
1B		ΔB	ΔB	500 - 999 V	Any part that passes after exposure to an ESD pulse of 500 V but fails after exposure to an ESD pulse of 1,000 V.
1C		ΔC	ΔC	1,000 - 1,999 V	Any part that passes after exposure to an ESD pulse of 1,000 V but fails after exposure to an ESD pulse of 2,000 V.
2	B	ΔΔ	ΔΔ	2,000 - 3,999 V	Any part that passes after exposure to an ESD pulse of 2,000 V but fails after exposure to an ESD pulse of 4,000 V.
3A		ΔΔΔA	ΔΔΔA	4,000 V - 7,999 V	Any part that passes after exposure to an ESD pulse of 4,000 V but fails after exposure to an ESD pulse of 8,000 V.
3B		ΔΔΔB	ΔΔΔB	8,000 V – 15,999	Any part that passes after exposure to an ESD pulse of 8,000 V but fails after exposure to an ESD pulse of 16,000.
Non-sensitive				> 15,999 V	Any part that passes after exposure to an ESD pulse above 16,000 volts.

E.4.2.1.1 ESDS. ESDS classification testing shall be done in accordance with test method 1020 of MIL-STD-750 and the applicable specification sheet (see 3.10.3.1). Devices shall be handled in accordance with the manufacturer's in-house control documentation. Devices that are classified non-sensitive are not required to be handled as ESD sensitive, and manufacturer's in-house control documentation plan is not required. Handling shall begin at lead clip or wire bond (e.g., for packages which do not have a lead shorting bar or do not have leads shorted together). Guidance for device handling is available in the JESD-625 document.

1/ ESD class designator 1 has been replaced with designators 0, 1A, 1B, and 1C; and ESD class designator 3 has been replaced by 3A and 3B ESD class designators as of 30 May 2006. Devices not yet ESD re-classified may continue to be marked as class 1, 2, or 3 until testing determines the appropriate class. The manufacturer may test and mark the tested level obtained under the column titled “Individual Park Marking”. After 30 May 2006 (for newly qualified products) the 1A, 1B, 1C, 2, 3A, and 3B designators shall be used and all currently qualified devices specified in the current revision of QPDSIS-19500 as ESD classes 1 or 3 shall be ESD tested by the respective manufacturers to determine correct device ESD classification within 1 year after this revision has been dated. Prior designation category devices previously classified by test as category B may be marked as class 2 (ΔΔ).

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E.4.2.2 Qualification by extension. Qualification of a structurally identical device, a series of devices from the same or different specification sheets, or qualification of a new package for a previously qualified die may be extended from a previously fully qualified device provided the following information and data are supplied to the qualifying activity:

a. Previously qualified device type, specification sheet number, and qualification reference number. When qualifying by extension, thermal impedance data shall be submitted for all specification sheets, in addition to the specification sheet the qualification testing was performed on.

b. Design and construction information on devices covered under different specification sheets.

c. Samples of structurally identical devices with certification that these samples are structurally identical to the previously qualified device.

d. Group A variables data on a sample plan of each structurally identical device type except for series of devices which shall be the sample plan of the highest and the lowest voltage types, or as the qualifying activity requires, or as specified in specification sheets covering groups of devices. Test samples of selected devices in a group or portion of a group shall be from the same inspection lot.

e. Results and variables data for each structurally identical device on all groups B and C electrical tests not specified in group A, including tests at temperature extremes.

f. All results and variables data on groups B and C tests as follows:

    (1) Data on any tests not required by the qualified device.

    (2) Data that is the result of tests performed at stress levels greater than those required for the qualified device.

    (3) Data for any tests requiring more exacting limits than those found for the qualified device.

g. Items E.4.2.2.d through E.4.2.2.f shall not be required if the qualifying activity can be assured that the previously fully qualified device at least meets all of the conditions and requirements for the proposed structurally identical device type, except for device type marking.

Qualification by extension does not necessarily imply conformance inspection coverage to all device types covered by the qualification by extension approval.

E.4.3 End-points. End-point electrical measurements shall be measured and recorded as applicable (e.g., if delta's are required) starting and after completion of all specified tests in the subgroups of groups B, C, D and E. Devices shall not be hot solder dipped between burn-in and post burn-in electricals. There shall be no additional high temperature process or test performed prior to delta electrical measurements. Pre-test electrical failures shall be replaced by acceptable devices. An engineering evaluation shall be performed and corrective action taken when necessary on all screened devices which fail pre-test. Resubmission of the lot to the failed parameter shall be initiated whenever operator error or mishandling is not found to be the cause for failure.

E.4.4 Selection of samples. All samples shall be randomly selected from the qualification inspection lot. MIL-STD-750 method 1071 condition C is prohibited for use on conformance inspection samples. Sample selection for group D testing shall be in accordance with table E-VIII and shall be from each wafer or from each inspection lot, as appropriate.

E.4.5 Identification of samples. The manufacturer's management representative may, at their option, mark or authorize the marking of each sample to be subjected to qualification testing in order to distinguish these devices from those not intended for qualification inspection.

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E.4.6 Qualification inspection lot release. The inspection lot from which the qualification samples are selected may be offered for delivery under contract after qualification approval has been granted provided screening and conformance inspection requirements are satisfied.

E.5 SCREENING

E.5.1 Screening. All JANS, JANTXV, and JANTX semiconductor devices (100-percent) shall have been subjected to, and passed, all the applicable screening tests (as specified in table E-IV) in the sequence shown and the applicable percent defective allowed (PDA) for the type of semiconductor and quality level (device class) specified. Devices which fail any test criteria in the screening flow shall be identified and controlled until removal from the lot. At the option of the manufacturer, the rejects may continue processing to allow the manufacturer to maintain traceability. The lot records shall identify the point of failure and the actual percent defective (PD) (as applicable). Any rejected devices shall be removed from the lot prior to shipment. Manufacturers shall have a procedure to address screening failures. Screening failures should be evaluated as necessary to determine root cause of failure to provide a means for continuous improvement. Except for JANS, the conditioning and screening tests performed as standard production tests need not be repeated when these are predesignated and acceptable to the qualifying activity as being equal to or more severe than specified herein provided the relative process conditioning sequence is maintained. All tests, preconditioning and screening operations, which were performed on the devices submitted for qualification inspections specified herein shall be performed on all devices subsequently submitted for conformance inspection (see E.6).

E.5.2 Percent defective allowable (PDA). Selected electrical parameters shall be designated in the specification sheet for screen 11 and 13, of table E-IV which shall be used for the PDA calculation. These parameters may also be compared to determine whether the change during burn-in (delta) is indicative of a lot stability problem. All burn-in pre-conditioning failures, either high temperature reverse bias (HTRB) or power burn-in, shall be counted towards the applicable PD unless the pre-conditioning is part of the manufacturer's qualified flow. A lot which fails PDA shall be subjected to an engineering evaluation.

E.5.2.1 JANTX and JANTXV PDA. The PDA for each inspection lot (or screening lot if the alternate flow is used) shall be 10 percent (for each burn-in) on all failures for the specified electrical parameters in screens 11 and 13a of table E-IV. Delta limits shall be defined in the specification sheets. For delta limits, the delta parameter values measured after burn-in (100-percent screening test) shall be compared with delta parameter values measured prior to that burn-in. Unless otherwise specified, inspection lots which exceed the 10 percent PDA may be resubmitted one time only to the burn-in operation failed. The PDA shall be 3 percent on the resubmitted inspection lot to each failed burn-in. If the combined burn-in PD's for the first submission exceeds 20 percent, or either of the resubmitted burn-in exceed the 3 percent PDA, the entire lot shall be unacceptable for any quality level.

E.5.2.2 JANS PDA**.** The PDA for each inspection lot shall be 5 percent (for each burn-in) on all failures for specified electrical parameters in screens 11 and 13a of table E-IV. Delta limits shall be defined in the specification sheets. For PDA delta limits, the delta parameter values measured after burn-in (100-percent screening test) shall be compared with delta parameter values measured prior to that burn-in. Unless otherwise specified, inspection lots which exceed the 5 percent PDA may be resubmitted one time only to the burn-in operation failed. The PDA shall be 3 percent on the resubmitted inspection lot to each failed burn-in. If the combined burn-in PD for the first submission exceeds 10 percent, or either of the resubmitted burn-in exceed the 3 percent PDA, the entire inspection lot shall be unacceptable for any quality level. A lot which fails the PDA shall be subjected to an engineering evaluation.

E.5.3 JANTX and JANTXV product. The procedure for testing and screening for JANTXV and JANTX devices shall be in accordance with tables E-IV, E-V, E-VIB, E-VII, figure E-1, and as specified in the applicable specification sheet.

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E.5.3.1 Alternate procedure for screening of JANTX and JANTXV types. JAN types may be processed and marked as JANTX and JANTXV types by the original part manufacturer on their own qualified product provided the following procedures are satisfied:

a. All devices to be proposed for JANTXV processing (except clear glass JANTXV diodes which shall be subjected to internal visual inspection before painting or marking) shall have been subjected to and passed JANTXV internal visual 100-percent screening prior to seal.

b. Groups A, B, and C inspection shall have met the JANTX and JANTXV level requirements in accordance with tables E-V, E-VIB, E-VII, figure E-2 , and the applicable specification sheet.

c. Screening shall be conducted in accordance with table IV, figure E-2, and the applicable specification sheet. All units failing these tests shall be removed from the lot and the quantity removed shall be noted in the lot history.

d. A sample of the screened devices shall be submitted to and pass the requirements of table E-V, subgroups 1 and 2 inspection (see table E-V and table E-VIA, subgroup 1) (see table E-VIB, subgroup 1) subsequent to the 100-percent screening (of the lot or separate portions thereof) as specified in E.5.3.1.c and as shown on figure E-2.

E.5.3.2 Bin and cell pre and post burn-in electrical measurements. Alternate methods to variables recording may be used to determine delta end-point requirements of JANTX and JANTXV burn-in provided the qualifying activity has granted written approval. When alternate methods to variables recording are used to determine delta end-point requirements, devices shall be separated into groups, each of which shall have maximum and minimum limits on the variable parameter(s). The difference in parameter limits for any group shall not exceed the delta requirements for the variable parameter(s).

E.5.3.3 Alternate procedures for qualification and conformance inspection where JAN is not covered by the specification sheet. When the JAN quality level is not included in the specification sheet, or at the option of the manufacturer, the alternate flow (see figure E-2) may be used for qualification and conformance inspection. The lot used shall be marked in accordance with 3.10.1, except the "JAN" designating symbol shall be replaced by "JANQ". Unless they are submitted to the flow that the lot was submitted to, these samples shall not be shipped.

E.5.3.4 Lead forming for JANTX and JANTXV. When lead forming is specified, it shall be followed by n = 116, c = 0 fine and gross seal tests, group A, subgroup 2, and external visual examination, n = 45, c = 0.

E.5.3.5 Laser Marking for JAN, JANTX, and JANTXV. When laser marking is performed, it shall be followed by external visual examination in accordance with MIL-STD-750 test method 2071, using minimum magnification of 30X, lot sampling, n = 22, c = 0. For JAN quality level product fine and gross testing, n = 22 c = 0 is required. Fine and gross leak screening (Test Method 1071, Screen 14) must be performed after laser marking operations have been completed. Alternate methods may be approved by the qualifying activity.

E.5.4 JANS product. The procedure for testing and screening of JANS devices shall be in accordance with tables E-IV, E-V, E-VIA, E-VII, figure E-3, figure E-4, and the applicable specification sheet.

E.5.4.1 PIND test for JANS devices. The inspection lot (or sublots) shall be submitted to 100-percent PIND testing a maximum of five times in accordance with test method 2052 of MIL-STD-750, test condition A. PIND prescreening shall not be performed. The lot may be accepted on any of the five runs if the percentage of defective devices is less than one percent (zero failures allowed for lots of less than 50 devices). All defective devices shall be removed after each run. Lots, which do not meet the one percent PDA on the fifth run, or exceed 25 PD cumulative, shall be rejected and resubmission is not allowed. These parts shall not be shipped as any other quality level. When calculating numbers of allowed failures using percentages, fractional values shall be increased to the next whole integer (see figure E-4).

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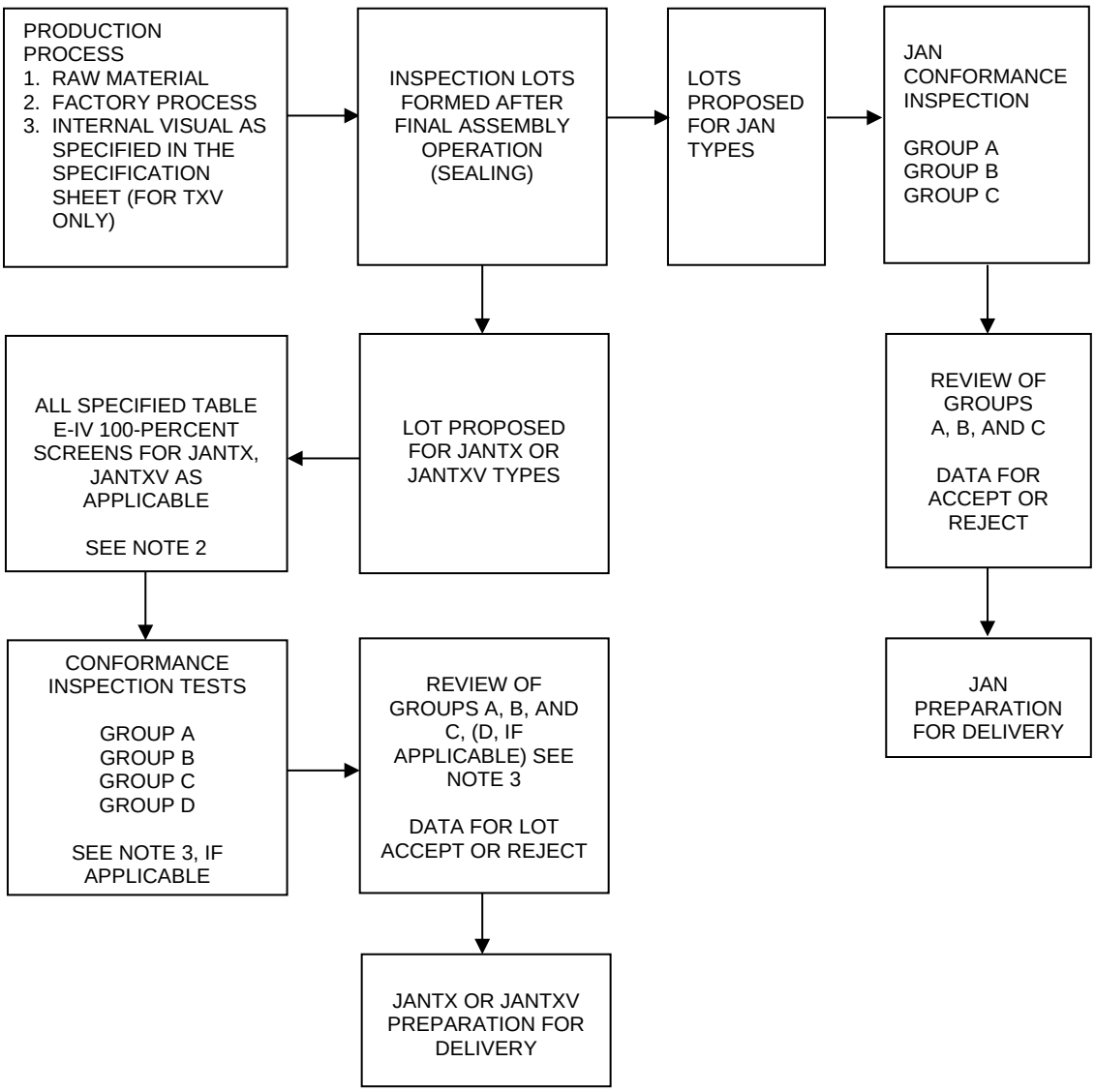
E.5.4.2 Lead forming for JANS . When lead forming is performed on JANS devices, it shall be followed by screen 7 of table E-IV, group A, subgroup 2, and external visual examination on 100-percent of the lot.

E.5.4.3 Burn-in socket verification for JANS . The electrical continuity between each device and the socket shall be verified prior to initiating burn-in (see MIL-STD-750 for details).

E.5.4.4 Laser Marking for JANS. When laser marking is performed on JANS devices, it shall be performed prior to screens 14, 15, and 16.

E.5.5 Failure evaluation for JANS Failures that occur during JANS screening shall be evaluated to determine if failure mode is the result of a latent (time dependent) defect, workmanship, or design weakness. This information shall be retained and presented to the qualifying activity, when requested, for review and determination if a failure trend is developing that needs corrective action. The manufacturer's procedure(s) shall define when a formal failure analysis is required to be performed.

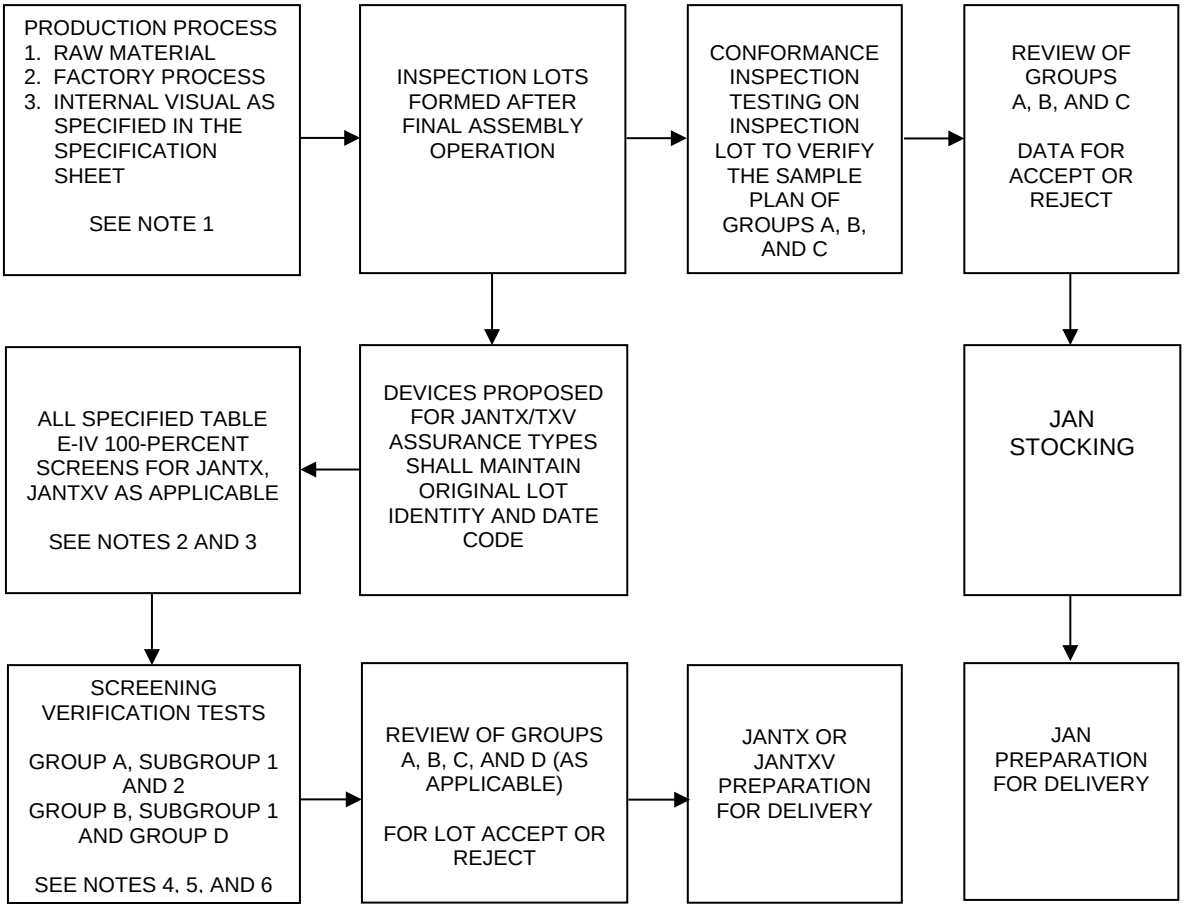
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NOTES:
1. All products to be proposed for JANTXV processing shall have been subjected to and passed JANTXV internal visual 100-percent screening at this step (except for clear glass JANTXV diodes which shall be subjected to internal visual inspection prior to painting and marking).
2. Order of the tests shall be performed as specified in table E-IV.
3. Group D inspection may be performed at any point prior to lot formation.

FIGURE E-1. Order of procedure diagram for JAN, JANTX, and JANTXV device types.

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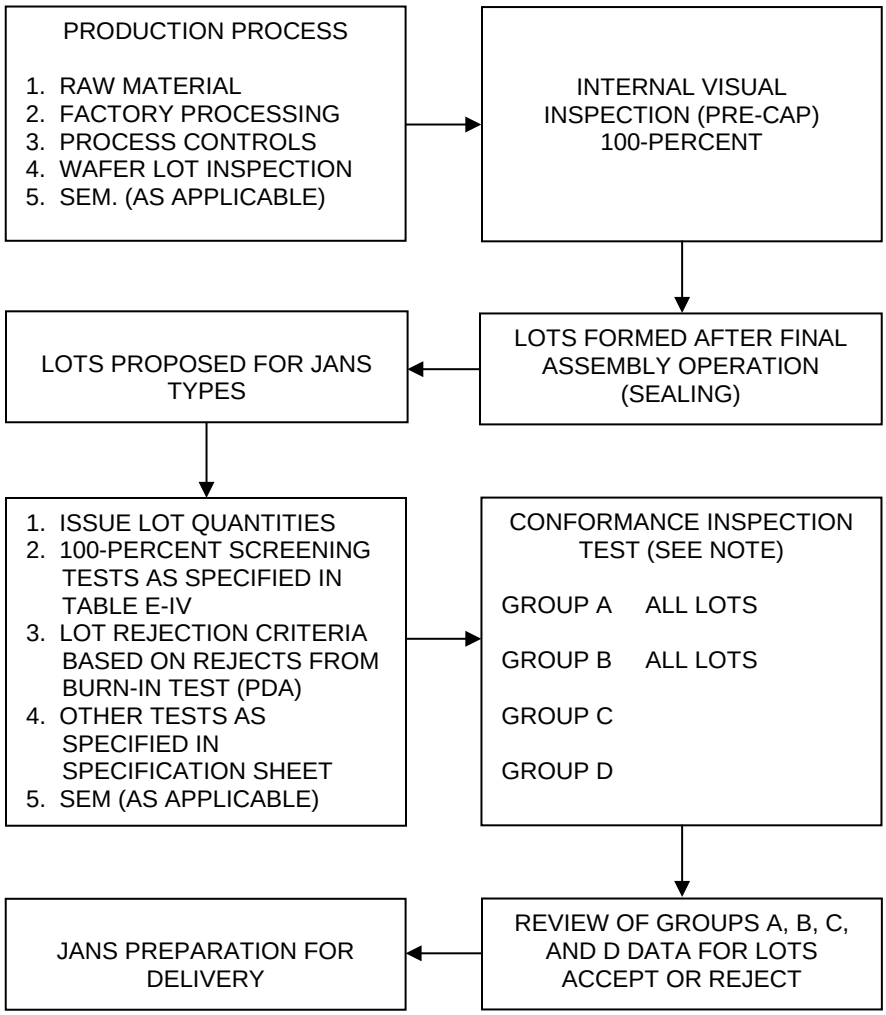


NOTES:

1. All product proposed for JANTXV processing shall have been subjected to and passed JANTXV internal visual 100-percent screening in accordance with table E-IV herein at this step (except for clear glass JANTXV diodes which shall be subjected to internal visual prior to body paint or mark).
2. If a JAN inspection lot is not processed in parallel with the material designated for JANTX and JANTXV, all groups A, B, C, and D testing shall be performed on a JANTX or JANTXV inspection lot as shown on figure E-1.
3. The order of all screening tests shall be performed as specified in table E-IV.
4. Test method 2026 of MIL-STD-750, omit steam ageing, sample size is 15 leads c = 0. This modified B1 testing cannot be used as a solderability inspection date.
5. Test method 1022 of MIL-STD-750, sample size is 15 devices c = 0.
6. If the inspection lot is composed of a collection of sublots, each sublot shall pass full group A inspection as specified.

FIGURE E-2. Alternate order of procedure diagram for JAN, JANTX, and JANTXV device types.

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NOTE: Group D testing may be performed at any point following the production process (see E.6.5).

FIGURE E-3. Order of procedure diagram for JANS.

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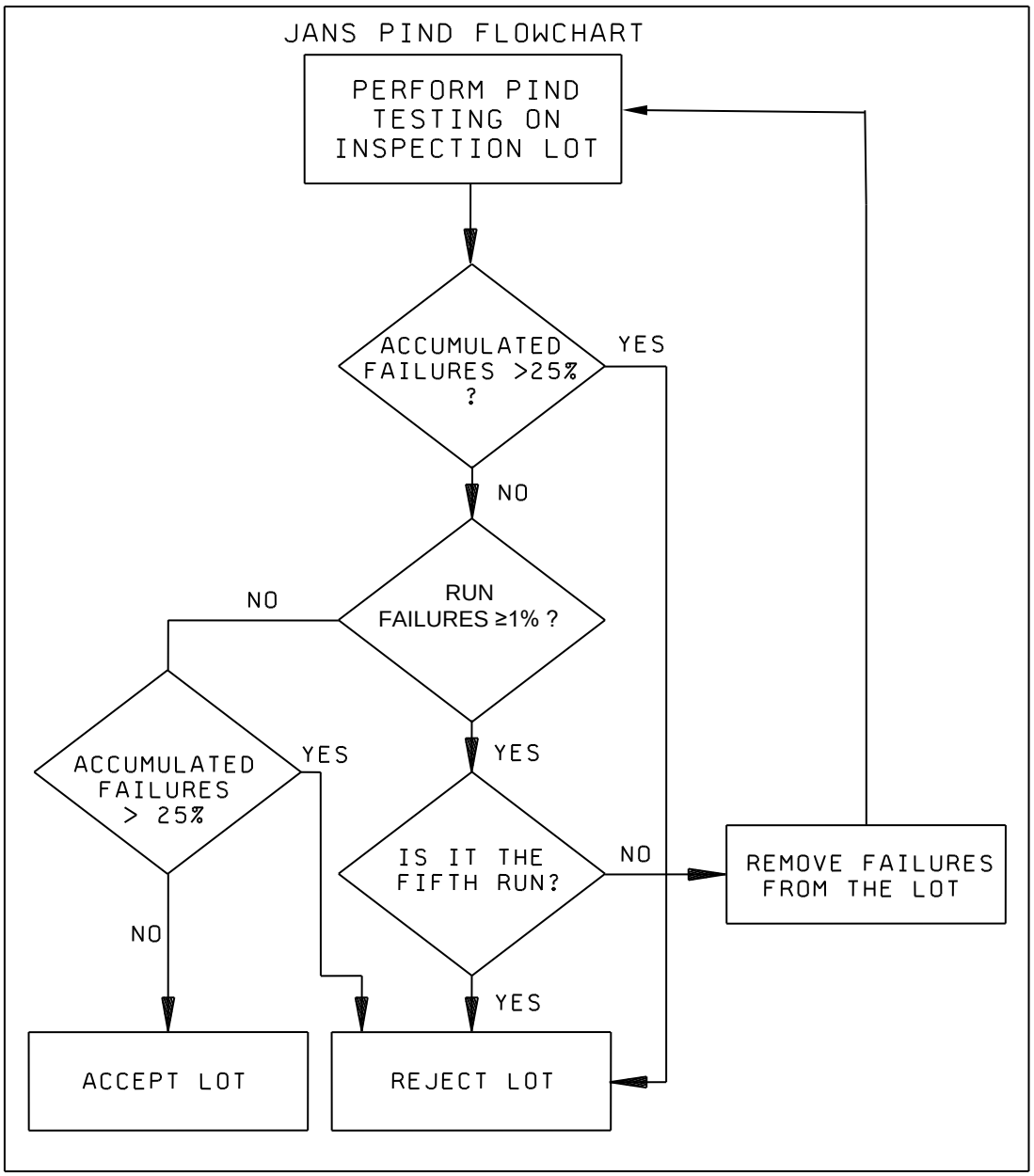


FIGURE E-4. JANS PIND flowchart.

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E.6 CONFORMANCE INSPECTION

E.6.1 Conformance inspection. Conformance Inspection shall be conducted in accordance with the requirements of groups A, B, and C for the specified quality level as well as group D to the applicable RHA level. If a lot is withdrawn in a state of failing to meet conformance requirements and is not resubmitted, it shall be considered a failed lot and reported as such. Each lot shall be subjected to groups A and B inspection. Successful completion of group C conformance for a given quality level shall satisfy the group C requirements for all quality levels and devices represented by the structurally identical group. The grouping of structurally identical devices (see E.3.2) shall be as agreed between the manufacturer and the qualifying activity. JANS devices shall not be used to represent the other quality levels. If a manufacturer elects to eliminate all or any conformance inspection steps substituting either a process monitor or statistical process control (SPC) procedures (when approved by the preparing activity and qualifying activity), the manufacturer is only relieved of the responsibility of performing the conformance inspection (see 4.4). The manufacturer still bears full liability for any failure that may result if these tests are performed at a later time. A manufacturers reliability program may be used in lieu of all or any conformance inspection when equivalent to or compliant with C.8.3. MIL-STD-750 test method 1071 condition C is prohibited.

TABLE E-I. Sample plans.

Maximum percent defective or lambda, accept number (c) (r = c + 1)	50	30	20	15	10	7	5	3	2
	Minimum sample sizes (for device-hours required for life test, multiply by 1,000).								
0	5	8	11	15	22	32	45	76	116
1	8	13	18	25	38	55	77	129	195

E.6.1.1 Nonconformance. Lots which fail subgroup requirements of groups A, B, C, or D may be resubmitted in accordance with the provisions of E.3.6. However, if the lot is not resubmitted or fails resubmission, the lot shall not be shipped and the JAN marking shall be obliterated or removed within 30 days. For additional guidance on group D failures see E.6.5. Manufacturers shall notify the qualifying activity when a conformance inspection lot is removed from consideration as a JAN branded inspection lot. When a group C conformance inspection lot is removed, manufacturers shall notify the qualifying activity within five business days. All catastrophic failures (opens, shorts) shall be evaluated. The manufacturer's procedure(s) shall define when a formal failure analysis is required to be performed.

E.6.2 Group A inspection. Group A inspection shall be performed on each inspection lot and shall consist of visual and mechanical inspection and electrical tests as specified in table E-V and the specification sheet. Group A inspection may be performed in any order. If an inspection lot is made up of a collection of sublots, each sublot shall pass group A inspection as specified. Unless the entire inspection lot has seen the same screening, devices which have received PIND (one pass condition A) may not be considered as candidates for this inspection. This does not apply to devices PIND tested as a process monitor.

E.6.3 Group B inspection. Group B inspection shall be performed on each inspection lot except for the specification sheets that specify the flow in table E-VIC. Group B shall be in accordance with table E-VIA, E-VIB, or E-VIC as applicable, and the specification sheet. Testing of one device type sublot in any subgroup or step shall be considered as complying with the requirements for that subgroup or step for all types in the lot. Different device types may be used for each subgroup except for JANS. All inspections, except for life tests, shall be applied only to completed and fully marked devices from lots which have been subjected to and passed the group A, subgroup 1, 2, and 3 requirements. When the final lead finish is solder, or any plating prone to oxidation at high temperature, the samples for life tests (table E-VIB, subgroups 3 and 6, and table E-VIA, subgroups 4, 5, and 7) and temp cycle (table E-VIA, subgroup 3, and table E-VIB, subgroup 2) may be pulled prior to the application of final lead finish. The test samples may be cleaned prior to the electrical end-point testing. An evaluation shall be performed on all failures to determine if the failure mode is the result of a latent (time dependent) defect, workmanship, or design weakness. Appropriate corrective action shall be performed as a result of the evaluation. All tests within a subgroup shall be performed in the order specified except table E-VIA, subgroup 2, and table E-VIB, subgroup 1.

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E.6.3.1 Lots shipped prior to group B completion. No lots shall be shipped prior to completion of group B without the approval of the qualifying activity. This provision is only to be requested for emergency procurement situations and it is expected that the user will provide a justification.

E.6.4 Group C inspection. Group C inspection shall be in accordance with table E-VII and shall include those tests specified which are performed periodically at 1-year intervals on at least one device type from each structurally identical device grouping (from the same or different specification sheet) in which the manufacturer has qualified device types. This inspection shall be applied only to completed and fully marked devices from lots which have been subjected to and passed the group A, subgroup 1, 2, and 3 requirements. When the final lead finish is solder, the life test subgroup may be pulled prior to the application of final lead finish. All tests within a subgroup shall be performed in the order specified. The test samples may be cleaned prior to the electrical end-point testing. An evaluation shall be performed on all failures to determine if the failure mode is the result of a latent (time dependent) defect, workmanship, or design weakness. Appropriate corrective action shall be performed as a result of the evaluation. Lots with an unscreenable failure mode shall be rejected. Unless all devices intended for manufacturing during that period will receive, as a minimum, the same screening, devices which have received PIND screening in accordance with E.5.4.1 may not be used to qualify the next group C inspection periods. A device type which fails a group C inspection shall not be accepted until the device type which failed, successfully completes the failed group C subgroup(s). Other device types from the same qualified group represented by the failed device type may be accepted provided group C inspection requirements have been satisfied for those device types.

Samples from subsequent lots of the device types in the structurally identical device grouping which failed group C resubmittal inspection, or are withdrawn in a failing state, shall then be subjected to all the tests in the subgroup in which the failure occurred, on a lot-by-lot basis until three successive lots pass the failed subgroup. The testing may then return to periodic testing.

E.6.4.1 Group C sample selection. Samples for subgroups in group C shall be chosen at random from the first lot submitted for conformance inspection during the specified group C inspection interval. Testing of one device type for each subgroup shall be considered as complying with the requirements for that subgroup for all structurally identical types (see E.3.2). A different device type(s) shall be tested at each successive inspection interval until all structurally identical device types qualified on the same or different specification sheet(s), from the same qualified line, have been tested, except power MOSFETs grouped by the same voltage as described in E.3.2. When none of the inspection lots passing group A of the first lot submitted contain the device type which is due to be tested, the samples for inspection shall be chosen from those types in the inspection lots being tested which have not been used for the longest time for group C inspection. The date code of the lot establishes (begins) the 1-year group C interval. Groups A and B shall also be completed on the group C inspection lot date code prior to the coverage being valid. Small lot samples may not be used to satisfy group C requirements for inspection lots of structurally identical devices. Small lot samples may only be used to satisfy group C for the inspection lot being sampled.

E.6.4.2 Lots shipped prior to group C completion. No lots shall be shipped prior to completion of group C without the approval of the qualifying activity. This provision is only to be requested for emergency procurement situations and it is expected that the user will provide a justification.

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E.6.5 Group D inspection. Group D inspection shall be performed in accordance with table E-VIII and the requirements of the specification sheet. Group D may be completed any time in the lot history (i.e., prior to lot formation, or before screening). Group D sample devices shall be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package and as a minimum, pass group A, subgroup 2 prior to irradiation. A sample which fails group D may be resubmitted once, in accordance with E.3.6.2. In lieu of resubmission, a lot or wafer which fails group D testing may be used as a non-RHA devices or may be certified at a lower RHA level if the group D data indicates the lot or wafer meets the lower level requirements. See E.4.1 for Group D qualification inspection requirements.

E.6.6 Group E inspection. Group E is a workmanship, ruggedness, and critical interface verification inspection. Group E testing, when specified on the specification sheet is required for initial qualification, re-qualification, and when group E requirements are changed. Whenever group E is re-performed due to changes in requirements (including sample size and bias condition modifications), only the affected subgroups need to be performed. The results of group E testing shall be submitted to the qualifying activity (by all manufacturers prior to shipment of product, as applicable). Product redesigns may be subjected to group E testing as required by the qualifying activity.

E.6.6.1 Group E testing requirements. Group E shall be performed in accordance with table E-IX and the specification sheet. All tests within a subgroup shall be performed in the order specified. An evaluation shall be performed on all failures to determine if the failure mode is the result of a latent (time dependent) defect, workmanship, or design weakness. Appropriate corrective action shall be performed and approved by the qualifying activity.

E.6.6.2 Alternate group E testing procedures. Manufacturers may use internal design verification or ongoing reliability assessment programs in lieu of group E, subgroups 1 and 2 only, provided this testing is equivalent to or more stressful than group E and is performed using the same design and construction (DLA Land and Maritime Form 36D) on file at the qualifying activity. This alternate testing shall be approved by the preparing activity and the qualifying activity.

E.6.7 Groups B, C, D, and E end-points. Post-test end-points specified in the specification sheet shall be measured for each device of the sample after completion of all specified tests in the subgroups. Pre-test electrical failures shall be replaced by acceptable devices. An engineering evaluation shall be performed and corrective action taken if necessary on all screened devices, which fail pre-test. Resubmission of the lot to the failed parameter shall be initiated whenever operator error or mishandling is not found to be the cause for failure. Except as specified or otherwise required, all life test (such as operation, storage, blocking) end-point measurements shall be performed within 96 hours after sample units have been subjected to and removed from required tests. All other end-point test measurements shall be made within 168 hours, or as specified. Additional measurements may be made at the discretion of the manufacturer. If end-point measurements can not be performed within the specified time, refer to the applicable MIL-STD-750 test method for the procedure to follow.

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TABLE E-II. RHA levels and requirements. 1/ 2/

Radiation level		
RHA designation	Total ionizing dose (RAD(Si)) 3/	Neutron fluence (N/Cm^2) 4/
M	3×10^3	
D	1×10^4	
P	3×10^4	
L	5×10^4	
R	1×10^5	
F	3×10^5	
G	5×10^5	
H	1×10^6	

1/ See E.6.5.
2/ The highest level may be qualified without qualifying any lower level.
3/ Test in accordance with test method 1019 of MIL-STD-750.
4/ Test in accordance with test method 1017 of MIL-STD-750. Unless otherwise specified in the specification sheet, the minimum neutron fluence shall be 2×10^{12} N/cm^2.

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TABLE E-III. Testing guidelines for changes to a qualified product. 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8/

Changes (see D.3.4.2)	JAN/TX/TXV Product Subgroups required to be performed and data submitted to DLA Land and Maritime	JANS Product Subgroups required to be performed and data submitted to DLA Land and Maritime	Samples to be submitted to qualifying activity in accordance with tables and subgroups herein
1. Die properties			
a. Construction technique (alloy, planar, mesa)	Group A, C6, D	Groups A, B, C, D, E	C6, 2 samples
b. Substrate, epitaxial properties	Group A, C6, D1	Group A, C6, D1	C6, 2 samples
c. Diffusion method (alloy, ion implant, diffusion)	Group A, C6, D1, D2	Group A, B5, D1, D2	C6, 2 samples
d. Surface deposition - passivation	Group A, test method 1039 of MIL-STD-750, cond A, C6, D2	Group A, B5, D2	C6, 2 samples
e. Surface deposition – front metal (material / process)	Group A, B2, and C6	Group A, B3, and C6	C6, 2 samples
f. Surface deposition – back metal (material / process)	Group A, B2, C6, and E4	Group A, B3,C5, C6, and E4	C6, 2 samples
g. Surface deposition – glassivation over metal (material / process)	Group A, test method 1039 of MIL-STD-750, cond A, C6, D2	Group A, B5, D2	C6, 2 samples
h. Surface deposition – Die protective coating (material / process)	Group A, test method 1039 cond A, C6	Group A, B3, C6	C6, 2 samples
i. Geometry - die size	Group A, C6, D, E4	Group A,B,C,D,E	C6, 2 samples
j. Geometry - die thickness	Group A, C6, E4, B4	Group A, C6, E4, B3	C6, 2 samples
k. Fab location move	Groups A, B, C, D, E, Notify qualifying activity	Groups A, B, C, D, E, Notify qualifying activity	One each B and C
l. Wafer diameter	Groups A, B, C, D, E, Notify qualifying activity	Groups A, B, C, D, E, Notify qualifying activity	
2. Package properties			
a. Package material / dimension change (base, lid, plug, lead, glass)	Groups B1, B2, B3, C1, C7, D2, E4	Group B1, B2, B3, Group C1-C7, D2 and E4	
b. Die attach method / material	Groups C3, C6, E1, E4	Groups B3, C3, C6, E4	C6, 2 samples
c. Bond wire material / diameter / process	Groups B2, B3, C3, B4	Groups B3, C3, C6	B3, 2 samples
d. Sealing technique / environment	Groups B2, B3, C3, C7, D2	Groups B3, C3, C7, D2	B3, 2 samples
e. Assembly location move	Notify qualifying activity	Notify qualifying activity	As required
f. Laser marking qualification	Groups C2, and, test methods 1021, and 1071 of MIL-STD-750	Groups C2, and, test methods 1021, and 1071, of MIL-STD-750	2 samples

1/ Acceptable supporting data may be submitted to reduce or eliminate required testing.
2/ When variable data is required for applicable groups A and C testing, data histograms providing acceptable parameter data summaries may be submitted in place of variables.
3/ If changes involve more than one device type from the same certified line, contact the qualifying activity to determine appropriate selection of device type(s) to be selected for testing.
4/ The qualifying activity may add or reduce testing if warranted by specification sheet requirements or unique design or process circumstances after notification of the manufacturer.
5/ Additional testing and evaluation in accordance with group E to establish confidence in the proposed change shall be performed as required by the qualifying activity (see E.6.6).
6/ New die design requires full qualification.
7/ For small die flow use group B, step 1 in lieu of B3 and C6 requirement; and use group A subgroup 1 (small die flow) in lieu of B1 and B2.
8/ Group D tests are only applicable for RHA devices.

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TABLE E-IV. Screening requirements

Screen	MIL-STD-750, method	Condition	JANS requirements	JANTXV requirements	JANTX requirements
1a. Die visual for glass diodes	2073	Condition B, die form prior to assembly	100-percent	Not applicable	Not applicable
1b. Internal visual (pre-cap) inspection			100-percent	100-percent	Not applicable
For diodes	2074 1/				
For power FETs	2069				
For microwave transistors	2070				
For transistors	2072				
2. High temperature life Nonoperating life (stabilization bake)	1032	T_{STG} ≤ maximum rated storage temperature t = as specified	Optional	Optional	Optional
3a. Temperature cycling	1051	20 cycles. No dwell time is required at +25°C. Test condition C or maximum storage temperature, whichever is less.	100-percent	100-percent	100-percent
3b. Surge (as specified) 2/	4066	Condition A or B, as specified	100-percent	100-percent	100-percent
3c. Thermal impedance (as specified) 2/		As specified	100-percent	100-percent	100-percent
Transistors,					
Power FETs	3161				
Bipolar	3131				
Diodes	3101				
IGBT	3103				
GaAs FET	3104				
4. Constant acceleration. Not required for stud devices and metallugically bonded diodes.	2006	Y_1 direction at 20,000 G minimum, except at 10,000 G minimum for devices with power rating of ≥10 watts at T_C = +25°C. The 1 minute hold time requirement shall not apply.	100-percent	Optional 3/	Optional 3/
5. PIND 4/	2052	Condition A	100-percent See E.5.4.1	Not applicable	Not applicable

See footnotes at end of table.

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TABLE E-IV. Screening requirements - Continued.

Screen	MIL-STD-750, method	Condition	JANS requirements	JANTXV requirements	JANTX requirements
6. Instability shock test (axial lead diodes only) 5/ a. FIST b. BIST	 2081 2082		 100-percent 100-percent	Not applicable	Not applicable
7. Hermetic seal 6/ a. Fine b. Gross 7/ 8/	 1071		 Optional Optional	 Optional Optional	 Optional Optional
8. Serialization		See 3.10.9.	100-percent	Not applicable	Not applicable
9. Interim electrical parameters		As specified	100-percent (read and record)	For case mounted rectifiers as specified.	For case mounted rectifiers as specified.
10. High temperature reverse bias (HTRB) a. For transistors	 1039	 Test condition A. 80 percent (minimum) of rated V_{CB} (bipolar), V_{GS}(FET) or V_{DS}(FET), as applicable.	 100-percent	 100-percent	 100-percent
b. For power FETs	1042	Test condition B. 80 percent (minimum) of rated V_{GS}	100-percent	100-percent	100-percent
c. For diodes and rectifiers	1038	Test condition A. Diodes (not required for LEDs, Zeners, and case mounted rectifiers) 80 percent minimum of rated V_R or V_{RWM} when dc conditions are specified. 95 - 100 percent of V_{RWM}, when half sine condition is specified.	100-percent 9/	100-percent	100-percent

See footnotes at end of table.

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TABLE E-IV. Screening requirements - Continued.

Screen	MIL-STD-750, method	Condition	JANS requirements	JANTXV requirements	JANTX requirements
11. Interim electrical and delta parameter for PDA (see E.5.2)		As specified but including all delta parameters as a minimum. When HTRB is performed leakage current shall be measured on each device before any other specified parametric test is made.	100-percent (Measure all specified parameters. Measure leakage current within 16 hours after removal of applied voltage in HTRB. Record those parameters which have a delta limit.) (See screen 13 of table E-IV.)	100-percent (Measure all specified parameters. Measure leakage current within 24 hours after removal of applied voltage in HTRB. Record those Parameters which have a delta limit.) (See screen 13 of table E-IV.)	100-percent (Measure all specified parameters. Measure leakage current within 24 hours after removal of applied voltage in HTRB. Record those Parameters which have a delta limit.) (See screen 13 of table E-IV.)
12. Burn-in		As specified.	100-percent	100-percent	100-percent
a. For bipolar transistors	1039	Test condition B.	240 hours (minimum)	160 hours (minimum)	160 hours (minimum)
b. For power FETs	1042	Test condition A.	240 hours (minimum)	160 hours (minimum) 10/	160 hours (minimum) 10/
c. For diodes, Zeners, and rectifiers	1038	Test condition B.	240 hours (minimum)	96 hours (minimum)	96 hours (minimum)
For case mount rectifiers		Condition A (HTRB), JANTX and JANTXV only.	Not applicable	48 hours (minimum)	48 hours (minimum)
		Condition B , for JANS	240 hours (minimum)	Not applicable	Not applicable
d. For thyristors 11/	1040		240 hours (minimum)	96 hours (minimum)	96 hours (minimum)
e. For bipolar small die transistors	1039	Condition B	240 hours (minimum)	160 hours (minimum)	160 hours (minimum)
f. For bipolar power transistors	1039	Condition B	240 hours (minimum)	160 hours (minimum)	160 hours (minimum)

See footnotes at end of table.

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TABLE E-IV. Screening requirements - Continued.

Screen	MIL-STD-750, method	Condition	JANS requirements	JANTXV requirements	JANTX requirements
13. Final electrical test (see E.5.2 For PDA) 12/		As specified.	100-percent	100-percent	100-percent
a. Interim electrical and delta parameters for PDA			Group A, subgroup 2, interim electrical and delta parameters. (read and record)	Group A, subgroup 2. Read and record interim electrical and delta parameters (see E.5.3.2).	Group A, subgroup 2. Read and record interim electrical and delta parameters (see E.5.3.2).
b. Other electrical parameters 13/			Group A, subgroup 3	Not applicable	Not applicable
14. Hermetic seal 6/	1071		100-percent	100-percent	100-percent
a. Fine		Omit for double plug diodes.			
b. Gross 7/ 8/					
15. Radiography	2076		100-percent 14/ 15/	Not applicable	Not applicable
16. External visual examination	2071	To be performed after complete marking and prior to lot acceptance	100-percent	Not applicable	Not applicable
17. Case Isolation 16/	1081	To be performed on all case isolated packages, as specified.	100-percent	100-percent	100-percent

See footnotes at end of table.

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TABLE E-IV. Screening requirements - Continued.

1/ Visual inspection (test method 2074 of MIL-STD-750) on clear glass diodes shall be performed any time prior to marking. Diodes assembled in packages other than glass shall be inspected to test method 2072.
2/ Shall be performed any time before completion of screen 13. Surge shall precede thermal impedance. Surge and thermal impedance are applicable only when specified in the screening table of the specification sheet.
3/ Constant acceleration shall be performed on gold ball bond devices and gold wire for germanium.
4/ PIND is not applicable to any device with external and internal pressure contacts (die to electrical contacts), optical coupled isolators, and double plug diodes. PIND screening may be performed any time after screen 4 when imposed by contract or order (see E.5.4.1).
5/ Omit BIST and FIST tests for double plug or case-mounted diodes. Omit FIST test for temperature compensated referenced diodes.
6/ Non-transparent glass encased double plug, noncavity axial lead diodes only may use test method 2068 of MIL-STD-750 in lieu of test method 1071 of MIL-STD-750.
7/ Conditions C is prohibited for MIL-PRF-19500 product. Condition D, fluorocarbon tests, are prohibited as a standalone test, but may be used as a supplemental test as applicable after dry gas hermeticity tests. When Condition D is utilized as a supplement it must be followed by a bake at 150°C for one hour to remove residual fluorocarbon. If helium is used after fluorocarbon the failure criteria must be tightened by one order of magnitude. Glass diodes shall not be painted until after seal tests. For clear glass diodes, utilize condition E for gross leak seal test.
8/ When condition E, or method 2068 of MIL-STD-750, is the required gross leak seal test, then it may be performed anytime after temperature cycle (screen 3a).
9/ For JANS only, Zener diodes shall be subjected to high temperature reverse bias at 80 - 85 percent of nominal V_Z for $V_Z > 10$ V. Omit test for devices with $V_Z \leq 10$ V. For JANS case mounted rectifiers condition A is required.
10/ Optional accelerated HTRB for power FETs in accordance with test method 1042 of MIL-STD-750, condition A, shall be 48 hours minimum at T_A = +175°C minimum. Initial use of this option is contingent upon subsequent completion of a one time 1,000 hour qualification in accordance with test method 1042 condition A, at T_A = +175°C minimum, and as specified on group E of the individual specification sheet; to be submitted with the initial qualification report. Alternate flow options shall not be used to qualify this accelerated HTRB option.
11/ For JANTX and JANTXV levels, full wave-blocking test shall replace power burn-in for all thyristors.
12/ For JANTX and JANTXV levels, tests previously performed 100-percent (surge, thermal impedance) need not be repeated in screen 13. For JANS, read and record thermal impedance shall be performed in screen 13 or anytime after serialization. Thermal impedance is not part of the PDA.
13/ Ninety-six hours post burn-in measurement time not applicable.
14/ The radiographic screen for JANS may be performed in any sequence after screen 8. The radiographic screen for packages whose construction contains materials such as: tungsten, molybdenum, copper tungsten etc.; where the package material interferes with the radiographic view of the die, including, but not limited to: Die, die mounting areas, die attach region, and lid seals shall use real time radiography in lieu of film, or equivalent, with the approval of the qualifying activity. An additional alternate view, as described in MIL-STD-750, Test Method 2076, Real Time Radiography paragraph specifying additional views shall be added when an initial view produces a completely black image.
15/ Conformance inspection may be initiated immediately prior to screen 15.
16/ May be performed anytime after plating. Test conditions not specified in the test method shall be provided by the supplier unless specified in the specification sheet.

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TABLE E-V. Group A inspection.

Subgroups	MIL-STD-750, method	JANS sample plan 1/	JAN, JANTX, JANTXV sample plan 1/
Subgroup 1 (all devices except small die flow) Visual and mechanical inspection (test method 2071 of MIL-STD-750)	 2071	15 devices c = 0	116 devices, c = 0 (JANTXV) 45 devices, c = 0 (JAN, JANTX)
Subgroup 1 (for small die flow only 2/ 3/)			
Visual and mechanical examination 4/	2071		116 devices, c = 0 (JANTXV) 45 devices, c = 0 (JAN, JANTX)
Solderability 4/	2026		15 leads, c = 0
Resistance to solvents 4/ 5/	1022		15 devices, c = 0
Salt atmosphere (corrosion), (Laser marked devices only. Not required for non-corrosive base metals.)	1041		6 devices, c = 0
Temperature cycling (air to air) 4/	1051		Test condition C, or maximum storage temperature, whichever is less, 25 cycles. 22 devices, c = 0
Electrical measurements (group A, subgroup 2)			
Hermetic seal 6/ Fine leak Gross leak	1071		22 devices, c = 0
Bond strength 4/	2037		Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs 11 wires, c = 0
Decap internal visual (design verification)	2075		4 devices, c = 0

See footnotes at end of table.

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TABLE E-V. Group A inspection – Continued.

Subgroups	MIL-STD-750, method	JANS sample plan 1/	JAN, JANTX, JANTXV sample plan 1/
Subgroup 2 DC (static) test at +25°C ±3 degrees C		116 devices c = 0 7/ 9/	116 devices 7/ c = 0
Subgroup 3 DC (static) tests at high (-0C, +10C) and low (+0C, -10C) specified temperatures.			116 devices 7/ 8/ c = 0
Subgroup 4 Dynamic tests at +25°C ±3 degrees C			116 devices 7/ 8/ c = 0
Subgroup 5 Safe operating area test (for transistors only): a. DC b. Clamped inductive (only when applicable) c. Unclamped inductive (only when applicable) End-point electrical measurements		45 devices c = 0 10/	45 devices c = 0
Subgroup 6 Surge current (for diodes/rectifiers only) End-point electrical measurements			22 devices c = 0
Subgroup 7 Selected static and dynamic tests			22 devices c = 0

See footnotes at end of table.

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TABLE E-V. Group A inspection – Continued.

1/ The specific parameters to be included for tests in each subgroup shall be as specified in the applicable specification sheet. Where no parameters have been specified in a particular subgroup or test within a subgroup, no group A testing is required for that subgroup or test to satisfy group A requirements. A single sample may be used for all subgroup testing. These tests are considered nondestructive and devices may be shipped.
2/ For resubmission of failed table E-V, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table E-V, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Not required for JANS devices.
4/ Separate samples may be used.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temperature cycling in addition to electrical measurements.
7/ If a device in the sample fails one or more test(s) in the subgroup(s) being sampled, each device in the (sub)lot represented by the sample shall be screened for the test(s) for which the sample failed. An alternate test method to remove the failure mode may be used after an engineering evaluation is performed. After the rescreening, a second sample (first resubmittal) using double the large lot sample size shall be tested to the original failed parameter. If the second sample fails, the lot shall be rejected. When processing devices using the small die flow, (TABLE E-VIC), A4 may be performed once per wafer lot.
8/ For small lot sampling plan, n = 45.
9/ All devices required by the specified sample plan shall be subjected to subgroups 2, 3, and 4 combined.
10/ All devices required by the specified sample plan shall be randomly selected from the devices subjected to subgroups 2, 3, and 4, and shall be subjected to subgroups 5, 6, and 7 combined.

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TABLE E-VIA. Group B inspections for JANS devices.

Inspections	MIL-STD-750, method	MIL-STD-750, condition	Qualification and large lot conformance inspection sample plan	Small lot conformance inspection
Subgroup 1 1/				
Physical dimensions	2066	Dimensions in accordance with case outline specified in specification sheets. Unless otherwise specified, lead dimensions are pre-solder dip or pre-secondary finish.	22 devices, c = 0	8 devices c = 0
Subgroup 2 1/		Separate samples may be used for each test.		
Solderability	2026	The sample plan applies to the number of leads inspected. A minimum of three devices shall be tested.	15 leads, c = 0	6 leads c = 0
Resistance to solvents	1022	Ink marked devices only. Not required if marking is etched into the device.	15 devices, c = 0	6 devices c = 0
Salt atmosphere (corrosion)	1041	Laser marked devices only Not required for non-corrosive base metals.	6 devices, c = 0	6 devices, c = 0
Subgroup 3				
Thermal shock (liquid-to-liquid)	1056	25 cycles, condition B (glass diodes only).	22 devices, c = 0	6 devices c = 0
Temperature cycling (air-to-air)	1051	Test condition C, or maximum storage temperature, whichever is less. (100 cycles).		
Surge	4066	As specified.		
Hermetic seal 2/	1071			
a. Fine		Fine leak not required for double plug diodes.		
b. Gross				
Electrical measurements		Group A, subgroup 2.		
Decap-internal visual (design verification) 3/	2075	Visual criteria in accordance with qualified design and internal visual precap criteria.	6 devices, c = 0	6 devices c = 0

See footnotes at end of table.

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TABLE E-VIA. Group B inspections for JANS devices - Continued.

Inspections	MIL-STD-750, method	MIL-STD-750, condition	Qualification and large lot conformance inspection sample plan	Small lot conformance inspection
Subgroup 3 - Continued				
Bond strength (wire or clip bonded devices only)	2037	Condition D.	22 wires or 11 devices, c = 0, (whichever requires the smaller number of devices.)	12 wires or 6 devices c=0 (whichever requires the smaller number of devices.)
SEM (for applicable designs) 4/	2077	See test method for sample size information.		
Die shear (excluding axial leaded devices)	2017		The same number of devices used for bond strength will also be used for die shear (minimum of six die).	
Subgroup 4				
Intermittent operation life	1037 or 1042	2,000 cycles, Devices with .008 inch or larger bond wires, 6,000 cycles. (Condition D is required when using TM1042)	22 devices, c = 0	12 devices c = 0
Hermetic seal 2/	1071			
a. Fine		Fine leak not required for double plug diodes.		
b. Gross				
Electrical measurements		Group A, subgroup 2 and as specified.		
Bond strength (wire or clip bonded devices only) 5/	2037	Condition D. The sample shall include a minimum of three devices and shall include all wire sizes.	11 wires, c = 0	11 wires, c = 0
Subgroup 5				
Accelerated steady-state operation life	1027	Bias conditions as specified. T_J = +275°C minimum (for 96 hours minimum) or T_J = +225°C minimum (for 216 hours minimum) or T_J = rated °C minimum (for 1,000 hours minimum).	22 devices, c = 0	12 devices c = 0
Electrical measurements		Group A, subgroup 2 and 3. 6/		

See footnotes at end of table.

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TABLE E-VIA. Group B inspections for JANS devices - Continued.

Inspections	MIL-STD-750, method	MIL-STD-750, condition	Qualification and large lot conformance inspection sample plan	Small lot conformance inspection
Subgroup 5 - Continued				
Schottky diodes, Case mount rectifiers	1038	T_J = rated T_J maximum (for 1,000 hours minimum).		
Electrical measurements		Group A, subgroup 2 and 3. 6/		
Accelerated steady-state gate stress power MOSFETS	1042	Condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours or T_A = 150°C, t = 48 hours.		
Electrical measurements		Group A, subgroup 2 and 3. 6/		
Accelerated steady-state reverse bias power MOSFETS	1042	Condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours or T_A = 150°C, t = 240 hours and as specified.		
Electrical measurements		Group A, subgroup 2 and 3. 6/		
Bond strength (wire or clip bonded devices only)	2037	As specified. Bond strength samples shall have passed accelerated steady-state operation life.	20 wires, c = 0	20 wires c = 0
Subgroup 6 7/				
Thermal resistance		As specified. Thermal resistance may be performed on a group C frequency whenever 100 percent thermal impedance is performed, except for power and case mounted devices.	22 devices, c = 0	8 devices c = 0
Diodes	3101 or 4081			
Transistors (bipolar)	3131			
Transistors (POWER FETs)	3161			
Thyristors	3181			
IGBT	3103			
GaAs FET	3104			

See footnotes at end of table.

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TABLE E-VIA. Group B inspections for JANS devices - Continued.

Inspections	MIL-STD-750, method	MIL-STD-750, condition	Qualification and large lot conformance inspection sample plan	Small lot conformance inspection
Subgroup 7 8/ High-temperature life (nonoperating) Electrical measurements	1032	340 hours minimum, $T_{STG(max)} = T_A$ Group A, subgroup 2	32 devices, c = 0	12 devices c = 0

1/ Electrical reject devices, from the same inspection lot, may be used for all subgroups, when electrical end-point measurements are not required. Other non-catastrophic rejected devices (i.e., PIND, X-ray) may be utilized for all subgroups. For subgroups with end-point measurements, the devices shall be screened to table E-IV through screen 13.
2/ Non-transparent glass encased double plug noncavity diodes only may use test method 2068 in lieu of 1071 of MIL-STD-750. This test may be performed after electrical measurements.
3/ Verification of metallurgical bond as defined in appendix A in its entirety shall be documented. (Photos are required with a scale or magnification identifier).
4/ This test may be performed at any time prior to lot formation. When performing SEM as part of a qualification or requalification, it shall include a cross section analysis. See E.3.1.2.1 for applicable designs.
5/ If sample is continued to satisfy the C6 requirement then bond strength may be performed after C6.
6/ Ninety-six hours post burn-in measurement time not applicable for A3.
7/ Thermal resistance may be performed on a group C frequency whenever 100-percent thermal impedance is performed, except for power and case mounted devices.
8/ Not required for power MOSFETs.

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TABLE E-VIB. Group B inspections for JAN, JANTX, and JANTXV devices.

Inspections 1/	MIL-STD-750		Sample plan	Small lot conformance inspection
	Method	Condition		
Subgroup 1 2/		Separate samples may be used for each test		
Solderability	2026	The sample plan applies to the number of leads inspected. A minimum of 3 devices shall be tested.	15 leads c = 0	4 leads c = 0
Resistance to solvents	1022	Not required if marking is etched into the device.	15 devices c = 0	3 devices c = 0
Salt atmosphere (corrosion)	1041	Laser marked devices only	6 devices, c = 0	6 devices, c = 0
Subgroup 2			22 devices c = 0	6 devices c = 0
Thermal shock (liquid-to-liquid)	1056	10 cycles, condition B, (glass diodes only).		
Temperature cycling (air-to-air)	1051	Test condition C, or maximum storage temperature, whichever is less. (45 cycles including screening)		
Surge	4066	As specified.		
Hermetic seal 3/	1071			
a. Fine leak		Fine leak not required for double plug diode.		
b. Gross leak				
Electrical measurements 4/		Group A, subgroup 2		
Subgroup 3 5/			45 devices c = 0	12 devices c = 0
Steady-state operation life 6/	1027	Bias conditions as specified, 340 hours (minimum)		
Electrical measurements		Group A, subgroup 2		
or				
Intermittent operation life 7/	1037	2,000 cycles (minimum)		
	1042	Condition D, 2,000 cycles (minimum)		
Hermetic seal 3/	1071			
a. Fine		Fine leak not required for double plug diodes.		
b. Gross				
Electrical measurements		Group A, subgroup 2		
Bond strength (wire or clip bonded devices only)	2037	Condition D. The sample shall include a minimum of three devices and shall include all wire sizes.	11 wires c = 0	11 wires c = 0

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TABLE E-VIB. Group B inspections for JAN, JANTX, and JANTXV devices – Continued.

Inspections 1/	MIL-STD-750		Sample plan	Small lot conformance inspection
	Method	Condition		
Subgroup 4 Decap internal visual (design verification)	2075	Visual criteria in accordance with qualified design.	1 device c = 0	1 device c = 0
Subgroup 5 Thermal resistance Diodes Transistors (bipolar) Transistors (power FETs) Thyristors IGBT GaAs FET	 4081 3131 3161 3181 3103 3104	 As specified. Thermal resistance maybe performed on group C frequency whenever 100-percent thermal impedance is performed except for power and case mounted devices.	15 devices c = 0	6 devices c = 0
Subgroup 6 8/ High-temperature life (non-operating) Electrical measurements	1032	340 hours minimum, $T_{STG(max)} = T_A$ Group A, subgroup 2	32 devices c = 0	12 devices c = 0

1/ An engineering evaluation shall be performed if there is a device failure. Corrective action shall be taken as necessary.
2/ Electrical reject devices from the same inspection lot, may be used for all subgroups when electrical end-point measurements are not required. Other non-catastrophic rejected devices (i.e., PIND, X-ray) may be utilized for all subgroups. For subgroups with end-point measurements, the devices shall be screened to table E-IV through screen 13.
3/ Non-transparent glass encased double plug non-cavity diodes only may use test method 2068 in lieu of 1071 of MIL-STD-750. This test may be performed after electrical measurements.
4/ Unless otherwise specified, omit delta parameters limits for low current gain (h_{fe}) and leakage measurements included in end-point measurements.
5/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340-hour or 2,000 cycle life tests may be continued on test to 1,000 hours or 6,000 cycles, as applicable, in order to satisfy the group C life test requirements; and bond pull may be performed after group C life test. End-point measurements shall be performed at either group B, subgroup 3 (340 hours or 2,000 cycles, as applicable) to satisfy group B lot acceptance or group C, subgroup 6 (1,000 hours or 6,000 cycles, as applicable) to satisfy group B and C lot acceptance. Bond strength may be performed after C6.
6/ T_J = 150°C (min) or rated T_J, whichever is less (except Schottky and power MOSFETs) for operation life.
7/ Intermittent operation life shall be performed on all case mounted devices.
8/ Not required for power MOSFETs.

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TABLE E-VIC. Group B inspections (small die flow only) for JAN, JANTX, and JANTXV devices.

Inspections 1/ 2/	MIL-STD-750		Sample plan
	Method	Condition	
Step 1			n = 45, c = 0
Steady-state operation life 3/	1026	1,000 hours minimum 4/ 5/	
Electrical measurements		Group A, subgroup 2	
or			
intermittent operation life	1037	Intermittent life 6,000 cycles	
Hermetic seal 6/	1071		
a. Fine		Fine leak not required for double plug diodes.	
b. Gross			
Electrical measurements		Group A, subgroup 2	
Step 2			n = 45, c = 0
HTRB	1048	48 hours minimum.	
Electrical measurements		Group A, subgroup 2	
Step 3			n = 22, c = 0
High-temperature life (non-operating),	1032	t = 340 hours, T_A = +200°C.	
Electrical measurements		Group A, subgroup 2	

1/ For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot.
2/ Small die shall be device types identified by the manufacturer based on the manufacturer's internal process capabilities and the expected business model for the product. Classification of devices as small die shall be approved by the procuring activity. Once a device type is approved a small die, it shall be processed in accordance with the conformance inspection small die flow identified herein and shall not be changed without qualifying activity approval.
3/ Test method 1026 of MIL-STD-750 is required for eutectic die attach and test method 1037 of MIL-STD-750 is required for solder die attach.
4/ T_J = 150 degrees C (min) or rated T_J, whichever is less (except Schottky and power MOSFETs) for operation life.
5/ The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
6/ Non-transparent glass encased double plug noncavity diodes only may use test method 2068 of MIL-STD-750, in lieu of 1071.

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TABLE E-VII. Group C periodic inspections (all quality levels) .

Inspections	MIL-STD-750		Sample plan	Small lot conformance inspection
	Method	Condition		
Subgroup 1			15 devices c = 0	6 devices c = 0
Physical dimensions 1/ (not required for JANS)	2066	Dimensions in accordance with case outline specified in specification sheets. Unless otherwise specified, lead dimensions are pre-solder dip or pre-secondary finish.		
Subgroup 2			22 devices c = 0	6 devices c = 0
Thermal shock (liquid to liquid)	1056	25 cycles, condition B.		
Temperature cycling (air-to-air)	1051	Test condition C, or maximum storage temperature, whichever is less. (45 cycles including screening).		
Terminal strength	2036	As specified.		
Surface mount end cap bond integrity	2038	As specified (Condition B for US devices)		
Hermetic seal 2/	1071			
a. Fine leak		Fine leak not required for double plug diodes.		
b. Gross leak				
Moisture resistance	1021	Omit initial conditioning.		
Electrical measurements		Group A, subgroup 2.		
Subgroup 3		Not required for disc packages or metallurgically bonded double plug devices or stud packaged devices.	22 devices c = 0	6 devices c = 0
Shock	2016	Nonoperating, 1,500 G's, 0.5 ms, 5 blows in each orientation, X1, Y1, and Z1 (Y1 only for axial glass diodes.)		
Vibration, variable frequency	2056			
Constant acceleration 3/	2006	1 minute minimum in each orientation. X1, Y1, and Z1 at 20,000 G's minimum, except at 10,000 G's minimum for devices with power rating of ≥ 10 watts. T_C = +25°C.		
Electrical measurements		Group A, subgroup 2.		

See footnotes at end of table.

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TABLE E-VII. Group C periodic inspections (all quality levels) – Continued.

Inspections	MIL-STD-750		Sample plan	Small lot conformance inspection
	Method	Condition		
Subgroup 4 Salt atmosphere (corrosion) 1/	 1041		15 devices c = 0	6 devices c = 0
Subgroup 5 Thermal resistance 4/ Diodes Transistors (bipolar) Transistors (power FETs) Thyristors IGBT GaAs FET	 4081 3131 3161 3181 3103 3104	 As specified.	15 devices c = 0	6 devices c = 0
Subgroup 6 5/ 6/ Steady-state operation life Electrical measurements or Intermittent operation life Hermetic seal 2/ a. Fine b. Gross Electrical measurements Bond strength 9/ or Blocking life 8/ Electrical measurements	 1026 1037 1042 1071 2037 1048 	Not required for disc packages. 1,000 hours minimum, bias conditions as specified. 7/ 8/ Group A, subgroup 2. 6,000 cycles minimum. Condition D, 6,000 cycles minimum. Fine leak not required for double plug diodes. Group A, subgroup 2. Condition D, .008 inch or larger wire or clip bonded devices only. The sample shall include a minimum of three devices and shall include all wire sizes. Group A, subgroup 2.	22 devices c = 0 11 wires, c = 0	12 devices c = 0 11 wires, c = 0
Subgroup 7 10/ Internal gas analysis	 1018	 To be performed on each structurally identical package family.	3 devices c = 0	3 devices c = 0

See footnotes at end of table.

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TABLE E-VII. Group C periodic inspections (all quality levels) – Continued.

1/ Electrical reject devices, from the same inspection lot, may be used for all subgroups when electrical end-point measurements are not required. Other non-catastrophic rejected devices (i.e., PIND, X-ray) may be utilized for all subgroups. For subgroups with end-point measurements, the devices shall be screened to table E-IV through screen 13. Salt atmosphere not required for Laser marked devices when devices from group B, or from group A for small die flow, salt atmosphere have been selected to satisfy group C inspection requirements.

2/ Non-transparent glass encased double plug noncavity diodes only may use test method 2068 of MIL-STD-750, in lieu of 1071. This test may be performed after electrical measurements.

3/ Not applicable to any devices with external and internal pressure contacts (die to electrical contacts), optical coupled isolators, and double plug diodes.

4/ Not required when performed in group B.

5/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, then the life test samples may be continued on test to 1,000 hours or 6,000 cycles, as applicable, in order to satisfy the group C life test requirements. End-point measurements shall be performed on either table E-VIA, group B, subgroup 4, or table E-VIB group B, subgroup 3 (340 hours or 2,000 cycles, as applicable) to satisfy group B (table E-VIA or table E-VIB) lot acceptance or group C, subgroup 6 (1,000 hours or 6,000 cycles, as applicable) to satisfy group B and C lot acceptance.

6/ Intermittent operation life shall be performed on all case mounted devices.

7/ T_J = 150°C (min) or rated T_J whichever is less (except schottky and power mosfets) for operation life.

8/ The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 22,000 device hours minimum, and the actual time of test is at least 340 hours.

9/ Required for JANS devices with .008 inch or larger bond wires only. Not required when JANS devices from group B bond pull inspection have been selected to satisfy group C inspection requirements.

10/ Internal gas analysis shall be performed on hermetic devices. An engineering evaluation shall be performed if there is a device failure to determine the moisture source (e.g. sealing environment, non hermetic device). The entire lot shall be rescreened in accordance with screen 14 herein (and resubmitted at 6/0.) Corrective action shall be taken as necessary. Subgroup 7 is not required for noncavity double plug devices.

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TABLE E-VIII. Group D inspection (RHA inspections). 1/

Test	MIL-STD-750		JANS		JANTXV	
	Method	Condition	Quantity/ (accept number)	Notes	Quantity (accept number)	Notes
Subgroup 1 2/						
Neutron irradiation	1017	+25°C				
Qualification and conformance inspection			11(0)	3/	(a) 11(0)	4/
End-point electrical parameters		As specified in accordance with specification sheet				
Subgroup 2 5/						
Steady-state total dose irradiation	1019	+25°C				
Qualification and conformance inspection			4(0) 2(0) 1(0)	6/ 8/ 9/	11(0)	7/
End-point electrical parameters		As specified in accordance with specification sheet				
Subgroup 3 10/						
Power transistor electrical dose rate test	3478	+25°C	11(0)	3/	11(0)	4/
End-point electrical parameters		As specified in accordance with specification sheet				

See footnotes at end of table.

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TABLE E-VIII. Group D inspection (RHA inspections) – Continued. 1/

1/ Parts used for one subgroup test may not be used for other subgroups but may be used for higher levels in the same subgroup. Unless testing is performed within the time limits of the test method, total exposure shall not be considered cumulative. Group D tests may be performed prior to device screening (see E.6.5).
2/ Unless by design, waive neutron tests for MOS devices, bipolar elements are an integral part of the device function.
3/ In accordance with wafer lot. If one part fails, seven additional parts may be added to the test sample with no additional failures allowed, 18 devices, c = 1.
4/ In accordance with inspection lot. If one part fails, seven additional parts may be added to the test sample with no additional failures allowed, 18 devices, c = 1.
5/ JANTXV devices shall be inspected using either the JANTXV quantity/accept number criteria as specified, or by using the JANS criteria on each wafer.
6/ For device types with greater than or equal to 4,000 die per wafer, selected from a random locations on the wafers.
7/ In accordance with inspection lot. If one part fails, 16 additional parts may be added to the test sample with no additional failures allowed, 27 devices, c = 1. For devices which require more than one bias, the sample size shall be 11(0) for each bias.
8/ For device types with greater than 500 and less than 4,000 die per wafer, selected from random locations on each wafer.
9/ For device types with less than or equal to 500 die per wafer, selected from random locations on each wafer.
10/ Upset testing during qualification on first conformance inspection shall be conducted when specified in contract or order when specified, the same devices may be tested in more than one subgroup.

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TABLE E-IX. Group E inspections (all quality levels).

Inspections	MIL-STD-750		Sample plan
	Method	Condition	
Subgroup 1			45 devices, c = 0 or as specified.
Thermal shock or	1056	100 cycles or as specified.	
Temperature cycling	1051	500 cycles minimum or as specified. Test condition C or max storage temp, whichever is less.	
Hermetic seal 3/	1071	As applicable.	
a. Fine leak		Fine leak not required for double plug diodes.	
b. Gross leak			
Electrical measurements		Group A, subgroup 2.	
Subgroup 2			45 devices, c = 0 or as specified.
Intermittent operating life	1037	As specified.	
Electrical measurements		Group A, subgroup 2.	
or			
Life test	1042	Condition A, B, C, or D.	
Electrical measurements		Group A, subgroup 2.	
or			
Steady-state operating life	1026	As specified.	
Electrical measurements		Group A, subgroup 2.	
or			
Blocking life	1048	As specified.	
Electrical measurements		Group A, subgroup 2.	
Subgroup 3 Not applicable			
Subgroup 4			
Thermal impedance curves (as applicable)	N/A	Each supplier shall submit a thermal impedance ($Z_{\theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal impedance. Thermal impedance curves (from $Z_{\theta JX}$ test pulse time to $R_{\theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal impedance screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\theta JX}$ shall also be submitted. The proposed specification maximum thermal impedance curve shall be submitted. The qualifying activity may approve a different $Z_{\theta JX}$ limit not to exceed the specification's thermal curve for conformance inspection end-point measurements as applicable. The supplier shall support (with applicable data) their $Z_{\theta JX}$ end-point limit proposal when it exceeds the screening $Z_{\theta JX}$ limit. A delta (read and record) $Z_{\theta JX}$ shall be determined by the manufacturer and approved by the qualifying activity for all case mounted devices for conformance inspection (intermittent life test and temperature cycling) end-point measurements. Any exceptions shall be justified to, and approved by, the qualifying activity. Equivalent data, procedures, or SPC plans may be used for part, or all, of the above requirements.	N/A

See footnotes at end of table.

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TABLE E-IX. Group E inspections (all quality levels) - Continued.

Inspections	MIL-STD-750		Sample plan
	Method	Condition	
Subgroup 4 - continued		The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening, and group A subgroup 2. The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.	
Subgroup 5 Barometric pressure (reduced) (required only on all devices with rated voltage > 200 V)	1001	As specified.	3 devices, c = 0 or as specified.
Subgroup 6 ESD	1020	As required by E.4.2.1.	11 devices.
Subgroup 7 Resistance to soldering heat 1/ Visual inspection Hermetic seal 3/ a. Fine leak b. Gross leak Electrical measurements	2031 1071	See test method 2031 of MIL-STD-750 and H.6 for package family and test conditions. 2/ As applicable. Fine leak not required for double plug diodes. Group A, subgroup 2.	3 devices, c = 0 or as specified.
Subgroup 8 Reverse stability (for bipolar transistors only)	1033	When specified. Condition A for devices > 400 V. Condition B for devices < 400 V.	45 devices, c = 0 or as specified.
Subgroup 9 Resistance to glass cracking (glass diodes only)	1057	Condition B. Step stress to destruction by increasing cycles or up to a maximum of 25 cycles. The results shall be available upon request.	45 devices.

1/ As an option, the manufacturer may submit data (alternate testing) to the qualifying activity for approval in lieu of performing specific soldering heat test conditions.
2/ After subjection to the test, failure of one or more specified end-point measurements or examinations, evidence of defects or damage to the case, leads, or seals shall be considered a failure. Damage to the marking caused by fixturing or handling during tests shall not be cause for device rejection.
3/ Non-transparent glass encased double plug noncavity diodes only may use test method 2068 of MIL-STD-750, in lieu of 1071. This test may be performed after electrical measurements.

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TABLE E-X. Minimum data requirements for JANS devices.1/, 2/

JANS data requirements
1. Wafer lot inspection results (see D.3.9.4.2).
2. SEM Inspection results and photographs(when applicable, see E.3.1.2.1).
3. Screening data: a. Travelers [Including quantities and dates. Test times (when applicable). Assembly travelers for screening steps that occur in assembly]. b. All recorded electrical data (see E.3.8). c. Summary of parts fallout (see E.3.8.1). d. Radiograph data (when required and if requested).
4. Conformance inspection Data [Groups A, B, C. Group D (when applicable)]: a. Travelers [Including quantities and dates. Test times (when applicable)]. b. All specified electrical data.

1/ These data requirements do not require the manufacturer to provide the required data unless specified in the order. The customer may specify additional data requirements in their order.
2/ Any group B inspection used to satisfy the JANS group C inspection requirements shall be included in the data requirements.

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MIL-PRF-19500M, appendix F, dated 22 October 1999, is hereby canceled without replacement.

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DISCRETE SEMICONDUCTOR DIE/CHIP LOT ACCEPTANCE

G.1 SCOPE

G.1.1 Scope. The purpose of this appendix is to establish minimum standards for screening and qualification of JANHC and JANKC unencapsulated discrete semiconductor devices (die/chips) for use in semiconductor devices. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

G.2 APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

G.3 DEFINITIONS

G.3.1 Date code. Seal week of the element evaluation (EE) packaged samples.

G.3.2 Die/chips. Unencapsulated discrete semiconductors. The term chip is interchangeable with the term die.

G.3.3 Manufacturer. Original wafer lot fabricator.

G.3.4 Identification. See 1.3 for identification.

G.3.5 Wafer lot. A wafer lot shall consist of only semiconductor wafers subjected to each and every process step as a group. Each wafer lot shall be assigned a unique identifier which provides traceability to all wafer processing steps. Rework provisions shall be in accordance with D.3.13.2. Wafer lot records shall identify all JANHC and JANKC device inspection lots formed from the wafer lot.

G.3.6 Inspection lot. An inspection lot shall consist of one device type from a single wafer lot.

G.4 REQUIREMENTS

G.4.1 General. Semiconductor die shall conform to the requirements contained herein. Additionally, die in packaged devices from a wafer (lot) that has been subjected to and passed the JANTXV screening, group A, and group B requirements, shall be considered as meeting the JANHC requirements of Table G-II, Subgroups 3 and 4. Die in packaged devices from a wafer (lot) that has been subjected to and passed the JANS screening, group A and JANS group B requirements, shall be considered as meeting the JANKC requirements of Table G-II, Subgroups 3 through 6. This allowance is only applicable if the die in question are identical (topography, metalization, etc) to the packaged die. The other Subgroups listed in Table G-II shall be performed as required. The screening and CI data with traceability to the wafer (lot) shall be provided as evidence of JANHC/JANKC element evaluation when procured by the customer.

G.4.2 Steady-state life and screening. Steady-state life time and temperatures shall be in accordance with table G-I. Semiconductor die shall be screened in accordance with table G-II, subgroups 1 and 2 (see G.5.2.1 and G.5.2.2).

G.4.3 Qualification. Qualification shall be based on the results of the first EE performed and submitted to DLA Land and Maritime.

G.4.3.1 Wafer fabrication. Qualification shall be performed by the basic plant or original wafer lot manufacturer from die manufactured in the same wafer fab that was used to qualify a PIN on the QML of the same function and technology. Qualification for JANKC can only be approved on die processed in a wafer fab that has been used to qualify a JANS part on the QML of the same technology and function. For the purpose of this specification, examples of function are: Signal transistor, fast recovery epitaxial diode (FRED), power transistor, Zener diode, rectifier or transient suppressor. The term technology may include: Diffusion metal oxide semiconductor (DMOS), V-groove metal oxide semiconductor (VMOS), diffused junction, alloy junction, junction field effect transistor (JFET) and Schottky.

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G.4.3.2 Facility. JANHC qualification shall only be granted to a manufacturer who has a MIL-PRF-19500 certified facility(basic plant). JANKC qualification will only be granted to a manufacturer who is MIL-PRF-19500 certified to manufacture JANS products.

G.4.4 Performance characteristics. The electrical performance characteristics of semiconductor die shall be as specified in the specification sheet.

G.4.5 Critical interfaces. The critical interfaces and physical dimensions of the semiconductor die shall be in accordance with the requirements of MIL-PRF-19500 and with G.4.5.1 through G.4.5.3. A completed DLA Land and Maritime Form 36D and die topography, including dimensions, pad locations, and metallization descriptions (die map) shall be made available for inspection to the qualifying activity prior to qualification. A unique critical interface identifier as part of the PIN shall be assigned based on any of the following differences:

a. Bond pad metal.

b. Backside metal.

G.4.5.1 Bonding pad. The bonding pad size, location, and electrical function shall be in accordance with the applicable specification sheet. Unless approved by the qualifying activity, the minimum bond pad dimensions shall be 3 mils.

G.4.5.1.1 Metallization integrity. The bonding pads shall be metallized and suitable for bonding as specified in the applicable specification sheet and shall meet the requirements of G.5.2.5.1.

G.4.5.2 Backing material. The backing material shall be as described in the specification sheet and meet the requirements of G.5.2.5.2.

G.4.5.3 Glassivation. Glassivation requirements of H.3.7 apply.

G.5 VERIFICATION PROVISIONS

G.5.1 General. EE of semiconductor die shall be performed at a facility with MIL-STD-750 laboratory suitability for the applicable test methods.

G.5.1.1 Responsibility for inspection. The supplier shall be responsible for the performance of all inspection requirements as specified herein and in the specification sheet.

G.5.1.2 Retention of records. The supplier shall maintain adequate records of all examinations, inspections, and tests performed in accordance with the requirements specified herein and the specification sheet. Records, including variables data, shall be retained in accordance with appendix D.

G.5.1.3 JANKC wafer lot inspection. For JANKC, the process monitors of D.3.9.4 shall apply.

G.5.1.4 Sequence of testing. Subgroups within a group of tests (see table G-II) may be performed in any sequence, but individual tests within a subgroup shall be performed in the sequence indicated.

G.5.1.5 Sample selection. Samples shall be randomly drawn from inspection lots. The sample size columns in the evaluation tables give minimum quantities to be evaluated with applicable accept number enclosed in parenthesis.

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G.5.1.6 Wafer traceability. For JANKC, wafer traceability shall be maintained on the inspection lot and the element evaluation.

G.5.2 Element evaluation. Die from each wafer lot shall be evaluated in accordance with table G-II.

G.5.2.1 Subgroup 1, 100-percent electrical test of die. Each die shall be electrically tested, which may be done at the wafer level provided all failures are identified and removed from the lot when the die are separated from the wafer. Test limits and conditions shall be chosen by the supplier to assure compliance with all the electrical characteristics specified by the specification sheet. This allows the supplier to assign test values or test details which differ from the specification sheet requirements.

G.5.2.2 Subgroup 2, 100-percent visual inspection of die. Each die shall be visually inspected to assure conformance with the die related requirements, test methods 2069, 2070, 2072, or 2073 of MIL-STD-750 as applicable. Qualified die may be stored at the manufacture's facility prior to 100-percent visual.

G.5.2.3 Subgroup 3.

G.5.2.3.1 Internal/die visual inspection. The die visual sample shall be randomly selected from die that have successfully completed subgroup 2 of table G-II.

G.5.2.3.2 Test sample preparation. Test samples shall be assembled in suitable packages using standard assembly procedures.

G.5.2.3.3 Packaged sample identification. The packaged sample shall be marked or labeled in such a manner to identify the following:

a. Serial numbers if required.

b. Device PIN.

c. Inspection lot number or date code.

G.5.2.3.4 Internal visual. Each sample may be visually inspected after assembly and prior to encapsulation to assure conformance with the applicable requirements, test methods 2069, 2070, 2072, or 2073 of MIL-STD-750.

G.5.2.4 Subgroup 4.

G.5.2.4.1 Subgroup 4. Each sample shall be processed in accordance with subgroup 4 of table G-II.

G.5.2.4.2 Class HC sample size. The JANHC sample will consist of at least ten die from each inspection lot.

G.5.2.4.3 Class KC sample size. The JANKC sample shall require three die from each wafer and a minimum of twenty two die from each inspection lot.

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G.5.2.5 Subgroups 5A and 5B.

G.5.2.5.1 Bond pull (5A). From each wafer lot, a sample of at least five die requiring ten bond wires minimum shall be selected.

G.5.2.5.1.1 Wire bond strength testing. Bond strength shall be performed in accordance with test method 2037 of MIL-STD-750, test condition D.

The die metallization shall be acceptable if no failure occurs. If only one wire bond fails, another sample may be selected in accordance with G.5.2.5.1 and subjected to subgroup 5A evaluation. If the second sample contains no failures, the bonding test results are acceptable and the lot of die is acceptable. If the second sample contains one or more failures, or if more than one failure occurs in the first sample, the lot of die shall be rejected.

G.5.2.5.2 Die shear (5B). Die shear shall be performed in accordance with test method 2017 of MIL-STD-750. If only one die fails, another sample may be selected and subjected to subgroup 5B evaluation. If the second sample contains no failures, the die shear test results are acceptable and the lot of dice is acceptable. If the second sample contains one or more failures, or if more than one failure occurs in the first sample, the lot of dice shall be rejected.

G.5.2.6 Subgroup 6, scanning electron microscope (SEM). To be performed on selected die designs in accordance with E.3.1.2.2. Sample selection and reject criteria shall be in accordance with, test method 2077 of MIL-STD-750.

G.5.2.7 Subgroup 7, radiation hardness assurance (RHA). RHA inspection shall be as specified in the specification sheet.

G.5.3 Sample acceptance. The lot is acceptable if it passes all the appropriate requirements of table G-II. If the test sample fails the criteria of the appropriate flow, the inspection lot shall be rejected. For JANKC lots represented by five wafers or more, any wafer whose assembled sample fails any of the requirements of subgroup 4, table G-II may be removed from the lot with no jeopardy to the rest of the lot if failure analysis determines that the failure mechanism is inherent to the removed wafers only. The inspection lot is rejected if more than 20 percent of the wafers had failures in subgroup 4 of table G-II If a failure is attributed to packaging or handling defects, Electrostatic discharge (ESD), equipment malfunction, or operator error, these samples shall be verified by failure analysis. Upon verification of such defects, the test sample may be replaced in accordance with appendix E.

G.5.3.1 Lots shipped prior to element evaluation completion. No lots shall be shipped prior to completion of the element evaluation without the approval of the qualifying activity. This provision is only to be requested for emergency procurement situations and it is expected that the user will provide a justification.

G.5.4 Storage. Die shall be stored in dry nitrogen or other inert atmosphere.

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G.6 PACKAGING

G.6.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

G.6.1.1 Packaging sensitivity. All semiconductor die shall be packaged in individually lidded containers. For ESD sensitive devices (classes 1 and 2), conductive or anti-static containers shall be required with an external conductive field shielding barrier. Stacking of containers without lids shall not be allowed. The supplier may submit an alternate procedure for packaging of die for approval by the using activity.

G.6.2 Container marking. The following information shall be marked on each container of semiconductor die:

a. PIN.

b. Applicable specification sheet number.

c. Manufacturer’s logo or designation symbol.

d. Lot identification code.

e. Quantity.

f. ESD symbol (if applicable).

g. Date code (see G.3.1).

h. Wafer identity for JANKC.

G.6.3 Certificate of conformance. The certificate of conformance shall be in accordance with 3.7

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TABLE G-I. Steady-state life time and temperature.

Option	Minimum time	Minimum junction temperature 1/ T_J
A	240 hours	175°C
B	500 hours	150°C
C	1,000 hours	125°C

1/ Example: If T_J of steady-state life test is 187.5°C ±12.5°C, then option A is used.

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TABLE G-II. Die element evaluation requirements.

Subgroup	Class		Test	MIL-STD-750		Quantity	Reference paragraph
	K	H		Method	Condition	(Accept no.)	
1	X	X	Electrical test			100-percent	G.5.2.1
2	X	X	Visual inspection	2069 2070 2072 2073		100-percent	G.5.2.2
3A	X	X	Internal/die visual inspection	2069 2070 2072 2073		10 (0) For class H 22 (0) For class K	G.5.2.3.1
3B	X	X	Sample assembly			10 pieces min for class H, 22 pieces min for class K	G.5.2.3.2 G.5.2.3.3
4						10 (0) For class H 22 (0) For class K	
	X	X	Temperature cycling	1051	C		
	X		Mechanical shock or	2016	Y1 axis direction		
			Constant acceleration	2006	Y1 axis direction		
	X	X	Electrical test (read/record)		Group A, subgroups 2, 3, 4		1/
	X	X	HTRB		Screen 10		2/
	X	X	Electrical test (read/record)		Group A, subgroup 2		1/ 3/
	X	X	Burn-in		Screen 12		2/
	X	X	Electrical test (read/record)		Group A, subgroup 2		1/ 3/
	X		Steady-state life				4/
			Transistors Power FETS Diodes/rectifiers	1039 1042 1038	B A A or B		

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TABLE G-II. Die element evaluation requirements – Continued.

Subgroup	Class		Test	MIL-STD-750		Quantity	Reference paragraph
	K	H		Method	Condition	(Accept no.)	
4 (Continued)	X		Electrical test (read/record)		Group A, subgroup 2, 3, 4		1/
5A	X	X	Wire bond evaluation	2037	As applicable	10 (0) wires or 20 (1) wires	G.5.2.5.1
5B	X	X	Die shear evaluation	2017		5 (0) or 10 (1)	G.5.2.5.2
6	X		SEM	2077	As applicable	See test method 2077	G.5.2.6 5/
7	X X		RHA Total dose Neutron irradiation	1019 1017		6/	G.5.2.7 5/

1/ Thermal impedance shall not apply.
2/ HTRB and burn-in shall be performed when specified on the applicable specification sheet.
3/ For JANHC only, if one device fails during any of the subgroup 4 tests following “electrical tests” (table E-V, group A, subgroups 2, 3, or 4), then 20 additional devices may be added to the element evaluation with no additional failures allowed, 30 devices, c = 1.
4/ Time and temperature requirements in accordance with table G-I.
5/ May be performed at any time.
6/ Sample size shall be in accordance with the specification sheet.

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CRITICAL INTERFACE AND MATERIALS

H.1 SCOPE

H.1.1 Scope. This appendix contains critical interface information which will assist manufacturers in producing devices which meet 3.2. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

H.2 APPLICABLE DOCUMENTS

H.2.1 General. The documents listed in this section are specified in sections H.3, H.4, or H.5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they shall meet all specified requirements of documents cited in sections H.3, H.4, or H.5 of this specification, whether or not they are listed.

H.2.2 Government documents.

H.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those~~,~~ cited in the solicitation or contract

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services.

(Copies of these documents are available online at http://quicksearch.dla.mil or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

H.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

H.3 CRITICAL INTERFACE AND CONSTRUCTION CRITERIA

H.3.1 Allowable alternate design, materials, and construction. See 3.2.2 herein.

H.3.2 Package. All packaged devices supplied under this specification shall be hermetically sealed unless otherwise specified in the applicable specification sheet. No organic or polymeric material shall be used as a package or package seal unless specifically allowed in the applicable specification sheet

H.3.3 Fungus-resistant material. External parts of the semiconductor device shall be inherently non-nutrient to fungus.

H.3.4 Metals. Internal surfaces shall be capable of resisting progressive degradation within a hermetically sealed package. External metal surfaces shall be corrosion resistant or shall be plated or treated to resist corrosion. Device package material shall be free of burrs and other potential particle contamination.

H.3.5 Screw threads. Standard screw threads listed in FED-STD-H28 shall be required for all semiconductor devices where screw threads are a mechanical requirement of the device.

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H.3.6 Internal conductors. Internal conductors which are in thermal contact with a substrate along their entire lengths (such as metallization strips, contact areas, and bonding interfaces) shall be designed so that no properly fabricated conductor shall experience, at device maximum rated current, a current density in excess of the values shown below for the applicable conductor material including allowances for worst case conductor composition, cross-sectional area, normal production tolerances on critical interface dimensions, and actual thickness at critical areas, such as steps in the elevation or contact windows:

Conductor material	Maximum allowable continuous current density (RMS for pulse applications)
Aluminum (99.99 percent pure or doped) without glassivation	2×10^5 amps/cm^2
Aluminum (99.99 percent pure or doped) with glassivation	5×10^5 amps/cm^2
Gold	6×10^5 amps/cm^2
All other (unless otherwise specified)	2×10^5 amps/cm^2

H.3.6.1 Wire bonds. Thermocompression wedge bonds shall not be utilized when aluminum wire is used. Unless otherwise specified in the specification sheet, bi-metallic (e.g. gold-aluminum) bonds at the die shall not be permitted. In specification sheets where bi-metallic bonds at the die are used, see D.3.9.5.b for bake requirements.

H.3.6.2 Die mounting. Pure glass shall not be used for device die mounting.

H.3.7 Silicon transistor metallization protective coating. All silicon transistors with maximum rating of less than 4 watts at T_C of +25°C, shall have an inorganic transparent protective overlay material on the active metallization (excluding the bonding pads). For JANS (overlay structures or expanded metallization) devices, the minimum deposited glassivation thickness shall be 3,500 Å of SiO_2 or 1,000 Å of Si_3N_4. The glassivation shall cover all electrical conductors on the chip except the bonding pads. For JANS (overlay structures or expanded metallization) devices, a minimum of 2 mils (0.050 mm) distance shall be maintained between all uncoated conducting paths, except where the functional performance parameters of the device require closer spacing.

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H.3.8 Critical interface restrictions. Unless it is part of the original design, the external surface of package, header, or flange shall be finished and not have any depression or cavity. External parts, elements, or coatings shall not blister, crack, (excluding glass meniscus), outgas, soften, flow, or exhibit defects that adversely affect storage, operation, or environmental capabilities of semiconductor devices. For JAN, JANTX, and JANTXV the use of silicone or organic material inside the packages shall only be allowed when approved by the qualifying activity. Desiccants shall not be used. For JANS devices, silicone or organic materials may only be used when specified by the specification sheet. Polymer impregnations (such as backfill) of the packages shall not be permitted.

WARNING: Packages containing beryllium oxide (BeO) shall be marked in accordance with 3.10.3.2 and shall not be ground, machined, sandblasted, or subjected to any mechanical operation which will produce dust containing any beryllium compound. Packages containing any beryllium compound shall not be subjected to any chemical process (such as etching) which will produce fumes containing beryllium or it's compounds.

H.3.8.1 Terminal feedthroughs. Methods used for electrical isolation of the terminal feedthroughs for flat packages (TO-254, TO-257, TO-258, TO-259, etc.) with leads .03 inches (0.762 mm) and larger shall employ materials that contain a minimum of 90 percent Al_2O_3 (ceramic).

H.3.9 Metallurgical bond for JANS axial diodes. All JANS diodes (excluding Schottky barrier and point contact ultra high frequency (UHF) devices) shall be metallurgically bonded at the interface of any mechanical connection within the assembly of the device (see appendix A for axial lead diodes).

H.4 PACKAGE FINISH

H.4.1 Package finish. External metallic package elements, including leads and terminals, shall meet the applicable environmental requirements without additional finishing, or shall be finished with a coating which conforms to one of the options listed in H.4.3 and table H-I. Pure tin may not be used to coat any surface nor shall it be used as an undercoat (see H 4.3.f)

H.4.2 Lead and terminal finish. In addition to the requirements of H 4.1, all leads and terminals, except those intended to be attached using threaded fasteners, shall be solderable in accordance with test method 2026 of MIL-STD-750. Combinations of pre-plate electroplate and/or electroless nickel shall not exceed 650 microinches total, and 350 microinches for leads intended to be lead formed. See table H-II for a guideline for standard lead finishes. If the customer requires a specific lead finish it shall be specified on the acquisition document.

H.4.3 Detail lead finish requirements.

a. For all devices mounted by leads or terminals coated by hot solder dipping, the coating shall extend to the seating plane. For devices which are to be connected by wires soldered to lugs or other terminals not used to mount the device, the solder shall cover an area extending .050 inches (1.27 mm) in all directions beyond the designed attachment area.

b. For leads with solder applied over a surface which is not compliant with table H-I, all non-compliant material shall be covered by solder to the package seal or point of lead emergence, or the lot shall pass test method 1041 of MIL-STD-750 salt atmosphere test with sample size of to 22 pieces, no failures allowed.

c. All devices which are solder dipped shall pass screen 14 of table E-IV and table E-V, subgroup 2, appendix E with a sample size of 116 pieces, no failures allowed.

d. All copper or copper clad leads that are to be plated with gold or silver shall first be coated with a barrier layer to prevent diffusion of the copper through the final lead finish.

e. Silver leads and silver cladding shall contain a minimum of 99.7 percent pure silver.

f. Tin based coatings shall be alloyed with a minimum of 3 percent lead which has been shown to inhibit the growth of tin whiskers. Pure tin shall not be used as an undercoat.

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TABLE H-I. Coating thickness and composition requirements.

Coating	Thickness microinch/micrometer		Coating composition requirements
	Minimum	Maximum 1/	
Hot solder dip (for round leads) 2/	60/1.52	NS	The solder bath shall have a nominal composition of Sn60 or Sn63. 3/
Hot solder dip (for all shapes other than round leads) 2/ 4/	200/5.08	NS	The solder bath shall have a nominal composition of Sn60 or Sn63. 3/ 5/
Tin plate (as plated) 6/	300/7.62	NS	Shall contain no more than 0.12 percent by weight co-deposited organic material measured as elemental carbon. See H.4.3. f.
Tin-lead plate (as plated) 4/ 6/	200/5.08	NS	Shall consist of 3 to 50 percent by weight lead (balance nominally tin) homogeneously co-deposited. Shall contain no more than 0.12 percent by weight co-deposited organic material measured as elemental carbon.
Tin dipping 4/	100/2.54		See H.4.3.f.
Gold plate	10/.254	225/5.72	Shall contain a minimum of 99.7 percent gold. Only cobalt shall be used as the hardener.
Silver plate	100/2.54	425/10.8	99.7 percent silver minimum.
Silver cladding	250		
Nickel plate (electroplate) 7/ 8/	50/1.27	350/8.89	The introduction of organic addition agents to nickel bath is prohibited. Up to 40 percent by weight cobalt is permitted as a co-deposit.
Nickel plate (electroless) 7/ 8/	50/1.27	250/6.35	The introduction of organic addition agents to nickel bath is prohibited.
Nickel cladding 9/	50/1.27	350/8.89	

1/ NS = not specified.
2/ See H.4.3.a and H.4.3.b.
3 The solder Sn concentration in the pot may range from 50 percent Sn to 70 percent Sn.
4/ For threaded stud packages and for terminals not intended for mounting the device only, the minimum coating thickness shall be 100 microinches/2.54 micrometers.
5/ As measured to the center of the flat.
6/ The maximum carbon content (and minimum lead content in tin-lead plate) shall be determined by the manufacturer on at least a quarterly basis. The determination of carbon and lead content may be made by any accepted analytical technique (e.g., for carbon: Pyrolysis, infrared detection (using an IR212, IR244 infrared detector or equivalent); for lead: X-ray fluorescence, as long as the assay reflects the actual content in the total deposited finish.
7/ The maximum specified thickness applies to the final coating, provided all previously deposited nickel layers have been annealed to eliminate the residual deposition stress.
8/ Combinations of pre-plate electroplate or electroless nickel and final electroplate nickel finishing shall not exceed 700 microinches/17.78 micrometers total. Combinations of pre-plate electroplate nickel and final electroless nickel plate finishing shall not exceed 600 microinches/15.23 micrometers total. For leads intended to be lead formed the total of all nickel plating layers shall not exceed 350 microinches/8.89 micrometers.
9/ Maximum nickel thickness applies only to lead material.

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TABLE H-II. Guideline for standard lead finishes.

SnPb, dipped or plated	Au plated.
Cylinder style with leads through the base: TO-18, TO-5, TO-39, TO-46, TO-72 etc. (JAN, TX, TXV, JANS as applicable)	Cylinder style with leads through the base: TO-18, TO-5, TO-39, TO-46, TO-72 etc. (JANS only)
Rectangular leaded devices: TO-254, TO-257.	Ceramic Surface Mount U, U1, U2, U3, U4, U5, UA, UB
Heat sink mounted TO packages: TO-3, TO-66, TO-59, TO-63	14 lead flat pack TO-86, all quality levels
Stud mounted rectifiers: DO-4, DO-5, DO-8, DO-9	
All axial and MELF SMD's. DO-35, DO-35UR, DO-7, DO-213	
10, 14 and 16 lead ceramic DIPs	

1/ This table is only provided for a guideline of typical case outline finishes, manufacturers may provide any finish specified in section H.4 herein. If the customer requires a specific lead finish it shall be specified on the acquisition document.

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H.5 PACKAGE OUTLINES AND IDENTIFIERS

Package identifier	Package outlines
T1	TO-254
T2	TO-205AF
T3	TO-257
T4A	TO-204AA (TO-3) (40 mil diameter leads)
T4B	TO-204AD (TO-3) (50 mil diameter leads)
T4C	TO-204AE (TO-3) (60 mil diameter leads)
T5A	TO-213AA (TO-66) (30 mil diameter leads)
T6A	TO-210AA (TO-59) (120q lead orientation, number 3 connected to case)
T6B	TO-210AA (TO-59) (90q lead orientation, all isolated from case)
T6C	TO-210AA (TO-59) (120q lead orientation, number 3 isolated from case)
T7A	TO-210AC (TO-61) (lead number 3 connected to case)
T7B	TO-210AC (TO-61) (all leads isolated from case)
T8	TO-258AA
U1	SMD-1
U2	SMD-2
U2A	SMD (2 leads connected to side, 1 pad on bottom, land pattern of SMD-2)
U2S	TO-276AC with carrier board option
U2L	TO-276AC with lead option
U3	SMD-.5
U4	SMD-.22
U5	LCC 16-20 pin
U6	LCC 28 pin
U7A	DO-217aa SLUGGER
U7B	DO-217ab
U8	SMD-.2
PE1	SOT23
BT	Button tab
D1	TO-254, no tab and lead formed for surface mount application
D3	TO-257, no tab and lead formed for surface mount application
D4	TO-254, No Tab
D5	TO-257, No Tab
FL	Surface mount diode with formed flat leads
U	LCC, 3 leads,6 leads, or unleaded device
UA	LCC 4 or 6 leads
UAC	LCC 4 or 6 leads with a ceramic lid
UB	LCC 4 leads
UBN	LCC 3 leads with a isolated lid
UBC	LCC 4 leads with a ceramic lid
UBCN	LCC 3 leads with a ceramic isolated lid
UMC	LCC 2 leads with a ceramic lid
UR	Surface mount diode with round endcaps
URS	Surface mount diode with one round and one square endcap
US	Surface mount diode with square endcaps
UTK1	ThinKey 1
UTK2	ThinKey 2
UTK3	ThinKey 3
UTK4	ThinKey 4

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H.6 PACKAGE FAMILY GROUPING FOR RESISTANCE TO SOLDERING HEAT (TM2031)

1.	Case mounted cans:	Condition A and (B or C).
2.	Lead mounted cans:	Condition A and (B or C).
3.	Axial leaded glass (tungsten) class I bond:	Condition A and (B or C).
4.	Glass surface mount (tungsten) class I bond:	Condition A, (B or C), and I.
5.	Glass surface mount (dumet) class II and III bond:	Condition A, (B or C), and I.
6.	Axial leaded glass (dumet) class II and III bond:	Condition A and (B or C).
7.	Case mounted packages with ceramic seals:	Condition A and (B or C).
8.	Dual in-line packages:	Condition A and (B or C).
9.	Flat packs:	Condition A and (B or C).
10.	UA and UB:	Condition (B or C) and I.
11.	U1 through U4, also known as SMD 1, SMD 2, SMD.5 and SMD.22:	Condition (B or C) and I.
12.	Studs are exempt from any soldering heat testing: A - Soldering iron. C - Topside wave solder. H - Vapor phase. I - Infrared.	

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